ACADEMIC REGULATIONS & SYLLABUS 2023 -24

Faculty of Computer Science & Applications

Smt. Chandaben Mohanbhai Patel Institute of Computer Applications

Bachelor of Computer Applications (B.C.A.) Programme

(Choice Based Credit System)

Charotar University of Science and Technology (CHARUSAT)
CHARUSAT Campus, At Post: Changa – 388421, Taluka: Petlad, District: Anand
Phone: 02697-247500, Fax: 02697-247100, Email: info@charusat.ac.in
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ACADEMIC RULES

To ensure uniform system of education, duration of under graduate programmes, eligibility criteria for and mode of admission, credit load requirement and its distribution between course and system of examination and other related aspects, following academic rules and regulations are recommended.

1. System of Education

The Semester system of education should be followed across the Charotar University of Science and Technology (CHARUSAT) at Master's levels. Each semester will be at least 90 working days duration. Every enrolled student will be required to take a specified load of course work in the chosen course of specialization and also complete a project/dissertation if any.

2. Duration of Programme

Undergraduate Programme	Bachelor of Computer Applications(BCA)
Minimum	6 Semesters (3 Academic years)
Maximum	10 Semesters (5 Academic years)

3. Eligibility & Mode of Admissions

Eligibility of a candidate and mode of admission to the programme will be according to the regulations for admission committee decided by Government of Gujarat from time to time.

4. Programme structure and Credits

A student admitted to a program should study the course and earn credits specified in the course structure.

5. Attendance

5.1 All activities prescribed under these regulations and listed by the course faculty members in their respective course outlines are compulsory for all students pursuing the courses. No exemption will be given to any student from attendance except on account of serious personal illness or accident or family calamity that may genuinely prevent a student from attending a particular session or a few sessions. However, such unexpected absence from classes and other activities will be required to be condoned by the Dean/Principal.

5.2 Student attendance in a course should be 80%.

6. Course Evaluation

6.1 The performance of every student in each course will be evaluated as follows:

6.1.1 Internal evaluation by the course faculty member(s) based on continuous assessment, for 30% of the marks for the course; and

6.1.2 Final examination by the University through written paper or practical test or oral test or presentation by the student or a combination of any two or more of these, for 70% of the marks for the course.

6.2 University Examination

6.2.1 The final examination by the University for 70% of the evaluation for the course will be through written paper or practical test or oral test or presentation by the student or a combination of any two or more of these.

6.2.2 In order to earn the credit in a course a student has to obtain grade other than FF.

6.2.3 Performance at Internal & University Examination will be done on the relative grading system.

7. Grading

The student's performance in any semester will be assessed by the Semester Grade Point Average (SGPA). Similarly, his performance at the end of two or more consecutive semesters will be denoted by the

Cumulative Grade Point Average (CGPA). The SGPA and CGPA are defined as follows:

Grading Scheme

Range of Marks (%)	≥80	≥75 <80	≥70 <75	≥65 <70	≥60 <65	≥55 <60	≥50 <55	<50
Letter Grade	**AA**	**AB**	**BB**	**BC**	**CC**	**CD**	**DD**	**FF**
Grade Point	**10**	**9**	**8**	**7**	**6**	**5**	**4**	**0**

SGPA = $\sum C_i G_i / \sum C_i$ where C_i is the number of credits of course i
G_i is the Grade Point for the course
i and i = 1 to n, n = number of courses in the
semester
CGPA = $\sum C_i G_i / \sum C_i$ where C_i is the number of credits of course i
G_i is the Grade Point for the course
i and i = 1 to n, n = number of courses of all
semesters
up to which CGPA is computed.

8. Detention Rule

A student will be promoted to next year
only if he/she has cleared all the courses of the year he/she is studying in. Awards of Degree.

8.1 Every student of the programme who fulfils the following criteria will be eligible for the award of the degree:

8.1.1 He should have earned at least minimum required credits as prescribed in course structure; and

8.1.2 He should have cleared all evaluation components in every course; and

8.2 The student who fails to satisfy minimum requirement of CGPA will be a lowed to improve the grades so as to secure a minimum CGPA for the award of degree. Only latest grade will be considered.

9. Award of Class

The class awarded to a student in the programme is decided by the final CGPA as per the following scheme:

Distinction:	**CGPA ≥7.5 & ≤10**
First class:	**CGPA ≥ 6.0 & <7.5**
Second Class:	**CGPA ≥ 5.0 & <6.0**
Pass Class:	**CGPA <5.0**

10. Transcript

The transcript issued to the student at the time of leaving the University will contain a consolidated record of all the courses taken, credits earned, grades obtained, SGPA, CGPA, class obtained, etc.

CHOICE BASED CREDIT SYSTEM

The choice based credit system provides flexibility in designing curriculum and assigning credits based on the course content and hour of teaching. The choice based credit system provides an opportunity for the students to choose courses from the prescribed courses comprising core, elective and open elective courses. The CBCS provides a cafeteria type approach in which the students can take courses of their choice and adopt an interdisciplinary approach to learning. The courses shall be evaluated on the grading system, which is considered to be better than the conventional marks system.

CBCS – Conceptual Definitions / Key Terms (Terminologies)

Types of Courses: The Programme Structure consist of 3 types of courses:

Foundation Courses, Core Courses and Elective Courses.

Foundation Course

These courses are offered by the institute in order to prepare students for studying courses to be offered at higher levels.

Core Courses

A Course which shall compulsorily be studied by a candidate to complete the requirements of a degree / diploma in a said programme of study is defined as a core course. Following core courses are incorporated in CBCS structure:

A. **University Core Courses(UC):**

   University core courses are compulsory courses which are offered across university and must be completed in order to meet the requirements of programme.

B. Programme Core Courses(PC):

Programme core courses are compulsory courses offered by respective programme owners, which must be completed in order to meet the requirements of programme.

Elective Courses

Generally, a course which can be chosen from a pool of courses and which may be very specific or specialized or advanced or supportive to the discipline of study or which provides an extended scope or which enables an exposure to some other discipline / domain or nurtures the candidates

proficiency / skill is called an elective course. Following elective courses are incorporated in CBCS structure:

A. University Elective Courses(UE):

The pool of elective courses offered across all faculties / programmes.

B. Institute Elective Course (IE)

Institute elective courses are those courses which any students of the University/Institute of a Particular Level (PG/UG) will choose as offered or decided by the University/Institute from time-to-time irrespective of their Programme /Specialization.

C. Programme Elective Courses(PE):

The programme specific pool of elective courses offered by respective programme.

Vision, Mission, PEOs, POs and PSOs

Vision

To become a leading institution in the field of computer applications and contribute in national efforts of computerizing public systems

Mission

To produce competent computer professionals with the ability to face future challenges.

PROGRAM EDUCATIONAL OBJECTIVES (PEOs)

The graduates will:

PEO1: Be able to understand the requirement of computing problem and implement an effective solution.

PEO2: Be able to successfully take up various available career options.

PEO3: Be able to continuously learn in their preferred domains.

PEO4: Be able to acquit themselves in ethical and professional manner while demonstrating effective skills.

PROGRAM OUTCOMES (POs)

The Graduate of Computer Science and Applications will be able to:

PO1: **Computational Knowledge**: Apply knowledge of computing fundamentals, computing specialization, mathematics, and domain knowledge to provide effective solution in the area of computing.

PO2: **Problem Analysis**: Identify, formulate, and solve complex computing problems reaching substantiated conclusions using fundamental principles of mathematics, computing sciences, and relevant domain disciplines.

PO3: **Design /Development of Solutions:** Design and evaluate solutions for computing problems, and design and evaluate systems, components, or processes that meet specified needs by considering several societal aspects.

PO4: **Modern Tool Usage:** Create, select, adapt and apply appropriate techniques, resources, and modern computing tools for any computing activities, with an understanding of the limitations.

PO5: **Professional Ethics:** Understand and commit to professional ethics, responsibilities, and norms of professional computing practice.

PO6: **Life-long Learning**: Recognize the need, and have the ability, to engage in independent learning for continual development as a computing professional.

PO7: **Project management:** Demonstrate knowledge and understanding of the computing and management principles and apply these to one's own work, as a member and leader in a team, to manage projects and in multidisciplinary environments.

PO8: **Communication Efficacy:** Communicate effectively with the computing community, and with society at large, about complex computing activities by being able to comprehend and write effective reports, design documentation, make effective presentations, and give and understand clear instructions.

PO9: **Societal and Environmental Concern:** Understand and assess societal and environmental issues within local and global contexts, and the consequential responsibilities relevant to professional computing practice.

P10: **Individual and Team Work:** Function effectively as an individual and as a member or a leader in diverse teams and in multidisciplinary environments**.**

P11: Innovation and Entrepreneurship: Identify a timely opportunity and using innovation to pursue that opportunity to create value and wealth for the betterment of the individual and society at large.

PROGRAM SPECIFIC OUTCOMES

At the end of the programme, the student should be able to

PSO1: Analyze, evaluate and formulate workable computing solutions to real world problems using computing methods.

PSO2: Apply proficiencies of communication, teamwork and management skills with spirit of entrepreneurship.

SALIENT FEATURES

- Four Years unique programme as per National Education Policy-2020.
- Exit option is available after every year (i.e 1^{st}, 2^{nd} and 3^{rd})
- Award of Degree:
    - **Certificate of Computing** (Candidate wish to Exit after the First Year)
    - **Diploma in Computer Applications (**Candidate wish to Exit after the Second Year)
    - **Bachelor of Computer Applications (**Candidate wish to Exit after the Third Year)
    - **Bachelor of Computer Applications(Honor’s)** (Candidate finishes all Four Years)
- Programme is designed based on Choice Based Credit System(CBCS).
- Right blend of fundamentals and latest technologies related courses.
- Exposure of humanities courses in every semester for holistic development of students.
- Interdisciplinary exposure in form of university elective courses.
- Unique methodologies in form of case study and project based learning for teaching-learning.
- Inclusion of environmental awareness through environment science courses.
- Coverage of emerging technologies through expert sessions, seminar and workshop.

Total Credits of Programme (8 Semesters): 196

TEACHING AND EXAMINATIONSCHEME FOR

BCA PROGRAMME

EFFECTIVE FROM ACADEMIC YEAR 2023-24

Teaching & Examination Scheme
Bachelor of Computer Applications (B.C.A.) Programme
(Choice Based Credit System)
Effective from Year 2023 - 24
Semester - I

Course Code	Course Title	Teaching Scheme				Examination Scheme						
		Contact Hours			Credit	Theory			Practical			Total
						Internal		External	Internal		External	
		Theory	Practical	Total		Case Study	Tests/ Criteria		Term work	Tests		
CA121 – 122 **OCBCA1001-1002**	Elective-I	3	-	3	3	10	20	70	-	-	-	100
CA123	Foundation of Programming	4	3	7	7	10	20	70	15	15	70	200
CA124	Digital Computer Fundamentals	4	-	4	4	10	20	70	-	-	-	100
CA125	Fundamentals of Web Designing	-	3	3	3	-	-	-	15	15	70	100
HS101.02 C	Communicative English	-	2	2	2	-		-	30		70	100
		11	08	19	19	300			300			600

Elective-I

Course Code	Course Title
CA121	Information Technology Trends
CA122	Business Organization and Principles of Management
OCBCA1001	Business to Business Marketing
OCBCA1002	Privacy and Security In Online Social Media

Teaching & Examination Scheme
Bachelor of Computer Applications (B.C.A.) Programme
(Choice Based Credit System)
Effective from Year 2023 - 24
Semester - II

Course Code	Course Title	Teaching Scheme				Examination Scheme						
		Contact Hours			Credit	Theory			Practical			Total
						Internal		External	Internal		External	
		Theory	Practical	Total		Case Study	Tests		Term work	Tests		
CA126-127 OCBCA1003-1005	Elective-II	3	-	3	3	10	20	70	-	-	-	100
CA128	Advanced Web Designing	-	3	3	3	-	-	-	15	15	70	100
CA129	Fundamentals of Object Oriented Programming	4	3	7	7	10	20	70	15	15	70	200
CA130	Database Fundamentals	4	3	7	7	10	20	70	15	15	70	200
HS201.02C	Liberal Arts	-	2	2	2	-		-	30		70	100
		11	11	22	22	300			400			700

Elective-II

Course Code	Course Title
CA126	Fundamentals of E-Commerce
CA127	Business Economics
OCBCA1003	Foundations of Cryptography
OCBCA1004	E-Business
OCBCA1005	Entrepreneurship Essentials

Teaching & Examination Scheme
Bachelor of Computer Applications (B.C.A.) Programme
(Choice Based Credit System)
Effective from Year 2023 - 24
Semester - III

Course Code	Course Title	Teaching Scheme				Examination Scheme						
		Contact Hours			Credit	Theory			Practical			Total
						Internal		External	Internal		External	
		Theory	Practical	Total		Case Study	Tests		Term work	Tests		
CA226-227	Elective - III	4	-	4	4	10	20	70	-	-	-	100
CA228	System Analysis and Design	3	-	3	3	10	20	70	-	-	-	100
CA229	Fundamentals of Data Structures and Algorithms	4	3	7	7	10	20	70	15	15	70	200
CA230	Fundamentals of Operating Systems	4	3	7	7	10	20	70	15	15	70	200
HS121.02 C	Creativity, Problem Solving and Innovation	-	2	2	2	-		-	30		70	100
CL144.01 C	Environmental Sciences	-	2	2	2	-		-	30		70	100
	University Elective – I **	-	2	2	2	-	-	-	30		70	100
		15	12	27	27	400			500			900

** Student will take any university elective offered by different institutions of university. CMPICA has decided to offer **Introduction to Web Designing** course for others.

Elective-III	
Course Code	**Course Title**
CA226	Computer Oriented Numerical Methods
CA227	Computer Oriented Management System

University Elective - I			
S.No.	**Course Code**	**Course Name**	**Department/Faculty**
1	CE281.01	Art of Programming	Engineering
2	CL281.01	Environmental Sustainability & Climate Change	Engineering
3	IT281.01	ICT Resources & Multimedia	Engineering
4	ME281.01	Engineering Drawing	Engineering
5	PH233.01	Fundamentals of Packaging	Pharmacy
6	PD260.01	Basic Laboratory Techniques	Applied Science
7	NR251.01	First Aid & Life Support	Nursing
8	PT191.01	Health Promotion & Fitness	Physiotherapy
9	BM231	Banking & Insurance	Management
10	EC281.01	Introduction to MATLAB Programming	Engineering
11	PD261	Astrophysics, Space and Cosmos-1	Applied Science
12	CL283	SDG Handprint Laboratory	Engineering
13	EE284	Python Programming	Engineering

Teaching & Examination Scheme
Bachelor of Computer Applications (B.C.A.) Programme
(Choice Based Credit System)
Effective from Year 2023 - 24
Semester - IV

Course Code	Course Title	Teaching Scheme				Examination Scheme						Total
		Contact Hours			Credit	Theory			Practical			
						Internal			Internal			
		Theory	Practical	Total		Case Study	Tests	External	Term work	Tests	External	
CA231-232	Elective - IV	-	3	3	3	-	-	-	10	20	70	100
CA233	Software Engineering	4	-	4	4	10	20	70	-	-	-	100
CA234	Advanced Database System	-	3	3	3	-	-	-	15	15	70	100
CA235	Open Source Technology	4	3	7	7	10	20	70	15	15	70	200
CA236	Computer Networks	4	-	4	4	10	20	70	-		-	100
HS111.02 C	Human Values and Professional Ethics	-	2	2	2	-		-	30		70	100
	University Elective – II **	-	2	2	2	-	-	-	30		70	100
		12	13	25	25	300			500			800

** Student will take any university elective offered by different institutions of university. CMPICA has decided to offer **Programming the Internet course** for others.

Elective-IV	
Course Code	**Course Title**
CA231	Fundamentals of Embedded Programming
CA232	Foundation of Multimedia

University Elective - II			
S.No.	**Course Code**	**Course Name**	**Department/Faculty**
1	EC282.01	Prototyping Electronics with Arduino	Engineering
2	CE282.01	Web Designing	Engineering
3	CL282.01	Basics of Environmental Impact Assessment	Engineering
4	IT284	Data Visualization	Engineering
5	ME282.01	Material Science	Engineering
6	PH238.02	Cosmetics in Daily Life	Pharmacy
7	NR261.01	Life Style Diseases & Management	Nursing
8	PT192.01	Occupational Health & Ergonomics	Physiotherapy
9	BM241	Health Care Management	Management
10	CA225	Programming the Internet	Computer Applications
11	EE287	MATLAB Programming	Engineering
12	EE288	Maintenance of Household Apparatus	Engineering

Teaching & Examination Scheme
Bachelor of Computer Applications (B.C.A.) Programme
(Choice Based Credit System)
Effective from Year 2023 - 24
Semester - V

Course Code	Course Title	Teaching Scheme				Examination Scheme						
		Contact Hours			Credit	Theory			Practical			Total
		Theory	Practical	Total		Internal		External	Internal		External	
						Case Study	Tests		Term work	Tests		
CA324-326	Elective-V	4	3	7	7	10	20	70	15	15	70	200
CA327	Object Oriented Programming using JAVA	4	3	7	7	10	20	70	15	15	70	200
CA328	Fundamentals of Visual Programming	4	3	7	7	10	20	70	15	15	70	200
CA329	Introduction to Multi-Paradigms Programming language	-	3	3	3	-	-	-	15	15	70	100
HS131.02 C	Communication and Soft skills	-	2	2	2				30		70	100
		12	14	26	26	300			500			800

Elective-V	
Course Code	**Course Title**
CA324	Document Databases
CA325	Foundation of Data Science
CA326	Advanced Embedded Programming

Teaching & Examination Scheme
Bachelor of Computer Applications (B.C.A.) Programme
(Choice Based Credit System)
Effective from Year 2023 - 24
Semester - VI

Course Code	Course Title	Teaching Scheme				Examination Scheme						
		Contact Hours			Credit	Theory			Practical			Total
						Internal		External	Internal		External	
		Theory	Practical	Total		Case Study	Tests		Term work	Tests		
CA330	Basics of Mobile Applications	3	3	6	6	10	20	70	15	15	70	200
CA331	Software Testing	-	3	3	3	-	-	-	15	15	70	100
CA332	Mini Project	-	15	15	15	-	-	-	100		200	300
HS132.02C	Contributor Personality Development	-	2	2	2	-		-	30		70	100
		03	23	26	26	100			600			700

Teaching & Examination Scheme
Bachelor of Computer Applications (B.C.A.) Programme
(Choice Based Credit System)
Effective from Year 2023 - 24
Semester - VII

Course Code	Course Title	Teaching Scheme				Examination Scheme						
		Contact Hours			Credit	Theory			Practical			Total
						Internal		External	Internal		External	
		Theory	Practical	Total		Case Study	Tests		Term work	Tests		
CA333	Online MOOC Course on Current Trends Information Technology #	-	-	-	03	-	-	-	-		-	-
CA334	Minor Project	-	24	24	24	-	-	-	200		400	600
		-	24	24	27	100			600			700

Teaching & Examination Scheme
Bachelor of Computer Applications (B.C.A.) Programme
(Choice Based Credit System)
Effective from Year 2023 - 24
Semester - VIII

<table>
<tr><th rowspan="4">Course Code</th><th rowspan="4">Course Title</th><th colspan="4">Teaching Scheme</th><th colspan="7">Examination Scheme</th></tr>
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<tr><th rowspan="2">Theory</th><th rowspan="2">Practical</th><th rowspan="2">Total</th><th colspan="2">Internal</th><th rowspan="2">External</th><th colspan="2">Internal</th><th rowspan="2">External</th></tr>
<tr><th>Case Study</th><th>Tests</th><th>Term work</th><th>Tests</th></tr>
<tr><td>CA335</td><td>Dissertation</td><td>-</td><td>24</td><td>24</td><td>24</td><td>-</td><td>-</td><td>-</td><td colspan="2">200</td><td>400</td><td>600</td></tr>
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DETAILED SYLLABUS

FOR

BCA PROGRAMME
(1st SEMESTER)

EFFECTIVE FROM
ACADEMIC YEAR 2023-24

Teaching & Examination Scheme
Bachelor of Computer Applications (B.C.A.) Programme
(Choice Based Credit System)
Effective from Year 2023 - 24
Semester - I

Course Code	Course Title	Teaching Scheme				Examination Scheme						
		Contact Hours			Credit	Theory			Practical			Total
		Theory	Practical	Total		Internal		External	Internal		External	
						Case Study	Tests/ Criteria		Term work	Tests		
CA121 – 122 **OCBCA1001-1002**	Elective-I	3	-	3	3	10	20	70	-	-	-	100
CA123	Foundation of Programming	4	3	7	7	10	20	70	15	15	70	200
CA124	Digital Computer Fundamentals	4	-	4	4	10	20	70	-	-	-	100
CA125	Fundamentals of Web Designing	-	3	3	3	-	-	-	15	15	70	100
HS101.02 C	Communicative English	-	2	2	2	-		-	30		70	100
		11	08	19	19	300			300			600

Elective-I	
Course Code	**Course Title**
CA121	Information Technology Trends
CA122	Business Organization and Principles of Management
OCBCA1001	Business to Business Marketing
OCBCA1002	Privacy and Security In Online Social Media

CA121: Information Technology Trends
(100 Marks)

Contact Hours: 03

Pre-requisite: None

Methodology & Pedagogy: During theory lectures emerging trends of information technology will be introduced to students. Emphasis will be given on acquisition, storage and processing of data to generate meaningful information. Students will be made familiar with applications related to information technology. Emerging trends and societal impacts of information technology will be discussed to students.

Outline of the Course:

Unit No.	Title of the Unit	Minimum Number of Theory Hours
1	Information Technology Essentials	06
2	Internet Applications	05
3	Internet Service Trends	07
4	Data Analytics Trends	06
5	GPS and GIS Trends	05
6	Emerging Trends and Societal Impact of IT	07

Total Hours (Theory): 36

Total Hours:36

Detail Syllabus:

Unit I: Information Technology Essentials **Hours 06**

Introduction to Information Technology, Acquisition of various types of data- Introduction to Number and Textual Data, Image Data, Audio and Video Data Numbers and Textual Data, Data Storage and organization.

Unit II: Internet Applications **Hours 05**

Internet: Introduction, Internet basics, Internet protocols, Internet addressing, Browser WWW, E-mail, telnet, FTP, application, benefits and limitation of internet, electronic conferencing, and teleconferencing.

Unit III: Internet Service Trends **Hours 07**

Fundamentals of Cloud Computing, Types of Cloud based Services, Varieties of Cloud Computing, Types of Cloud Computing, Advantages and Disadvantage Cloud Computing, Cloud Computing Providers E-commerce, M-Commerce.

Unit IV: Data Analytics Trends **Hours 06**

Types of Data – Qualitative and Quantitative, Data Analytics basics, Types of Analytics, Common technologies used in data analytics, Tools for Data Analytics

Unit V: GPS and GIS Trends **Hours 05**

What is GPS, Key uses of GPS, how it is applied, what is GIS, Key Uses of GIS, which industries use GIS, Future of GPS and GIS.

Unit VI: Emerging Trends and Societal Impact of IT **Hours 07**

Internet of Things, Virtual and Augmented Reality, Industry 4.0, Image Processing Introduction to Societal Impact, Social Use of World Wide Web, Privacy, Security and Integration of Information, Disaster Recovery, Intellectual Property Right, Career in IT

Core Books:

1. V Rajaraman : Introduction to Information Technology, 3rd Edition, PHI Learning Private Limited, 2018.
2. P.T. Joseph, S.J.: E-Commerce An Indian Perspective, th Edition, PHI Learning Private Limited, 2019.
3. Thomas Erl, Z. Mahmood and R. Puttini: Cloud Computing Concepts, Technology & Architecture, Pearson Publication, 2014

Reference Books:

1. Syed Muhammad Fahad Akhtar, Big Data Architect's Handbook, Packt Publishing ltd, 2018.
2. R. Kalakota & M. Robinson, e-Business 2.0, 2nd Edition, Pearson Publisher, 2012

Web References:

1. https://blog.k2datascience.com/the-basics-of-data-analytics-77e5cc7ea741 [For Data Analytics]
2. https://www.investopedia.com/terms/c/cloud-computing.asp [For Cloud Computing]
3. https://files.eric.ed.gov/fulltext/ED536788.pdf [For Data Analytics]
4. https://nobelsystemsblog.com/facts-between-gis-gps/ [For GPS,GIS]

Course Outcomes: Upon successful completion of the course, students will be,

CO1 :	**Understand the essentials of Information Technology.**
CO2 :	Aware about Internet based applications and their uses in real life.
CO3 :	Understand the trends of Internet Services.
CO4 :	Understand basics of data analytics, GPS and GIS related technologies.
CO5 :	Understand the Emerging trends and Societal Impact of Information Technology.

Course Outcomes Mapping:

Unit No.	Unit Name	Course Outcomes				
		CO1	CO2	CO3	CO4	CO5
1	Information Technology Essentials	√				
2	Internet Applications		√	√		
3	Internet Service Trends			√		
4	Data Analytics Trends				√	
5	GPS and GIS Trends				√	√
6	Others Emerging Trends and Societal Impact of IT					√

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO1	PSO2
CO1	3	1	1	-	-	2	-	-	-	-	-	2	2
CO2	3	1	2	1	-	3	-	-	-	-	2	2	2
CO3	3	1	2	1	-	3	-	-	-	-	2	2	2
CO4	3	1	2	1	-	3	-	-	-	-	2	2	2
CO5	3	1	2	1	2	2	-	-	2	-	-	2	2

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

CA122: Business Organization and Principles of Management

(100 Marks)

Contact Hours: 03.

Pre-requisite: None

Methodology & Pedagogy: This syllabus is designed to impart the basic knowledge of major components of Management, namely Planning, Organizing Business in modern era, Staffing, Leading, Controlling using different techniques.

Outline of the course:

Unit No.	Title of the Unit	Minimum Number of Hours
1	Forms of Business Ownership	6
2	Business Services	7
3	Emerging Modes of Business	4
4	Principles of Management	8
5	Business Ethics and Its Social Responsibilities	4
6	Entrepreneurship Development	7

Total Hours (Theory): 36

Total Hours: 36

Detailed Syllabus:

Unit -I: Forms of Business Ownership **Hours: 06**

Sole proprietorship, Joint Hindu Family Business – meaning, features, merits and demerits, Partnership – meaning, types, registration, types of Partners, Co – Operative societies – types, merits and limitations, Company – Private Ltd, Public Ltd –merits, limitations, Starting a business – Basic factors, Choice of forms of business organizations.

Unit -II: Business Services **Hours: 07**

Nature and types of Business services – Banking, Insurance, Transportation, Warehousing, communication.

Meaning, role, Functions of Banking, Insurance, Postal and Telecom services, Warehousing, Transport, Hospitality and Tourism Services.

Unit -III: Emerging Modes of Business **Hours: 04**

E – business – Meaning, Scope and benefits, Resource required for successful E – Business implementation, On – line transactions, payment mechanism, Security and safety of business transactions, Outsourcing – Concept, need and scope.

Unit -IV: Principles of Management Hours: 08

Definition and nature of management, purpose of management, managerial functions at different levels, management- art or science, planning, planning process & purpose, steps of planning, goal setting, decision making, organizing function, organizing process, formal and informal organizations, centralization and decentralization organization, staffing function, staffing process, recruitment, selection, training, performance appraisal, directing function, concepts of leading, motivating, communicating, qualities of leaders, controlling, control process, control techniques.

Unit -V: Business Ethics and Its Social Responsibilities Hours: 04

Concept of social responsibility, cases for social responsibility, responsibility towards different interest groups, owners, investors, employees, consumers, government, community, public in general, business ethics – concept and elements, business and environmental protection.

Unit -VI: Entrepreneurship Development Hours: 07

The concept and introduction, personality and mindset of an entrepreneur, difference between an entrepreneur, intrapreneur and manager/executive, entrepreneurial eco-system, types and functions of an entrepreneur- entrepreneurial motivation, entrepreneurship development programs, business idea generation, business plan and detailed project report, funding and support aspects.

Core Books:

1. Harold Koontz & Heinz Weihrich: Essentials of Management, Ninth Edition, Tata McGraw Hill Publishing, 2012.
2. Dr. F. C. Sharma, Dr. D. Chandra, Anju Agarwal: Business Organization, SBPD Publications, 2020.

Reference Books:

1. Tripathy & Reddy: Principles of Management, fifth edition, Tata McGraw Hill Publishing, 2015.
2. Robbins , Decenzo & Coulter: Fundamentals of Management, eighth edition, Pearson Education, 2013.
3. Y.K. Bhushan: Fundamentals of Business Organisation, Sultan Chand Publicatio, 2018.

Web References:

1. https://www.dynamictutorialsandservices.org/2018/10/entrepreneurship-development-notes.html [Entrepreneurship Development]
2. https://leverageedu.com/blog/emerging-modes-of-business/ [Emerging Modes of Business]
3. https://www.managementstudyguide.com/management_functions.htm [Functions of Management]

Course Outcomes: Upon successful completion of the course, students will be,

CO1 :	Students will be able to understand different types of business proprietorship.
CO2 :	Students will get the knowledge of functionality of various online business services.
CO3 :	Students will learn the important concepts of management.
CO4 :	To be able to understand the ethics and social responsibilities of business towards various entities.
CO5 :	Students will learn the concepts required to be a good entrepreneur.

Course Outcomes Mapping:

Unit No.	Unit Name	Course Outcomes				
		CO1	**CO2**	**CO3**	**CO4**	**CO5**
1	Forms of Business Ownership	√				
2	Business Services		√			
3	Emerging Modes of Business		√			
4	Principles of Management			√		
5	Business Ethics and Its Social Responsibilities				√	
6	Entrepreneurship Development					√

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO1	PSO2
CO1	-	-	-	-	1	3	-	-	-	2	2	-	1
CO2	-	2	-	2	1	3	2	-	-	2	2	-	2
CO3	-	2	-	-	1	3	-	2	1	3	2	-	2
CO4	-	-	-	-	2	2	-	-	3	3	-	-	2
CO5	-	-	-	-	-	3	-	-	-	2	2	-	2

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation "-"

OCBCA1001: **Business to Business Marketing (B2B)**

Description:

This is course OCBCA1001 – Business To Business Marketing (B2B) is offered from SWAYAM as noc23_mg120– Business To Business Marketing (B2B).

Credit and Week:

Teaching Scheme	Week	Marks	Credit
	12	100	3

About the course:

Business to Business Marketing (B2B) has been one of the most discussed topics in marketing. It involves all kinds of industries be it manufacturing or service oriented ones. It encompasses small, medium and large industries and creates employment for large number of people. Business organizations have been constantly trying to optimize their processes and products in order to provide better products and services to other firms and thus constant innovation in such organizations has been the norm. With growing technology, better transport facilities and advanced research it has grown at a much faster pace in comparison to others.

This course aims at building a strong theoretical base as well as enhances the learner's knowledge on designing successful business strategies and programs. As a part of this course, case studies from across the globe would be used to bring in an experiential learning among the learners.

Pre-requisites:

12th, the Course will aim at building Fundamentals of Industrial marketing at a Beginner's level

Industry support:

This course will be extremely helpful for anyone who is working in the industry in various capacities. It will be helpful for employees of large companies, Public and Private sector undertakings and SME'S

Course layout:

Week 1: Introduction to B2B Marketing: Business marketing, Classifying goods for the business market, Business market customers, Market structure, Environment and Characteristics of Business Marketing, Strategic role of marketing, Commercial enterprises, Commercial and institutional customers, B2B vs B2C Marketing.

Week 2: Organizational Buying and Buyer Behaviour: Organizational buyers' decision process - A Stepwise Model and A Process Flow Model, Organizational and business markets - Government as a customer - Commercial enterprises - Commercial and institutional customers, Value analysis, Buygrid framework, Strategic procurement.

Week 3: B2B Marketing Strategy: Strategy making and strategy management process, Industrial product strategy– Managing Products for Business Markets-Managing Services for Business Markets-Managing Business Market channels The Growth-Share Matrix, Multifactor Portfolio Matrix, The Balanced Scorecard.

Week 4: B2B Marketing STP: Market Segmentation, bases for segmenting business markets, basic framework of segmentation, choosing target segments and positioning.

Week 5: Business Marketing Communications- B2B Advertising, Digital marketing,- Trade shows, exhibitions, business meets - Managing the sales force - Deployment analysis, Direct marketing.

Week 6: Demand forecasting: industrial market, Forecasting- meaning, importance and relevance, issues related to forecasting, forecasting measurement models, sales force forecasting, estimating segment demand, Collaborative approach to estimate demand, qualitative and quantitative forecasting methods.

Week 7: Product management: (existing and new) in industrial market, role of product in the industrial market, new product development, industrial product life cycle, product evaluation matrix, techniques for indentifying new products QFD, perceptual mapping, reverse engineering, fish bone diagram, role of service and maintenance in industrial markets, customer experience life cycle, service quality.

Week 8: Pricing: Pricing strategies; The pricing policy; Price on the Internet; Financial marketing, competitive bidding, commercial terms and conditions, role of leasing.

Week 9: Buyer seller relationship, types of relationships, transactional and collaborative relationships, influencing industrial customers, role of service in industrial markets. CRM.

Week 10: B2B marketing research, challenges in B2B research, developing a marketing information system, role of qualitative research techniques in B2B research.

Week 11: Business marketing channels and participants - Channel design and management decisions -B2B logistics management, types of industrial middlemen And intermediaries, marketing logistics and physical distribution.

Week 12: Strategic decision making in industrial markets, strategic planning at corporate levels, allocation of resources, portfolio analysis, developing SBU'S objectives and goals, implementing and controlling marketing plan. Marketing through electronic commerce.

Books and references

Text Books / Basic Material

1. Business Market Management Understanding, Creating and Delivering Value By James C. Anderson, Das Narayandas, James A. Narus and D.V.R. Seshadri Pearson, 2010 3rd edition
2. Business Marketing Management b2b By Hutt and Speh South-Western CENGAGE, Learning www.cengagebrain.com 2013.
3. B2B Brand Management By Kotler and Pfoertsch Springer www.springer.com 2006
4. Business Marketing: Text and Cases By Krishna K Havaldar, McGrawhill Publications, 2014 4th edition.

Criteria to get a certificate:
Average assignment score = 25% of average of best 8 assignments out of the total 12 assignments given in the course.
Exam score = 75% of the proctored certification exam score out of 100.

Final score = Average assignment score + Exam score.

YOU WILL BE ELIGIBLE FOR A CERTIFICATE ONLY IF AVERAGE ASSIGNMENT SCORE >=10/25 AND EXAM SCORE >= 30/75. If one of the 2 criteria is not met, you will not get the certificate even if the Final score >= 40/100.

OCBCA1002: **Privacy and Security in Online Social Media**

Description:

This is course OCBCA1002 – Privacy and Security in Online Social Media is offered from SWAYAM as noc23_cs13– Privacy and Security in Online Social Media

Credit and Week:

Teaching Scheme	Week	Marks	Credit
	12	100	3

About the course:

With increase in the usage of the Internet, there has been an exponential increase in the use of online social media and networks on the Internet. Websites like Facebook, YouTube, LinkedIn, Twitter, Flickr, Instagram, Google+, Foursquare, Pinterest, Tinder, and the likes have changed the way the Internet is being used. However, widely used, there is a lack of understanding of privacy and security issues on online social media. Privacy and security of online social media need to be investigated, studied and characterized from various perspectives (computational, cultural, psychological, etc.). Student completing the course will be able to appreciate various privacy and security concerns (spam, phishing, fraud nodes, identity theft) on Online Social Media and Student will be able to clearly articulate one or two concerns comprehensively on one Online Social Media, this will be achieved by homework.

Pre-requisites:

Basic / Intermediate programming course. Understanding of Python will be necessary for the course. Should be able to quickly learn APIs, and to collect data from social networks.

Industry support:

Any company which is interested in social media / networks data will be interested in recruiting the students finishing the course.

Course layout:

Week 1: What is Online Social Networks, data collection from social networks, challenges, opportunities, and pitfalls in online social networks, APIs

Week 2: Collecting data from Online Social Media.

Week 3: Trust, credibility, and reputations in social systems

Week 4: Trust, credibility, and reputations in social systems

Week 5: Online social Media and Policing

Week 6: Information privacy disclosure, revelation and its effects in OSM and online social networks

Week 7: Phishing in OSM & Identifying fraudulent entities in online social networks

Week 8: Refresher for all topics

Week 9 to 12: Research paper discussion

Books and references

Text Books / Basic Material

1. https://cdn.iiit.ac.in/cdn/precog.iiit.ac.in/teaching/psosmonnptel/2023/slides/3.2-2021.pdf
2. Social Media Security by Micheal Cross, publisher : syngress
3. https://cse.usf.edu/dsg/data/publications/papers/privacy_survey_imrul.pdf

Criteria to get a certificate:
Average assignment score = 25% of average of best 8 assignments out of the total 12 assignments given in the course.
Exam score = 75% of the proctored certification exam score out of 100

Final score = Average assignment score + Exam score

YOU WILL BE ELIGIBLE FOR A CERTIFICATE ONLY IF AVERAGE ASSIGNMENT SCORE >=10/25 AND EXAM SCORE >= 30/75. If one of the 2 criteria is not met, you will not get the certificate even if the Final score >= 40/100.

CA123: Foundation of Programming
(200 Marks)

Contact Hours: 07

Pre-requisite: None

Methodology & Pedagogy: During theory lectures the emphasis will be given on theoretical concepts of flow charts and algorithms. Structure of C programming language and different elements of C programming will be explained in lectures. During practical sessions the concepts learned in theory lectures will be implemented.

Outline of the course:

Unit No.	Title of the Unit	Minimum Number of Hours	
		Theory	Practical
1	Programming Basics	8	36
2	Introduction to C	10	
3	Expressions and Control Statements	9	
4	Arrays and Strings	7	
5	Introduction to Functions	5	
6	Structure, Enumeration datatypes and Pointers	9	

Total Hours (Theory): 48

Total Hours (Lab): 36

Total Hours: 84

Detail Syllabus:

Unit I: Programming Basics **Hours 08**

Introduction to algorithm, Key features of algorithm, Introduction to flow charts, Significance of flow chart, Advantages and limitation of flow chart, Introduction to programming languages, Introduction to editor, compiler, and translator

Unit II: Introduction to C **Hours 10**

Introduction, characteristics of C, Structure of C Program, writing first C program, Files used in C program, Compiling and Executing C program; Basic data types in C, User define data types, Variables in C, Declaring and initializing variables in C, Constants, Input / Output statements in C; Operators - arithmetic, relational, logical, assignment, increment - decrement, conditional, Bitwise and special, comma operator, size-of operator, operator precedence chart.

Unit III: Expressions and Control Statements **Hours 09**

Arithmetic expressions, evaluation of expressions, type conversions in expressions, operator precedence and associativity, mathematical functions. Type conversion and casting; Introduction to decision control statements, Conditional and branching statements, loop, nested loop, Break, continue and go to statement.

Unit IV: Array and Strings **Hours 07**

Arrays: One-dimensional, two-dimensional, Handling of Character Strings: Declaring and initializing string variables, reading string from terminal, writing string to screen, string handling functions, table of strings.

Unit V: Introduction to Functions **Hours 05**

Need for user-defined functions, the form of c function, return values and their types, calling a function, category of functions, handling of non-integer functions, nesting of Functions, recursion, functions with arrays, storage class.

Unit VI: Structure, Enumeration datatypes and Pointers **Hours 09**

Understanding the Computer's Memory, Introduction to Pointers, Declaring Pointer Variables, Pointer Expressions and Pointer Arithmetic, Null Pointers, Generic Pointers, passing arguments to Function using Pointers, Pointers and Arrays, Passing an Array to a Function, Difference between Array name and Pointer, Pointers and Strings, Array of Pointers.

Core Books:

1. ReemaThareja: Computer Fundamentals and Programming in C, 2nd Edition, Oxford University press, 2012
2. E. Balagurusamy: Programming In C, 6th Edition, 2012
3. PradipDey and Manas Ghosh. “Programming in C”, Second Edition, Oxford University Press.

Reference Books:

1. Herbert Schildt: The complete reference C, Fourth edition, Tata McGraw Hill,2000
2. YashwantKanetkar: Let us C, 13th Edition, BPB publication, 2013.
3. PradipDey, Manas Ghosh: Programming in C, 4th Edition, Oxford University press , 2012.
4. Brian W. Kerighan and Dennis M. Ritchie : The C Programming Language,Prentice Hall.

Web References:

1. https://cprogrampracticals.blogspot.com/p/basic-concepts.html [For basic c programing practical]
2. https://www.programtopia.net/c-programming/docs/operators-expressions [For operators and expressions in C]
3. https://www.javatpoint.com/functions-in-c [For functions]
4. https://www.programiz.com/c-programming [For basic c programing practical]
5. https://www.javatpoint.com/c-pointers [pointers in c]

Course Outcomes: Upon successful completion of the course, students will be,

CO1 :	Be able to solve problems using algorithms and flowcharts.
CO2 :	Learn about the structure of C program and able to develop simple programs using the C Programming language with data types, variables and constants.
CO3 :	Gain the knowledge of different conditional and iterative statements and expressions.
CO4 :	Arrays of numbers and characters as well as manipulating with them.
CO5 :	Be able to understand user define function. Also Understand the concept of Dynamic memory management. Learn Programing-using Array with Pointer and Structure.

Course Outcomes Mapping:

Unit No.	Unit Name	Course Outcomes				
		CO1	**CO2**	**CO3**	**CO4**	**CO5**
1	Programming Basics	√				
2	Introduction to C		√			
3	Expressions and Control Statements			√		
4	Arrays and Strings		√	√	√	
5	Introduction to Functions		√	√		√
6	Structure, Enumeration datatypes and Pointers		√			√

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	2	2	3	2	2	1	2	2	2	1	1	1	3	3
CO2	3	3	3	3	3	3	3	3	3	3	3	3	3	3
CO3	2	2	3	3	2	2	2	2	2	1	1	1	3	2
CO4	3	3	3	3	3	3	3	3	3	2	3	3	3	2
CO5	3	3	3	3	3	3	3	3	3	2	3	3	3	2

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation "-"

CA124: Digital Computer Fundamentals
(100 Marks)

Contact Hours: 04

Pre-requisite: None

Methodology & Pedagogy: In order to achieve the course objectives, students will be introduced to digital technologies. Various digital modules used to create digital computer devices like gates, flip flops, decoder, encoder etc. are to be studied. Mathematical base is created to understand the organization of digital computer.

Outline of the course:

Unit No.	Title of the Unit	Minimum Number of Hours
1	Digital Systems and Binary Numbers	10
2	Boolean Algebra and Logic Gates	10
3	Combinational Circuit Designing	8
4	Sequential Circuit Designing	8
5	Memory Interfaces and Data Representation	6
6	The Arithmetic Logic Unit and Control Unit	6

Total Hours (Theory): 48

Total Hours (Lab)

Total Hours: 48

Detailed Syllabus:

Unit - I: Digital Systems and Binary Numbers: Hours 10

Introduction and Evolution of Digital computers, Major components of digital computer, Interfaces and Buses, compiler, interpreter, Assembler, Introduction to various number systems (Decimal, Binary, Octal, Hexadecimal, BCD), Conversion of number system, Binary operations, signed binary numbers, 1's complement and 2's complement.

Unit - II: Boolean Algebra and Logic Gates: Hours 10

Fundamental concepts and rules of Boolean algebra, , Boolean Algebra Laws, Boolean expressions and its simplification with and without Truth Tables, Canonical form and standard form, D-Morgan's Theorem, Basic duality laws, Derivation of Boolean expressions, Introduction to logic gates (AND, OR, NOT, NAND, NOR, XOR, XNOR), Implementation of Boolean expressions using logic gates, Sum of product and Product of Sum forms, Universal gates, Implementation of other gates using universal gates, Karnaugh Map method for two, three, four and five variables, K-map with don't care conditions, Tabulation method, determination and selection of prime applicants.

Unit - III: Combinational Circuit Designing: Hours 08

Introduction to combinational circuits, Design procedures, Adders, Subtractors, code conversion circuits, Analysis procedures, Multilevel NAND circuits, Multilevel NOR circuits, EX-OR functions, Binary adder and Subtractors, Decimal adder and Subtractors, Magnitude comparator, Decoder and encoders, Multiplexer and de-multiplexer.

Unit - IV: Sequential Circuit Designing: Hours 08

Introduction to Sequential circuits, Concepts of Flip-flops, triggering flip-flops, Clock signals, Analysis of clock sequential circuits, State Reduction and assignments, Flip flop excitation table, Design procedure of asynchronous and synchronous counters (Ripple counter, Binary counter, BCD counter), Design of registers, Shift registers, Timing sequences.

Unit - V: Memory Interfaces and Data Representation: Hours 06

Memory, Types of ROM, Design of ROM, Types of RAM, Magnetic disc memory, Magnetic tape, Digital recording techniques, Linear-select memory organization, Types of digital codes (Gray code, 8421 code, Alphanumeric codes), Parity checking codes, Floating point representation, Fixed point representation.

Unit - VI: The Arithmetic Logic Unit and Control Unit: Hours 06

Basic operations of Arithmetic Logic Unit, Construction of the Arithmetic Logic Unit, Representation of Instruction word, Control registers, Instruction and execution cycles of Control registers, Sequence operation of Control Registers.

Core Books:

1. M. Morris Mano: Digital Logic and Computer Design, Third Edition, Pearson Education, 2016.
2. Thomas C. Bartee: Digital Computer Fundamentals, Sixth Edition, Tata McGraw Hill Publishing, 2012.
3. M. Morris Mano: Computer System Architecture, Third Edition, Pearson Education, 2011.

Reference Books:

1. Andrew S. Tanenbaum: Structured Computer Organization, Fourth Edition, Pearson Education, 2005.
2. Abert Paul Malvino and Jerald A. Brown: Digital Computer Electronics, Third Edition, Tata McGraw Hill Publishing, 2008.

Web References:

1. https://www.tutorialspoint.com/digital_circuits/index.htm [Digital Circuits Tutorial]
2. https://www.geeksforgeeks.org/digital-electronics-logic-design-tutorials/ [Digital Electronics and Logic Design Tutorials]
3. https://studymaterialz.in/digital-logic-and-computer-design-by-morris-mano/ [Digital Logic and Computer Design book]
4. https://cupola.gettysburg.edu/oer/1/ [Digital Circuit Projects]

Course Outcomes: Upon successful completion of the course, students will be,

CO1 :	To get the idea about digital system and numbering system.
CO2 :	To study logic gates for digital circuit designing.
CO3 :	Learn to design various combinational circuits
CO4 :	Learn to design various sequential circuits
CO5 :	To learn memory design concepts and to study computer Arithmetic Logic Unit and Control Unit

Course Outcomes Mapping:

Unit No.	Unit Name	Course Outcomes				
		CO1	CO2	CO3	CO4	CO5
1	Digital Systems and Binary Numbers	√				
2	Boolean Algebra and Logic Gates		√			
3	Combinational Circuit Designing			√		
4	Sequential Circuit Designing				√	
5	Memory Interfaces and Data Representation					√
6	The Arithmetic Logic Unit and Control Unit					√

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO1	PSO2
CO1	3	2	2	2	-	3	2	-	-	2	1	3	2
CO2	3	2	2	2	-	3	2	-	-	2	1	3	2
CO3	3	3	3	2	-	2	2	-	-	2	2	3	2
CO4	3	3	3	2	-	2	2	-	-	2	2	3	2
CO5	3	3	3	2	-	3	2	-	-	2	2	3	2

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation "-"

CA125: Fundamentals of Web Designing
(100 Marks)

Contact Hours: 03

Pre-requisite: Basic concepts of web

Methodology & Pedagogy: This course focuses on providing hands-on experience to students for design and develop entire web sites using several web designing tools and HTML scripting language.

Outline of the course:

Week No	Practical	Description
1- 2	Basic Construction of an HTML & HTML5 Page, HTML Text Editors, HTML Building blocks. Adding Content : Basic HTML tags – Text formatting tags, heading tags, paragraph tags, division tag, Span tags, Break line tag, Binding space in HTML	Student can learn basic concepts of web page design. Students get familiar with different tags and attributes which is used to add the text in a web page
3 - 4	Controlling Font size and color, Adding Table – Table tag, grouping and merging table rows and columns. Adding List – ordered, unordered and definition list	Student can able to change the font styles and color in more attractive way. Also, they can able to display the information in tabular & List format
5 - 6	Adding Image, Adding links - Changing link colors, Email link, Embedding audio and video, Adding Field sets.	Student can add multimedia like image, audio, video in a web page. Student can explore various internal and external linking methods.
7 - 8	Adding Form: Form tag, Action, Method. Basic form controls – Textbox, Password field, Radio Button, Checkbox, Multiple line text area, file upload, Various types of buttons – submit, reset, image, simple button,	Student can design the various form elements for interaction with the web page users.
9 - 10	HTML5 input elements - Color, Number, Date, Month, week, Time, datetime-local, Email, url, range, search, <datalist>, <output>, <progress>, <meter>	Student can apply HTML5 input element to make form more attractive
11 - 12	Regular expression, HTML5 new attributes – Placeholder, Required, Pattern, Autocomplete, Autofocus, novalidate, formnovalidate, formaction, formmethod, spellcheck, contenteditable.	Student can explore the basic HTML5 new attributes and also apply validation on form submission

Total Hours (Practical): 36
Total Hours: 36

Core Books:

1. Powel Thomas: HTML - The Complete Reference, 3rd Edition, McGraw-Hill Education – Europe,2018.
2. DT Editorial Services: HTML 5 Black Book,2nd Edition, Dreamtech Press, 2016
3. Matt West: HTML 5 Foundations, Wiley publication, 2013
4. Farrar: HTML Example book, BPB,2007.

Reference Books:

1. David DuRocher : HTML and CSS QuickStart Guide, Paperback,2021
2. Laura Lemay, Rafe Colburn, Jennifer Kyrnin : Mastering Html, Css & Javascript Web Publishing, Paperback, 2016.
3. Whyte: Basic HTML, 2nd Edition, Payne-Gallway, Oxford, 2003.
4. Shelly Woods: HTML introductory concepts and techniques, 5th Edition, Course Technology,2009.
5. Jon Dukett: Beginning Web Programming with HTML, XHTML and CSS, Wrox Publication.

Web References:

1. www.w3schools.com/html [For HTML tutorials]
2. html.net/tutorials/html/ [For HTML tutorials]
3. www.htmlgoodies.com/ [For HTML resources]
4. https://developer.mozilla.org/en-US/learn/html [For HTML best practices]

Course Outcomes: Upon successful completion of the course, students will be,

CO1 :	Able to understand basic construction of HTML & HTML5 document and started using basic tags.
CO2 :	Able to Add content using various tags and format the content in table & list.
CO3 :	Able to design links and multimedia elements such as image, audio, video.
CO4 :	Able to design a web page form using HTML & HTML5.
CO5 :	Able to apply validation using HTML5 attributes.

Course Outcomes Mapping:

Week No.	Practical Title	Course Outcomes				
		CO1	CO2	CO3	CO4	CO5
1	Basics of Hyper Text Markup Language	√				
2	Controlling font and adding Tables & Lists		√	√		
3	Adding Image, audio, video and Links			√		
4	Adding Form and Basic Form Controls				√	√
5	Adding HTML5 new Elements		√			√
6	**Adding HTML5 new attributes & apply validation on form control**		√		√	√

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO1	PSO2
CO1	1	2	1	3	2	2	1	1	-	1	1	1	1
CO2	-	1	2	3	1	3	1	-	-	2	2	2	2
CO3	1	1	2	3	1	3	1	-	-	2	2	2	2
CO4	1	1	2	3	1	1	1	1	-	1	-	1	1
CO5	2	2	3	3	2	3	3	2	1	2	2	3	2

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

HS 101.02 C: Communicative English

(100 Marks)

Contact Hours: 30 Hours

Pre-requisite courses: Grammar and Punctuation (Free Online Course for Beginners) https://www.coursera.org/learn/grammar-punctuation

Outline of the Course:

Unit No.	Title of the unit	Minimum number of hours
1.	Introduction to Communicative English	03
2.	Communication Functions	06
3.	Basic Communication Skills I – Listening	03
4.	Basic Communication Skills II – Reading	03
5.	Basic Communication Skills III – Speaking	06
6.	Basic Communication Skills IV – Writing	06
7.	Developing Vocabulary	03

Total Hours (Practical): 30

Total Hours: 30

Detailed Syllabus:

Unit I :	**Introduction to Communicative English**	**03 Hours**
	English as a Window Language; Varieties of English: British English, American English, Indian English; Language Variations; Importance of English for Academic and Professional Development; Strategies for Language Acquisition; Formal VS Informal English	
Unit II :	**Communication Functions**	**06 Hours**
	Greeting and Introducing; Making Requests and Asking for Information; Expressing Likes and Dislikes; Seeking Permission; Giving and Taking Advice; Describing People, Place, Things; Retelling Past Events; Comparing and Contrasting; Persuading; Describing Cause and Effect	
Unit III :	**Basic Communication Skills I – Listening**	**03 Hours**
	Importance of Listening as a Language Skill; Basic Listening Skills; Types of Listening; Barriers to Listening; Strategies for Effective Listening; Listening Practice	
Unit IV :	**Basic Communication Skills II – Reading**	**03 Hours**
	Importance of Reading as a Language Skill, Reading Strategies: Skimming, Scanning, Intensive Reading, Extensive Reading, Strategies for Effective Reading Comprehension, Reading Practice	
Unit V :	**Basic Communication Skills III – Speaking**	**06 Hours**
	Importance of Speaking as a Language Skill, Basic Speaking Skills; Paralanguage for Effective Speaking; Strategies for Oral Communication; Extempore and Public Speaking	
Unit VI :	**Basic Communication Skills IV – Writing**	**06 Hours**
	Importance of Writing as a Language Skill; Process of writing: Prewriting, Drafting, Revision, Editing, Publication; Seven C's of Writing; Sentence Construction – Complex, Compound, Paragraph Development; Letter Writing (Academic Context)	

Unit VII : **Developing Vocabulary** **03 Hours**

High Frequency Vocabulary (Every day and Academic use); Words Often Confused and Misused; Useful Phrasal Verbs, Idioms and Proverbs, Homonyms and Homographs; Lexical Range, Word Games

Text book:

1. Sanjay Kumar and PushpLata (Second Edition, 2015), Communication Skills, Oxford University Press, New Delhi
2. M V Rodriques (2013), Effective Business Communication, Concept Publishing Company (P) Ltd., New Delhi
3. Krishna Mohan and Meera Banerji (2010), Developing Communication Skills, Macmillan Publications India Ltd., New Delhi

Reference book:

1. Mohan and Meenakshi Raman (2006), Effective English Communication, Mcgraw-Hill Publishing Company Limited, New Delhi
2. Geoffrey Leech & Jan Swartvik (1994), A Communicative Grammar of English, Longman Publications, New York
3. Jones Leo (1979), Functions of English, Cambridge University Press, UK
4. European Journal of Language and Literature Studies Vol.1 Nr, 1 April 2015
5. English for Academic Purpose: A Tool for Enhancing Students' Proficiency in English Language Skills

Web material:

1. https://www.futurelearn.com/courses/language-assessment
2. https://www.coursera.org/learn/importance-of-listening?#syllabus
3. https://www.futurelearn.com/courses/english-academic-study

Course Outcome (COs): At the end of the course, the students will be able to

CO1	Communicate with people using English language functions including greetings, introductions, making and responding to requests, suggestions, invitations and apologies, conducting simple transactions in shops and offices, asking for and giving directions, etc
CO2	Go through a text and identify specific and global information.
CO3	Become more knowledgeable about speaking strategies and speak effectively using appropriate words, expression, tone and pronunciation.
CO4	Be aware about various reading strategies and read and comprehend academic and non-academic prose (text).
CO5	Write systematically using nuances of writing
CO6	Express their opinion and likes and dislikes, advice and convince others in a more polite and accepted way.

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO1	PSO2
CO1	-	-	-	-	-	-	-	-	3	-	-	-	3
CO2	-	-	-	-	-	-	-	-	3	-	-	-	-
CO3	-	-	-	-	-	-	-	-	3	-	-	-	-
CO4	-	-	-	-	-	-	-	-	3	-	-	-	-
CO5	-	-	-	-	-	-	-	-	3	-	-	-	-
CO6	-	-	-	-	-	2	-	-	3	-	-	-	-

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

DETAILED SYLLABUS

FOR

BCA PROGRAMME
(2nd SEMESTER)

EFFECTIVE FROM
ACADEMIC YEAR 2023-24

Teaching & Examination Scheme
Bachelor of Computer Applications (B.C.A.) Programme
(Choice Based Credit System)
Effective from Year 2023 - 24
Semester - II

Course Code	Course Title	Teaching Scheme				Examination Scheme						
		Contact Hours			Credit	Theory			Practical			Total
						Internal		External	Internal		External	
		Theory	Practical	Total		Case Study	Tests		Term work	Tests		
CA126-127 OCBCA1003-1005	Elective-II	3	-	3	3	10	20	70	-	-	-	100
CA128	Advanced Web Designing	-	3	3	3	-	-	-	15	15	70	100
CA129	Fundamentals of Object Oriented Programming	4	3	7	7	10	20	70	15	15	70	200
CA130	Database Fundamentals	4	3	7	7	10	20	70	15	15	70	200
HS201.02C	Liberal Arts	-	2	2	2	-		-	30		70	100
		11	11	22	22	300			400			700

Elective-II	
Course Code	**Course Title**
CA126	Fundamentals of E-Commerce
CA127	Business Economics
OCBCA1003	Foundations of Cryptography
OCBCA1004	E-Business
OCBCA1005	Entrepreneurship Essentials

CA126: Fundamentals of Electronic Commerce (100 marks)

Contact Hours: 03

Pre-requisite: None

Methodology & Pedagogy: During theory lectures, the fundamental concepts regarding Ecommerce, E-Payment, E-Security, E-branding and E-Data Interchange, CRM, ERP will be discussed. Students will be required to carry out case studies using the concepts and techniques they have learnt during theory sessions.

Outline of the course:

Unit No.	Title of the Unit	Minimum Number of Theory Hours
1	Introduction to E-commerce	07
2	E- Marketing	06
3	E-Security and E-Payment	07
4	Mobile Commerce	05
5	CRM and ERP	07
6	E-Commerce Applications	04

Total Hours (Theory):36

Total Hours:36

Detail Syllabus:

Unit I: Introduction to E-commerce **Hours: 07**

Emergence and use of Internet, origin of web and world wide web, advantages and disadvantages of E-commerce, Features of E-Commerce, Electronic Commerce over the Internet, Types of E-commerce.

Unit II: E- Marketing: **Hours:06**

Traditional Marketing, Online Marketing, E- advertising, Internet Marketing Trends and strategies, E-branding.

Unit III: E-Security and E-Payment: **Hours:07**

Security on the Internet, E-Business risk management issues, Digital token based E-Payment System, Properties of Electronic Cash, Digital Signature.

Unit IV: Mobile Commerce: **Hours:05**

Definition of M-Commerce, Features of M-Commerce, Advantages and Disadvantages of M-Commerce, Areas of M-Commerce Applications, Payment Systems and Models in M-Commerce.

Unit V: CRM and ERP **Hours: 07**

Customer Relationship Management, CRM capabilities and Customer life cycle, Introduction to ERP, Reasons for the growth of the ERP Market, Advantages and Disadvantages of ERP.

Unit VI: E-Commerce Applications **Hours:04**

Retail and Wholesale, Finance, Manufacturing, Online Booking, Online Publishing, Digital Advertising, Digital Shopping, Digital Media, Auctions, A case study need to be discussed for these application. Hands-on experience in setting up and managing an e-commerce store. Developing a business plan for an ecommerce startup.

Core Books:

1. P.T.Joseph, S.J.: E-Commerce- An Indian Perspective, 3rd Edition, PHI learning Private Limited.
2. Kamlesh K. Bajaj, Debjani Nag: E-Commerce, The Cutting Edge of Business, 2nd Edition, McGraw-Hill Education.

Reference Books:

1. Anita Rosen: The e-commerce Question and Answer Book: A Survival Guide for Business Managers, 2nd Edition, Amacom.
2. Janice Reynolds: The Complete E-Commerce Book: Design, Build & Maintain a Successful Web-based Business, 2nd Edition, CRC Press.
3. By Philippe Humeau&Matthieu Jung : The White Book of Ecommerce Solutions, NBS System.
4. IshitaLahiri and Sujit Kumar Ghose : Principals of Marketing and E-Commerce

Web References:

1. http://www.ecommercetutorial.net/ [For Tutorial]

Course Outcomes: Upon successful completion of the course, students will be,

CO1 :	To understand basic of Electronic Commerce and its Application
CO2 :	Getting knowledge of E-marketing and its Strategies.
CO3 :	Learn E-security and E-Payment in Detail and its term. Also, learn Encryption and decryption in basic.
CO4 :	To gain understanding of Mobile Commerce and its Application through real life examples.
CO5 :	To learn concepts of Customer Relationship Management and various solutions of E-Commerce (Software and Framework).

Course Outcomes Mapping:

Unit No.	Unit Name	Course Outcomes				
		CO1	CO2	CO3	CO4	CO5
1	Introduction to E-commerce	√				
2	E- Marketing	√	√			
3	E-Security and E-Payment		√	√		
4	Mobile Commerce		√	√		
5	CRM and ERP				√	
6	Application of E-Commerce					√

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	2	2	3	2	2	1	2	2	2	1	1	1	3	3
CO2	3	3	3	3	3	3	3	3	3	3	3	3	3	3
CO3	2	2	3	3	2	2	2	2	2	1	1	1	3	2
CO4	3	3	3	3	3	3	3	3	3	2	3	3	3	2
CO5	3	3	3	3	3	3	3	3	3	2	3	3	3	2

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

CA127-Business Economics
(100 Marks)

Contact Hours: 03

Pre-requisite: None

Methodology & Pedagogy: During theoretical lessons, real-world examples will be used to discuss economic topics such as demand, supply, production, cost function, market structure, factor pricing, and microeconomics. Students will be offered appropriate case studies in order to provide actual exposure to theoretical economic ideas.

Outline of the course:

Unit No.	Title of the Unit	Minimum Number of Hours
1	Fundamentals of Economics	05
2	Basic Elements of Demand and Supply	06
3	Theory of Production and Cost	07
4	Analysis of Revenue	07
5	Market Failures and Price Regulation	05
6	Macro-Economic Environment and National Income Accounting	06

Total Hours (Theory): 36

Total Hours: 36

Detail Syllabus:

Unit - I: Fundamentals of Economics **Hours: 05**

Fundamentals of Economics, Needs, wants, nature of economics, scope of economics, business economics, laws of economics, business cycle, difference between Micro and Macro Economics.

Unit - II: Basic Elements of Demand and Supply **Hours: 06**

Demand and supply, types of demand, determinants of demand and supply, law of demand and supply, balance between demand and supply, demand and supply schedule, demand function.

Unit - III: Theory of Production and Cost **Hours: 07**

Theory of Production- Factors of Production, Basic Concepts, Production Function, Producer's Equilibrium- Least-Cost Factor Combination and Output Maximization for a given Level of Outlay. Theory of Costs- Basic Concepts, Short-run Total Cost, Short-run Average and Marginal Cost, Relationship between Average and Marginal Cost, Average and Marginal Cost in the Long-run.

Unit- IV: Analysis of Revenue Hours: 07

Basic Concepts of Revenue, Revenue, Relationship between Average and Marginal Revenue, Concept of Market and Main Forms of Market; Equilibrium of the Firm- Meaning, Objectives of the Firm, Monopoly and its features, Monopolistic Competition and Oligopoly.

Unit - V: Market Failures and Price Regulation Hours: 05

Market failures and need for regulation, Regulations and market structure, Firm behavior, Price regulation.

Unit - VI: Macro-Economic Environment and National Income Accounting Hours: 06

Basic macro- economic concepts –aggregate, demand aggregate supply, money, income employment consumption savings and investment. National Income Accounting- concepts and methods of national income, different methods of national income accounting.

Core Books:

1. P.L Mehta: Managerial Economics, Sultan Chand and Sons Publication, 2014
2. Keat Paul , K Young Philip , Erfle Steve, Banerjee Sreejata: Managerial Economics, seventh edition, Pearson Education, 2017.
3. N. Gregory Mankiw: Principles of Economics, sixth edition, South-Western, Cengage Learning, 2012.

Reference Books:

1. Geetika, Ghosh & Choudhury: Managerial Economics, third edition, Mc Graw Hill Education, 2005.
2. Mote V.L., Samuel Paul and G.S. Gupta: Managerial Economics Concepts and Cases, Tata McGraw Hill Publishing Company Ltd., New Delhi, 2001.
3. G. S. Gupta: Macro Economics Theory and Applications, Second Edition, Tata McGraw Hill, 2006.

Web References:

1. https://www.sciencedirect.com/topics/economics-econometrics-and-finance/economic-law [Economic Law]
2. https://www.toppr.com/guides/business-economics-cs/basic-elements-of-demand-and-supply
   [Basic Elements of Demand and Supply]
3. https://corporatefinanceinstitute.com/resources/knowledge/economics/national-income-accounting/ [National Income Accounting].

Course Outcomes: Upon successful completion of the course, students will be,

CO1 :	Able to Interpret the various basic economic principles**.**
CO2 :	Able to understand theory of demand and supply.
CO3 :	Able to deal with balancing between cost and revenue
CO4:	Able to learn the causes and solution of market failures.
CO5:	Able to understand the basics of macroeconomics and accounting.

Course Outcomes Mapping:

Unit No.	**Unit Name**	**Course Outcomes**				
		CO1	**CO2**	**CO3**	**CO4**	**CO5**
1	Fundamentals of Economics	√				
2	Basic Elements of Demand and Supply		√			
3	Theory of Production and Cost			√		
4	Analysis of Revenue			√		
5	Market Failures and Price Regulation				√	
6	Macro-Economic Environment and National Income Accounting					√

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO1	PSO2
CO1	3	1	1	-	-	2	-	-	1	-	2	2	1
CO2	3	1	2	-	-	2	-	-	1	-	2	2	1
CO3	3	1	2	-	-	2	-	-	1	-	2	2	1
CO4	3	1	2	-	-	2	-	-	1	-	2	2	1
CO5	3	1	1	2	-	2	-	-	1	-	2	2	1

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation "-"

OCBCA1003: Foundations of Cryptography

Description:

This is course OCBCA1003 – **Foundations of Cryptography** is offered from SWAYAM as noc24_cs01 – Foundations of Cryptography**.**

Credit and Week:

Teaching Scheme	Week	Marks	Credit
	12	100	3

About the course:

The course provides the basic paradigm and principles of modern cryptography. The focus of this course will be on definitions and constructions of various cryptographic objects. We will try to understand what security properties are desirable in such objects, how to formally define these properties, and how to design objects that satisfy the definitions. The aim is that at the end of this course, the students are able to understand a significant portion of current cryptography research papers and standards.

Pre-requisites:

There are no pre-requisites for this course. However, it is expected that the students who are interested to take this course have had some exposure to a basic course on discrete mathematics, algorithms, or theory of computation. Having said that, I ensure that a significant effort will be made from my side to simplify the overall presentation of the course and make it easily accessible.

Industry support:

The course will be relevant for any IT related company.

Course layout:

Week 1: Course Overview, Symmetric-key Encryption, Historical Ciphers, Perfect Security and Its Limitations.

Week 2: Computational Security, Semantic Security and Pseudorandom Generators (PRGs)

Week 3: Stream Ciphers, Provably-secure Instantiation of PRG, Practical Instantiation of PRG, CPA-security and Pseudo-random Functions (PRFs).

Week 4: CPA-Secure Ciphers from PRF, Modes of Operations of Block Ciphers, Theoretical Constructions of Block Ciphers and Practical Constructions of Block Ciphers

Week 5: DES, AES and Message Authentication Codes (MAC).

Week 6: Information-theoretic Secure MAC, Cryptographic Hash Functions, Ideal-Cipher Model, Davies-Meyer construction and Merkle-Damgård Paradigm.

Week 7: Birthday Attacks on Cryptographic Hash Functions, Applications of Hash Functions, Random Oracle Model and Authenticated Encryption.

Week 8: Generic Constructions of Authenticated Encryption Schemes, Key-exchange Problem, One-way Trapdoor Functions and Cyclic Groups.

Week 9: Discrete-Logarithm Problem, Computational Diffie-Hellman Problem, Decisional Diffie-Hellman Problem, Elliptic-Curve Based Cryptography and Public-Key Encryption.

Week 10: El Gamal Encryption Scheme, RSA Assumption, RSA Public-key Cryptosystem, KEM-DEM Paradigm and CCA-security in the Public-key Domain

Week 11: CCA-secure Public-key Hybrid Ciphers Based on Diffie-Hellman Problems and RSA-assumption, Digital Signatures, RSA Signatures and Schnorr Identification Scheme.

Week 12: Schnorr Signature, Overview of TLS/SSL, Number Theory, Interactive Protocols and Farewell.

Books and references

Text Books / Basic Material

1. Introduction to Modern Cryptography by Jonathan Katz and Yehuda Lindell.
2. Cryptography Theory and Practice by Douglas Stinson.

CRITERIA TO GET A CERTIFICATE

Average assignment score = 25% of average of best 8 assignments out of the total 12 assignments given in the course.
Exam score = 75% of the proctored certification exam score out of 100

Final score = Average assignment score + Exam score

YOU WILL BE ELIGIBLE FOR A CERTIFICATE ONLY IF AVERAGE ASSIGNMENT SCORE >=10/25 AND EXAM SCORE >= 30/75.

If one of the 2 criteria is not met, you will not get the certificate even if the Final score >= 40/100.

OCBCA1004: E-Business

Description:

This is course OCBCA1004 – **E-Business** is offered from SWAYAM as noc24_mg16 – E-Business**.**

Credit and Week:

Teaching Scheme	Week	Marks	Credit
	12	100	3

About the course:

The Internet has changed the way companies carry out their businesses. The primary objective of this course is to introduce concepts, tools and approaches to electronic business to the post-graduate and undergraduate students. Further, the subject will help the students to develop skills to manage businesses in the digital world. The course will cover following aspects of E-Business Systems.

- Part 1: Foundations of E-Business systems
- Part 2: Infrastructure
- Part 3: Functional Areas
- Part 4: Decision Support for E-Business Systems

The course provides a balance approach including concepts from technology and management.

Pre-requisites: NIL

Industry support: All the companies.

Course layout:

Week 1: Introduction to E-Business

Week 2: Making Functional Areas E-Business Enabled : Value chain and supply chain, inter and intra organizational business processes, ERP

Week 3: Making Functional Areas E-Business Enabled : E-Procurement.

Week 4: Making Functional Areas E-Business Enabled : E-marketing, E-Selling, E-Supply Chain Management

Week 5: Technologies for E-Business: Internet and Web based system

Week 6: Technologies for E-Business: Security and payment systems

Week 7: Technologies for E-Business: Supply chain integration technologies (EDI, RFID, Sensors, IoT, GPS, GIS)

Week 8: Technologies for E-Business: Supply chain integration technologies (Web services and cloud)

Week 9: Decision Support in E-Business: Web analytics

Week 10: Decision Support in E-Business: Customer behavior modeling

Week 11: Decision Support in E-Business: Auctions

Week 12: Decision Support in E-Business: Recommender systems

BOOKS AND REFERENCES

1. Management Information Systems: Managing the Digital Firm, Laudon and Laudon, Pearson

2. Scaling for E-Business, Menasce & Almeida, PHI

3. eBusiness & eCommerce – Managing the Digital Value Chain, Meier & Stormer, Springer

4. eBook is available in springerlink.com

5. Some reference books, Internet Resources, and Research Papers

CRITERIA TO GET A CERTIFICATE

Average assignment score = 25% of average of best 8 assignments out of the total 12 assignments given in the course.
Exam score = 75% of the proctored certification exam score out of 100

Final score = Average assignment score + Exam score

YOU WILL BE ELIGIBLE FOR A CERTIFICATE ONLY IF AVERAGE ASSIGNMENT SCORE >=10/25 AND EXAM SCORE >= 30/75.

If one of the 2 criteria is not met, you will not get the certificate even if the Final score >= 40/100.

OCBCA1005: Entrepreneurship Essentials

Description:

This is course OCBCA1005 – **Entrepreneurship Essentials** is offered from SWAYAM as noc24_ge15 – Entrepreneurship Essentials.

Credit and Week:

Teaching Scheme	Week	Marks	Credit
	12	100	3

About the course:

The course provides foundational knowledge on various aspects of entrepreneurial venture creation and management during its life-cycle. It has been designed to address multidisciplinary audiences. The objective of the course is to teach key issues faced by entrepreneurs and managers at different stages of the life-cycle of an enterprise and is relevant both for aspiring entrepreneurs and for decision makers in established enterprises. Topics can be classified in some major themes such as : Making a choice to create an entrepreneurial venture, current trend of technology entrepreneurship, how to start a start-up, identifying opportunities, factors driving competitive advantages, organizational structure, basic knowledge of financial statements and project report, introductory knowledge on marketing management, human resource management, & strategic management, risk analysis, legal aspect of business, how to raise fund during life-cycle of a new ventures.

Pre-requisites: All students with education up to higher secondary can take up this course.

Industry support: All start-ups, companies in emerging technology domains, companies adopting 'corporate entrepreneurship', professionally managed companies

Course layout:

Week 1: Introduction

Dhirubhai Ambani & Sofia

Myths & Realities about entrepreneurship

entrepreneurial qualities

Why start-ups fail?

Week 2: Mission, vision, entrepreneurial qualities – I

Mission, vision, entrepreneurial qualities – II

Value proposition

Business Model canvas

Business model generation

Week 3: Competitive advantage

Lean start-up – 1

Lean start-up – 2

Team and early recruit

Legal forms of business

Week 4: Marketing management 1

Marketing management 2

Market research –I

Market research –II

Market research –Example

Week 5: Introduction to financial statements

Profit & Loss statement

Balance sheet

Cash flow

Example – 1

Example – 2

Cost-volume-profit & Bread-Even analysis

Capital budgeting

Week 6: Business plan-I
Business plan-II
Pitching
Go-to-market strategies Does & Don'ts

Week 7: How to innovate
Design Thinking
Design-Driven Innovation, Systems thinking
Open innovation, TRIZ
How to start a start-up?

Week 8: Government incentives for entrepreneurship (1 lecture)
Incubation, acceleration
Funding new ventures – bootstrapping, crowd sourcing,
angel investors, VCs, debt financing (3), due diligence
Legal aspects of business (IPR, GST, Labour law)

Week 9: Cost, volume, profit and break-even analysis
Margin of safety and degree of operating leverage
Capital budgeting for comparing projects or opportunities
Product costing
Product pricing

Week 10: Funding new ventures – bootstrapping, crowd sourcing,
Angel investors, VCs, debt financing (3), and due diligence
Incubation and acceleration
Government incentives for entrepreneurship
Project cost and Financial Closure

Week 11: Dos & Donts in entrepreneurship
Growth Hacking
Growth Strategy
Legal aspects of business (IPR, GST, Labor law)
Negotiation skill

Week 12: Human Resource management in startups
Pivoting
Entrepreneurial cases
Risk assessment and analysis

Strategy management for entrepreneurial ventures
Factors driving success and failure of ventures
Concluding remarks

BOOKS AND REFERENCES

1. Effective Entrepreneurial Management: Strategy, Planning, RiskManagement, and Organization - Robert D. Hisrich • VelandRamadani, Springer (2017)

2. Entrepreneurship- Theory, Process Practice –by Kuratko &Hodgetts, Thompson South-Western Publication

3. Entrepreneurship –by Robert D. Hisrich (Edition-9)

CRITERIA TO GET A CERTIFICATE

Average assignment score = 25% of average of best 8 assignments out of the total 12 assignments given in the course.
Exam score = 75% of the proctored certification exam score out of 100

Final score = Average assignment score + Exam score

YOU WILL BE ELIGIBLE FOR A CERTIFICATE ONLY IF AVERAGE ASSIGNMENT SCORE >=10/25 AND EXAM SCORE >= 30/75.

If one of the 2 criteria is not met, you will not get the certificate even if the Final score >= 40/100.

CA128: Advance Web Designing

(100 Marks)

Contact Hours: 03

Pre-requisite: CA123 – Fundamentals of Web Designing

Methodology & Pedagogy: This subject focuses on hands-on experiments to create dynamic and interactive web sites using HTML5, CSS3 and JavaScript.

Outline of the course:

Week No	Practical	Description
1- 2	CSS Styling the web - Applying CSS Building blocks – CSS Selectors, Cascade & inheritance, CSS Properties, CSS Box model.	Student can apply various types of CSS with its basic properties.
3 - 4	Creating Layout - Header, footer, navigation menu, and sub-navigation. Floats, Flexbox, Grids, Positioning, Multiple column layout.	Student can create different layout using semantic elements of HTML5 & CSS classification properties.
5 - 6	Creating Image Gallery and Animation - 2D/3D Transformation, Transitions and Animation, CSS Media query	Student can create layout image gallery & animation.
7 - 8	JavaScript – Adding Internal & External JavaScript, Working with variables, datatype – Number, String, Boolean, Array, Object, Operators, Conditional statements, Looping statements.	Student can write script for basic programs using condition, Loop, array and object.
9 - 10	Adding Dialog boxes in HTML page – Alert, Confirm and Prompt. Creating user defined function, Adding Mouse and Keyboard Event in HTML page, Manipulating String and Date.	Student can able to use in-build and user defined functions of JavaScript and make user interaction using various Events
11 - 12	JavaScript Document Object Model hierarchy – Create find and manipulate HTML Element using Objects and methods. Form validation, Applying Style using JavaScript. Creating New window, Accessing & manipulating History of HTML Pages.	Student can explore the DOM Hierarchy and apply various objects to make web page dynamic with user interactions.

Total Hours (Practical): 36
Total Hours:36

Core Books:

1. Julie C. Meloni, Jennifer Kyrnin : HTML, CSS, and JavaScript All in One, 3rd Edition, Pearson, 2018.
2. Joseph R. Lewis, Meitar Moscovitz : AdvancED CSS, W3C CSS Working Group, Apress, 2020.
3. Peter Gasston: The Book of CSS3: A Developer's Guide to the Future of Web Design, No Starch Press, April 2011.
4. MArijin Haverbeke : Eloquent Javascript – A modern Introduction to JavaScript, 3rd Edition, 2018
5. Michael Morrison: Head First JavaScript, O'Reilly, 2014

Reference Books:

1. Andy Budd, Emil Björklund: CSS Mastery 3rd Edition, Apress - Advance Web Standard Solutions.
2. Karl Barksdale, Shane Turner: HTML and Java Scripts Basics, 4th Edition, Course Technology,2006.
3. Jon Dukett: Beginning Web Programming with HTML, XHTML and CSS, Wrox Publication.
4. Richard York: Beginning CSS: Cascading Style sheets for Web Design, Wrox Press (Wiley Publishing), 2005.

Web References:

1. https://www.tutorialspoint.com/css/css3_tutorial.htm [For CSS Basics]
2. https://www.tutorialrepublic.com/css-tutorial/ [For CSS3 BOX Model & Animation]
3. https://javascript.info/ [For modern JavaScript tutorials]
4. https://www.tutorialsteacher.com/javascript/javascript-hoisting [JavaScript Basics]

Course Outcomes: Upon successful completion of the course, students will be,

CO1 :	Able to apply CSS & its basic properties in web pages.
CO2 :	Able to create HTML, CSS layout.
CO3 :	Able to apply the transition, transformation and animation in web page.
CO4 :	Able to write client side script – JavaScript to perform basic program
CO5 :	Able to use DOM objects for dynamic web pages

Course Outcomes Mapping:

Week No.	Practical Title	Course Outcomes				
		CO1	CO2	CO3	CO4	CO5
1	CSS Styles & Properties	√				
2	CSS Layout	√	√			
3	CSS Animation & Transition	√		√		
4	Adding client side script – JavaScript				√	
5	Handling function and events in JavaScript				√	
6	Accessing and manipulating DOM in JavaScript				√	√

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO1	PSO2
CO1	1	2	3	3	2	1	2	1	1	2	1	1	1
CO2	1	3	3	3	2	2	1	1	1	2	3	2	1
CO3	1	2	2	3	2	1	1	1	1	2	2	1	1
CO4	2	2	3	3	2	1	1	1	2	1	2	1	1
CO5	2	2	3	3	2	1	2	1	2	1	2	2	2

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

CA129: Fundamentals of Object Oriented Programming (200 Marks)

Contact Hours: 07

Pre-requisite: CA123: Foundation of Programming.

Methodology & Pedagogy: During the Theory sessions concept will be covered with real world examples to understand the application and usage of object oriented programming over procedural programming. During Practical sessions, students will implement the concepts using various assignments and case study.

Outline of the Course:

Unit No.	Title of the Unit	Minimum Number of Hours	
		Theory	Practical
1.	Introduction to Object Oriented Programming	06	36
2.	Classes and Objects	08	
3.	Polymorphism	09	
4.	Inheritance	09	
5.	Exception Handling and Type cast operators	09	
6.	Managing I/O formats and operations	07	

Total Hours (Theory): 48
Total Hours (Lab): 36
Total: 84

Detailed Syllabus:

Unit - I: Introduction to Object Oriented Programming Hours:06

Structured programming vs. object oriented programming, Introduction to C++: structure of a C++ program Tokens, inline function, pass by reference, default arguments, OOP characteristics.

Unit - II: Classes and Objects Hours:08

Define class, objects, visibility modes, static members, friend function, Constructors and Destructors, Default Constructor, Copy constructors, Parameterized Constructor, this keyword.

Unit - III: Polymorphism Hours:09

Introduction of Static Binding, Function Overloading and ambiguity raised in function overloading,;Operator overloading through member function and friend functions, type conversions from primitive to class, class to primitive and class to class.

Unit - IV: Inheritance Hours:09

Objective of Inheritance, types of inheritance, impact of access modes during inheritance, Virtual base classes, Dynamic Binding, Virtual functions, Pure virtual functions, Abstract class, Early vs. Late binding.

Unit - V: Exception Handling and Type Cast Operators Hours: 09

Introduction to Exception handling, catching class types, using multiple catch statements, exception handling options, creating custom exceptions, introduction to cast operators, const cast, static cast, reinterpret cast, dynamic cast.

Unit - VI: Managing I/O Formats and Operations Hours:07

Understand and apply concepts from various header files like iostream. Iomanip ,fstream, etc. opening and closing file, reading and writing text files and binary files, performing random access on files, handle command line arguments.

Core Books:

1. Herbert Schildt: C++: A Beginner's Guide, 2nd Edition Paperback, McGraw-Hill,2003.
2. E Balagurusamy: Object Oriented Programming with C++ Paperback, McGraw Hill India,2017.

Reference Books:

1. Yashavant Kanetkar: Let Us C++, Paperback, BPB Publications,2020.
2. Robert Lafore: Object Oriented Programming in C++, 4th Edition, Sams Publications,2002.
3. Bruce Eckel:Thinking in C++, Volume 1, 2nd Edition, Pearson Education,2006.

Web References:

1. https://www.geeksforgeeks.org/c-plus-plus/ [For Entire Syllabus]
2. https://google.github.io/styleguide/cppguide.html

[For coding standards and concepts]

3. http://www.tutorialspoint.com/cplusplus/cpp_basic_input_output.htm

[For Practicals]

Course Outcomes: Upon successful completion of the course, the students will:

Co1 :	Understand the basic structure of C++ program and understand the concept of object oriented programming characteristics.
Co2 :	Understand the concept of Constructor and Destructor and able to create class and objects.
Co3 :	Understand the need of Polymorphism and implement static polymorphism
Co4 :	Understand the purpose of Inheritance, identify relations between classes, implement inheritance and perform dynamic binding.
Co5 :	Able to handle the runtime errors or exceptions in C++ and use type cast operators. Able to read input from device, file, command line arguments and apply formats to input and output.

Course Outcomes Mapping:

Unit No.	Unit Name		Course Outcomes			
		Co1	Co2	Co3	Co4	Co5
1	Introduction to Object Oriented Programming	√				
2	Classes and Objects		√			
3	Polymorphism			√		
4	Inheritance				√	
5	Exception Handling and Type cast operators					√
6	Managing I/O formats and operations					√

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO1	PSO2
CO1	2	2	2	2	2	2	2	2	2	2	2	3	2
CO2	3	3	3	2	2	2	3	2	2	2	2	3	2
CO3	3	2	3	2	2	2	3	2	2	2	2	3	2
CO4	3	3	3	2	2	2	3	2	2	2	2	3	2
CO5	3	2	3	2	2	2	2	2	2	2	2	3	2

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

CA130: Database Fundamentals
(200 Marks)

Contact Hours: 07

Pre-requisite: Basic knowledge of working with computer

Methodology & Pedagogy: During the theory lectures, concepts for data storage will be discussed. Also DBMS and RDBMS concepts will be discussed in depth. The database applications used in the real world will be discussed with necessary examples. During the laboratory hours' students will implement the concepts that are discussed during lecture.

Outline of the course:

Unit No.	Title of the Unit	Minimum Number of Hours	
		Theory	Practical
1	Introduction to Database System	04	36
2	Data Modeling using Entity Relationship Model	08	
3	Relational Data Modeling	08	
4	Database Design Methodology	08	
5	Schema Definition, Constraints, Queries	10	
6	Advanced Query Processing	10	

Total Hours (Theory): 48

Total Hours (Lab): 36

Total Hours: 84

Detail Syllabus:

Unit I: Introduction to Database System **Hours: 04**

Basic Concepts: data, **information**, **file**, database, database systems, database management system Purpose and advantages of Database management system (over file systems), **Applications of DBMS** Various data models – ER Model and Relational Model, Three level architecture, Structure of DBMS, Database actors and workers

Unit II: Data Modeling using Entity Relationship Model **Hours: 08**

Entity Types, Entity Sets, Attributes, Keys, Relationship Types, Relationship Sets, Roles and Structural Constraints, Weak Entity Types, ER Diagram – Notations and fundamentals, Concept of Generalization and Specialization, Associative Entity, Conversion from ER Model to Relational Model

Unit III: Relational Data Modeling **Hours: 08**

Relational Model Concepts, Relational Model Constraints, Relational Model Schemas, Update Operation, Transactions, Dealing with Constraint Violations, Relational Algebra and Calculus, Difference between ER Model and Relational Model.

Unit IV: Database Design Methodology Hours: 08

Functional Dependency and Normalization for Database – Informal Design Guidelines for Relational Schemas, Functional Dependencies, Normal Forms (1NF, 2NF, 3NF, BCNF, 4NF, 5NF), Codd Rules

Unit V: Schema Definition, Constraints, Queries
Hours: 10

Table Fundamentals: Basic Data types, Create Table Command, Viewing Data in the tables, sorting, Insert, Delete and Update Statements in SQL, Modifying the structure of tables, renaming table, truncating table, destroying table.; Data Constraints and Functions: - Pseudo columns, Null values, TAB table, DUAL table Operators, Data constraints, Type of data constraints, modifying constraints, working with data dictionary and use of USER_CONSTRAINTS Functions – introduction, merits and demerits, types of functions Numeric functions, Character functions, Date functions, Conversion functions, Aggregate functions

Unit VI: Advanced Query Processing Hours: 10

Union, Intersect and Minus Clause, Except Nested Queries – Introduction, Co-related Nested Queries, Set Comparison Operators Aggregate Operators, Group By clause with Rollup and Cube, having clause, Order by clause Null Values – Comparison using Null Values, Logical Connectivity – AND, OR and NOT, Using Like predicate and IN, EXISTS, NOT EXISTS, ANY Operators, Impact on SQL constructs, Joins (Inner Join, Outer Join, Self-Join, Equi Join, Cross Join) Complex Integrity Constraints in SQL – constraints over single table, domain constraints and distinct types Creation and manipulation of database objects – indexes, views, sequences and synonym.

Core Books:

1. RamezElmasri, Shamkant B. Navathe : Fundamentals of Database Systems , 5th Edition, , Pearson Publication.
2. Ivan Bayross : SQL, PL/SQL – The programming Language Oracle.
3. Ramkrishnan, Gehrke : Database Management Systems, 3 rd Edition, McGrawHill Publication.

Reference Books:

1. Silberschatz, Korth, Sudarshan : Database System Concepts, 5th Edition, McGraw Hill.
2. C.J.Date, a Kannan, S Swaminathan : An Introduction to Database Systems, 8th Edition, Pearson Education,(Equivalent Reading).
3. Scoot Urban : Oracle 9i, PL/SQL Programming, Oracle Press.
4. S. K. Singh : Database Systems: Concepts, Design and Applications, Pearson Education
5. Peter Rob, Carlos Coronel: Database Systems: Design, Implementation and Management, 7th Edition, Cengage Learning, 2007.
6. Anjali, Amisha, Roopal and Nirav : Practice book on SQL and PL/SQL.
7. Leon and Leon : Database management Systems, Vikas Publication.

Web References:

1. https://www.javatpoint.com/er-model-vs-relational-model [For difference of ER and Relation model]
2. http://www.microsoftvirtualacademy.com/trainingcourses/databasefundamentals#?fbid=tbZ92pOp_Tt [For overall subject]
3. http://www.ntu.edu.sg/home/ehchua/programming/sql/Relational_Database_Design.html [for relational database design]
4. http://docs.oracle.com/cd/A97335_02/apps.102/a81358/05_dev1.htm [For ER Diagram]
5. http://plsql-tutorial.com/[For PL/SQL]

Course Outcomes: Upon successful completion of the course, students will be,

CO1 :	**Understand basic concepts regarding data, database systems and various data models.**
CO2 :	Understand various data modeling techniques and entity relationship models.
CO3 :	Learn about relational data model and related concepts.
CO4 :	Gain insights into database design, SQL and normalization concepts.
CO5 :	Advanced query processing

Course Outcomes Mapping:

Unit No.	Unit Name	Course Outcomes				
		CO1	CO2	CO3	CO4	CO5
1	Introduction to Database System	√				
2	Data Modeling using Entity Relationship Model		√			
3	Relational Data Modeling			√		
4	Database Design Methodology				√	
5	SQL, Schema Definition, Constraints, Queries					√
6	Advanced Query Processing					√

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO1	PSO2
CO1	2	3	1	1	-	3	-	-	-	-	-	2	-
CO2	1	3	3	3	-	3	1	-	-	-	-	1	5
CO3	3	3	3	3	-	3	2	-	-	2	-	3	2
CO4	-	3	3	3	-	3	3	-	-	3	3	-	3
CO5	2	3	3	3	2	3	3	-	-	3	3	2	3

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation "-"

HS201.02 C: PAINTING

(100 Marks)

Contact Hours: 30 Hrs.

Pre-requisite: None

Instructional Methods and Pedagogy: Teaching will be practical based on the hands on experiences, live and interactive sessions. It will run in the workshop mode. Four Workshops (each of a day) will be organized during the semester.

Evaluation: The students will be evaluated continuously in the form of their consistent performance throughout the semester. There is no theoretical evaluation. There is just practical evaluation. The evaluation (practical) is schemed as 30 marks for internal evaluation and 70 marks for external evaluation.

Internal Evaluation: Students' performance in the course will be evaluated on a continuous basis through the following components:

Sl. No.	Component	Number	Marks per incidence	Total Marks
1	Participation	-	05	05
2	Performance/ Activities	-	05	05
3	Project	-	15	15
4	Attendance	-	05	05
Total				**30**

External Evaluation: University Practical examination will be for 70 marks and will test the performance, activities and creative presentations of the students with reference to the course selected**:**

Sl. No.	Component	Number	Marks per incidence	Total Marks
1	Viva / Practical	-	70	70
Total				**70**

Outline of the course:

Unit No.	Title of the unit	Minimum number of hours
1	An Introduction to Painting	02
2	Drawing from Nature and Object	04
3	Colour Design and Colour Value	06
4	Composition and Perspective	06
5	Figure Drawing and Proportion	04
6	Sketching	04
7	Contemporary Issues in Painting	04

Practical Hours: 30 Hrs.
Total hours: 30 Hrs.

Detailed Syllabus:

Unit – I	**An Introduction to Painting**	**02 Hours**

- An Introduction to Painting
- Principles of Composition
- Medium and Techniques of Painting
- History of Painting: Folk Indian Painting / Western Painting
- 2D and 3D Painting

Unit – II	**Drawing from Nature and Object**	**04 Hours**

- Objects of Drawing: Nature and Manmade / Artificial Objects
- Drawing Still / Live Objects
- Drawing from Memory
- Drawing from Life

Unit - III	**Light and Shade**	**06 Hours**

- **Color Theory**:
  Color wheel (primary/secondary, complementary), transparency/opacity, hue, value (intensity, brightness), chroma (saturation, purity) & temperature (warm/cold)
- **Color Contrast & Attributes**:
  Interaction, harmony, psychology/mood, culture & expression
- **Media Characteristics & Surfaces**:
  Acrylic, oil, paper, wood & canvas (primed/unprimed)

Unit – IV	**Composition and Perspective**	**06 Hours**

- **Composition:**
  Space, movement, balance, asymmetry, rhythm, shapes, proportion & lighting
- Perspective: An approximate reproduction
- **Types of Perspectives:**
- Linear Perspective, One-point Perspective, Two-point Perspective, Three-point Perspective, Four-Point Perspective

Unit – V	**Figure Drawing and Proportion**	**04 Hours**

- Proportions of the Human Body
- Three views – **Anterior (front), Lateral (side) and Posterior (back)**
- Fundamental Proportion – The Big Three

Unit – VI	**Sketching**	**04 Hours**

- Sketching and Freehand
- Sketching Techniques
- Sketch and Drawing Medium

Unit VII	**Contemporary Issues in Painting** • Contemporary Indian Art • Pioneers of Contemporary Indian Art • Contemporary Issues in Painting	**04 Hours**

Text books/Reference books:

1. A.N. Hodge (2007) *The History of Art: The Essential Guide to Painting Through the Ages* Arcturus Foulsham, London
2. David Lewis (1984) Pencil *Drawing Techniques* Watson-Guptill Publications, New York
3. John C Van Dyke L H D (1895) *A Text Book of the History of Painting* Longmans Green and Co., London
4. Sarah Parks (2014) Drawing *Secrets Revealed - Basics How to Draw Anything-* North Light Books, Ohio

Web Material

1. https://conceptartempire.com/what-are-the-fundamentals/
2. https://www.creativebloq.com/art/painting-techniques-artists-31619638
3. https://www.thesprucecrafts.com/perspective-in-paintings-2578098
4. https://www.thehansindia.com/posts/index/Hans/2016-05-19/Origin-of-painting-in-India/229138

Course Outcome (COs): At the end of the course, the students will be able to:

CO1	have cultivated a sense of creativity.
CO2	be appreciative of art history, art criticism and aesthetics.
CO3	be able to recognize the elements of arts in painting.
CO4	have better cognizance and association of meaning of colors, shapes, and composition.
CO5	be able to acknowledge the principles of painting as in design and colors, concept, medium and formats.
CO6	have instantaneous painting experience about designing, lights, shades and colors and such other important aspects.

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO1	PSO2
CO1	-	-	-	-	-	-	-	-	3	-	-	-	-
CO2	-	-	-	-	-	-	-	-	3	-	-	-	-
CO3	-	-	-	-	-	-	-	-	3	-	-	-	-
CO4	-	-	-	-	-	-	-	-	3	-	-	-	-
CO5	-	-	-	-	-	2	-	-	3	-	-	-	-
Co6	-	-	-	-	-	-	-	-	3	-	-	-	-

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

HS202.02 C: PHOTOGRAPHY

(100 Marks)

Contact Hours: 30 Hrs.

Pre-requisite: None

Instructional Methods and Pedagogy: Teaching will be practical based on the hands on experiences, live and interactive sessions. It will run in the workshop mode. Four Workshops (each of a day) will be organized during the semester.

Evaluation: The students will be evaluated continuously in the form of their consistent performance throughout the semester. There is no theoretical evaluation. There is just practical evaluation. The evaluation (practical) is schemed as 30 marks for internal evaluation and 70 marks for external evaluation.

Internal Evaluation: Students' performance in the course will be evaluated on a continuous basis through the following components:

Sl. No.	Component	Number	Marks per incidence	Total Marks
1	Participation	-	05	05
2	Performance/ Activities	-	05	05
3	Project	-	15	15
4	Attendance	-	05	05
Total				**30**

External Evaluation: University Practical examination will be for 70 marks and will test the performance, activities and creative presentations of the students with reference to the course selected**:**

Sl. No.	Component	Number	Marks per incidence	Total Marks
1	Viva / Practical	-	70	70
Total				**70**

Outline of the course:

Sr. No.	Title of the unit	Minimum number of hours
1	An Introduction to Photography	03
2	Camera and Operating System	05
3	Light and Shade	10
4	Composition	09
5	Contemporary Issues in Photography	03

Practical Hours: 30 Hrs.
Total hours: 30 Hrs.

Reference Books

1. Maria Short, Sri-Kartini Leet and Elisavet Kalpaxi (2020) *Context and Narrative in Photography*-Routledge, New York.
2. Kevin Los (2010) *Improve Your Photography: 50 Essential Digital Photography Tips & Techniques* Digital Photography Series, New York.

Web Material

1. https://photographylife.com/photography-basics
2. https://artofvisuals.com/the-basics-of-photography-introduction-to-photography-tutorials/
3. https://www.photographytalk.com/beginner-photography-tips/camera-basics-for-beginners
4. https://www.makeuseof.com/tag/5-photography-tips-for-beginners/
5. https://annamcnaught.com/blog/basics-of-photography

Course Outcome (COs): At the end of the course, the students will be able to:

CO1	Understand, appreciate and demonstrate innovative approach, beauty and acute acumen in the area of photography
CO2	Develop photography skills and become familiar with the functions and importance of the visual elements of nature and artificial objects
CO3	Become independent thinkers who will contribute inventively and critically to culture through the making of art photography
CO4	Have thorough understanding and acute sense of light and shade, composition, and presentation of a piece of an art
CO5	Experiment and Represent the cultivated sense and skills in Photography to the mass
CO6	Prepare an impressive portfolio encompassing holistic approach to art and other the areas of study

	PO1	PO2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO10	PO11	PSO 1	PSO 2
CO1	-	-	-	-	-	-	-	-	3	-	-	-	-
CO2	-	-	-	-	-	2	-	-	3	-	-	-	-
CO3	-	-	-	-	-	-	-	-	3	2	-	-	-

CO4	-	-	-	-	-	2	-	-	3	-	-	-	-
CO5	-	-	-	-	-	-	-	-	3	-	-	-	-
CO6	-	-	-	-	-	2	-	-	3	-	-	-	-

Course Articulation Matrix:

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation "-"

HS205.02 C: MEDIA AND GRAPHIC DESIGN

(100 Marks)

Contact Hours: 30 Hrs.

Pre-requisite: None

Instructional Methods and Pedagogy: Teaching will be practical based on the hands on experiences, live and interactive sessions. It will run in the workshop mode. Four Workshops (each of a day) will be organized during the semester.

Evaluation: The students will be evaluated continuously in the form of their consistent performance throughout the semester. There is no theoretical evaluation. There is just practical evaluation. The evaluation (practical) is schemed as 30 marks for internal evaluation and 70 marks for external evaluation.

Internal Evaluation: Students' performance in the course will be evaluated on a continuous basis through the following components:

Sl. No.	Component	Number	Marks per incidence	Total Marks
1	Participation	-	05	05
2	Performance/ Activities	-	05	05
3	Project	-	15	15
4	Attendance	-	05	05
Total				**30**

External Evaluation: University Practical examination will be for 70 marks and will test the performance, activities and creative presentations of the students with reference to the course selected**:**

Sl. No.	Component	Number	Marks per incidence	Total Marks
1	Viva / Practical	-	70	70
Total				**70**

Outline of the course:

Sr. No.	Title of the unit	Minimum number of hours

1	An Introduction to Media and Graphic Design	03
2	Layout and Design	07
3	Form and Space	06
4	Computer Graphics	04
5	Fonts	04
6	Basic Print Media	03
7	Contemporary Issues in Graphic Design	03

Practical Hours: 30 Hrs.
Total hours: 30 Hrs.

Detailed Syllabus:

Unit – I	**An Introduction to Media and Graphic Design**	**03 Hours**

- Creating Art, Art in Context and Art as Inquiry
- History of Graphic Design
- Constructional, Representational, and Simplification Drawing

Unit – II	**Layout and Design**	**07 Hours**

- Layout, Design and Aesthetics
- Elements of Design
- Principles of Design:
  Harmony, Balance, Rhythm, Perspective, Emphasis, Orientation, Repetition and Proportion
- Impact/function of Design
- Indigenous design practices
- Role of design in the changing social scenario

Unit – III	**Form and Space**	**06 Hours**

- Types of Forms: Man-made, Nature
- Types of Space: Negative and Positive
- Composition of Form and Space to create Layout
  Exploring Creativity

Unit – IV	**Computer Graphics**	**04 Hours**

- An Introduction to Graphic Software
- Flash, Coreldraw, Illustrator and Photoshop
- Pre-press Process

Unit – V	**Fonts**	**04 Hours**

- Construction of Type
- Anatomy of Type
- Visual Language
- Creating Logo and Symbol

Unit – VI	**Basic Print Media**	**03 Hours**

- An Introduction to Press and its Development Phases
- Types of Press
- Types of Printing Technologies
- Post-press Processes

Unit – VII	**Contemporary Issues in Graphic Design**	**03 Hours**

- Present Day Graphic Designs
- Contemporary Designers and their Contribution
- Major Contemporary Issues in Graphic Design

Reference Books

1. Ellen Lupton and Jennifer Cole Philips (2008) Graphic *Design: The New Basics*, Princeton Architectural Press, New York
2. Ryan Hembree (2011) *The Complete Graphic Designer: A Guide to Understanding Graphics and Visual Communication* Rockport Publishers, Beverly

Web Material

1. https://99designs.com/blog/tips/graphic-design-basics/
2. https://edu.gcfglobal.org/en/beginning-graphic-design/fundamentals-of-design/1/
3. https://ultragraphicsmt.com/10-basic-concepts-to-improve-your-graphic-design/
4. https://www.jcsocialmedia.com/graphic-design/

Course Outcome (COs): At the end of the course, the students will be able to:

CO1	have cultivated a sense of creativity.
CO2	be appreciative of art and designs, art criticism and aesthetics.
CO3	be able to recognize the elements of arts in graphic design.
CO4	have better cognizance and association with the meaning of designs, shapes, colors, print and medium.
CO5	be able to design graphics using computer softwares like Photoshop, CorelDraw, and Illustrator.

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO 8	PO9	PO10	PO11	PSO1	PSO2
CO1	-	-	-	-	-	-	-	-	2	-	-	-	-
CO2	-	-	-	-	-	-	-	-	2	-	-	-	-
CO3	-	-	-	-	-	3		-	2	-	-	-	-
CO4	-	-	3	-	-	-	-	-	2	-	-	-	-
CO5	-	-	-	3	-	-	-	-	2	-	-	-	-

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation "-"

HS209.02 C DRAMATICS

(100 Marks)

Contact Hours: 30 Hrs.

Pre-requisite: None

Instructional Methods and Pedagogy: Teaching will be practical based on the hands on experiences, live and interactive sessions. It will run in the workshop mode. Four Workshops (each of a day) will be organized during the semester.

Evaluation: The students will be evaluated continuously in the form of their consistent performance throughout the semester. There is no theoretical evaluation. There is just practical evaluation. The evaluation (practical) is schemed as 30 marks for internal evaluation and 70 marks for external evaluation.

Internal Evaluation: Students' performance in the course will be evaluated on a continuous basis through the following components:

Sl. No.	Component	Number	Marks per incidence	Total Marks
1	Participation	-	05	05
2	Performance/ Activities	-	05	05
3	Project	-	15	15
4	Attendance	-	05	05
Total				**30**

External Evaluation: University Practical examination will be for 70 marks and will test the performance, activities and creative presentations of the students with reference to the course selected**:**

Sl. No.	Component	Number	Marks per incidence	Total Marks
1	Viva / Practical	-	70	70
Total				**70**

Outline of the course:

Sr. No.	Title of the unit	Minimum number of hours
1	Introduction to Drama	06
2	History of Drama and Contemporary Theatre	06
3	Theatre Design and Techniques	06
4	Technicalities of Stage Performance	08
5	Contemporary Trends in Drama	04

Practical Hours: 30 Hrs.

Total hours: 30 Hrs.

Detailed Syllabus:

Unit – I **Introduction to Drama** **06 Hours**

- Introduction to performing arts
- Drama - An art, a socializing activity, & a way of learning
- Form of Drama
- Elements of Drama
- Types of Drama

Unit – II **History of Drama and Contemporary Theatre** **06 Hours**

- Important world dramatists & drama–from Greek to modern
- Evolution of contemporary theatre in the context of developments in Indian theatre
- Major Movements in Drama

Unit – III **Theatre Design and Techniques** **06 Hours**

- Theatre Architecture
- Stage craft: Set, light, costume, make up, sound, props
- Theatre techniques: from selection of script to final performance

Unit – IV **Technicalities of Stage Performance** **08 Hours**

- Selection of plot and character
- Improvisation
- Movement
- Voice, Speech, Imagination
- Character Development
- Scene Enactment

Unit – V **Contemporary Trends in Drama** **04 Hours**

- New Tendencies in theatre
- Drama and Society
- Using drama for Social Change and Education

Reference Books

1. Bridget Panet (2009) Essential Acting: A Practical Handbook for Actors, Teachers and Directors Routledge, New York.
2. Tom Bancroft (2012) Character Mentor: Learn by Example to Use Expressions, Poses, and Staging to Bring Your Characters to Life Focal Press, New York

Web Material

1. https://entertainism.com/elements-of-drama
2. https://www.rcboe.org/cms/lib/GA01903614/Centricity/Domain/5069/the%20elements%20of%20drama.pdf
3. https://marrzipandrama.co.nz/the-6-elements-of-drama/
4. https://www.backstage.com/magazine/article/theater-trends-broadway-2019-66726/

Course Outcome (COs): At the end of the course, the students will be able to:

CO1	be aware about the concept of performing art and its nuances.
CO2	display a working knowledge of historic of drama, its development and current trends in dramatics.
CO3	demonstrate skills in the technical/design preparation and execution of a theatre performance.
CO4	demonstrate the ability to work collaboratively.
CO5	develop essential transferable skills in various relevant areas of the theatre.

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO1	PSO2
CO1	-	-	-	-	-	-	-	-	2	-	-	-	-
CO2	-	-	-	-	-	2	-	-	2	-	-	-	-
CO3	-	-	-	-	-	-	-	-	2	-	-	-	-
CO4	-	-	-	-	-	-	-	-	2	3	-	-	-
CO5	-	-	-	-	-	-	-	-	2	-	-	-	2

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

HS210.02C - CONTEMPORARY DANCE
(100 Marks)

Contact Hours: 30 Hrs.

Pre-requisite: None

Instructional Methods and Pedagogy: Teaching will be practical based on the hands on experiences, live and interactive sessions. It will run in the workshop mode. Four Workshops (each of a day) will be organized during the semester.

Evaluation: The students will be evaluated continuously in the form of their consistent performance throughout the semester. There is no theoretical evaluation. There is just practical evaluation. The evaluation (practical) is schemed as 30 marks for internal evaluation and 70 marks for external evaluation.

Internal Evaluation: Students' performance in the course will be evaluated on a continuous basis through the following components:

Sl. No.	Component	Number	Marks per incidence	Total Marks
1	Participation	-	05	05
2	Performance/ Activities	-	05	05
3	Project	-	15	15
4	Attendance	-	05	05
Total				**30**

External Evaluation: University Practical examination will be for 70 marks and will test the performance, activities and creative presentations of the students with reference to the course selected**:**

Sl. No.	Component	Number	Marks per incidence	Total Marks
1	Viva / Practical	-	70	70
Total				**70**

Outline of the course:

Sr. No.	Title of the unit	Minimum number of hours
1	Introduction to dance	04
2	Types of Dance	06
3	Basic Elements of Dance	04
4	Technical Skills in Professional Contemporary	06
5	Contemporary Trends in Dance	10

Practical Hours: 30 Hrs.

Total hours: 30 Hrs.

Detailed Syllabus:

Unit – I	**Introduction to dance**	**04 Hours**

- Dance as a Performing Art
- Dance as a Medium of Expression
- History and Development of Dance

Unit – II	**Types of Dance**	**06 Hours**

- Western dance and classical dance
- Salsa, rumba, hip hop, tap dance, belly dance, etc.
- Indian Classical Dance forms: Odissi, Bharatanatyama, Kathak, Kathakali, Kuchipudi etc.
- Other Regional dance forms in India

Unit – III	**Basic Elements of Dance**	**04Hours**

- Movements of different parts of a body for Expression
- Concepts of: Nritya, Laya and Taal

Unit – IV	**Technical Skills in Professional Contemporary Dance**	**06 Hours**

- Dance technique: alignment, balance, co-ordination, flexibility and control
- Expressive / presentation skills: Dynamic energy, physical engagement with the given material and stage, etc.
- Skills and processes of rehearsal and production:
- physical energy, stamina and athleticism

Unit – V	**Contemporary Trends in Dance**	**10 Hours**

- Prevalent trends and techniques in contemporary dance
- Future trends in contemporary dance form
- On Stage Performance

Reference Books

1. Helene Scheff, Marty Sprague and Susan Mc Greeve. (1939) *Experiencing dance from student to dance artist*. Human kinetics, New York
2. Robin Grove, Catherine J. Stevens, Shirley McKenzie (2005) Thinking in Four Dimensions: Creativity and Cognition in Contemporary Dance. Melbourne university press, Melbourne

Web Material

1. https://www.liveabout.com/what-is-contemporary-dance-1007423#:~:text=Contemporary%20dance%20is%20a%20style,body%20through%20fluid%20dance%20movements.
2. https://dancemagazine.com.au/2014/01/whats-contemporary-dance-days/
3. https://www.frontiersin.org/articles/10.3389/fpsyg.2019.00071/full
4. https://www.thehindu.com/features/friday-review/dance/what-is-contemporary-dance/article5096022.ece

Course Outcome (COs): At the end of the course, the students will be able to:

CO1	be able to develop ability to express through the form of dance.
CO2	have enhanced aesthetic sensitivity.
CO3	have improved concentration, mental alertness, quick reflex action, and physical agility.
CO4	be able to express a natural way human feelings and expressions by creating harmony.
CO5	be able to deliver contemporary dance performance.

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO1	PSO2
CO1	-	-	-	-	-	-	-	-	3	-	-	-	-
CO2	-	-	-	-	-	-	-	-	3	-	-	-	-
CO3	-	-	-	-	-	3	-	-	3	-	-	-	-
CO4	-	-	-	-	-	-	-	-	3	-	-	-	-
CO5	-	-	-	-	-	-	-	-	3	-	-	-	3

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation "-"

DETAILED SYLLABUS

FOR

BCA PROGRAMME
(3rd SEMESTER)

EFFECTIVE FROM
ACADEMIC YEAR 2023-24

Bachelor of Computer Applications (B.C.A.) Programme
(Choice Based Credit System)
Effective from Year 2023 - 24
Semester - III

Course Code	Course Title	Teaching Scheme				Examination Scheme						Total
		Contact Hours			Credit	Theory			Practical			
						Internal		External	Internal		External	
		Theory	Practical	Total		Case Study	Tests		Term work	Tests		
CA226-227	Elective - III	4	-	4	4	10	20	70	-	-	-	100
CA228	System Analysis and Design	3	-	3	3	10	20	70	-	-	-	100
CA229	Fundamentals of Data Structures and Algorithms	4	3	7	7	10	20	70	15	15	70	200
CA230	Fundamentals of Operating Systems	4	3	7	7	10	20	70	15	15	70	200
HS121.02 C	Creativity, Problem Solving and Innovation	-	2	2	2	-		-	30		70	100
CL144.01 C	Environmental Sciences	-	2	2	2	-		-	30		70	100
	University Elective – I **	-	2	2	2	-	-	-	30		70	100
		15	12	27	27	400			500			900

** Student will take any university elective offered by different institutions of university. CMPICA has decided to offer **Introduction to Web Designing** course for others**.**

Elective-III	
Course Code	**Course Title**
CA226	Computer Oriented Numerical Methods
CA227	Computer Oriented Management System

University Elective - I			
S.No.	**Course Code**	**Course Name**	**Department/Faculty**
1	CE281.01	Art of Programming	Engineering
2	CL281.01	Environmental Sustainability & Climate Change	Engineering
3	IT281.01	ICT Resources & Multimedia	Engineering
4	ME281.01	Engineering Drawing	Engineering
5	PH233.01	Fundamentals of Packaging	Pharmacy
6	PD260.01	Basic Laboratory Techniques	Applied Science
7	NR251.01	First Aid & Life Support	Nursing
8	PT191.01	Health Promotion & Fitness	Physiotherapy
9	BM231	Banking & Insurance	Management
10	EC281.01	Introduction to MATLAB Programming	Engineering
11	PD261	Astrophysics, Space and Cosmos-1	Applied Science
12	CL283	SDG Handprint Laboratory	Engineering
13	EE284	Python Programming	Engineering

CA226-Computer Oriented Numerical Methods

(100 Marks)

Contact Hours: 04

Pre-requisite: None

Methodology & Pedagogy: During theory lectures illustrations of certain real world problems requiring numerical computations inference will be explained. Numerical methods will be introduced as tools for solving such problems also be emphasized.

Outline of the course:

Unit No.	Title of the Unit	Minimum Number of Hours
1	Introduction to Numerical Methods	6
2	Interpolation	7
3	Numerical Integration and Differentiation	8
4	Solution of Linear Equation	10
5	Solution of Differential Equations	9
6	Solution of Partial Differential Equations	8

Total Hours (Theory): 48

Total Hours: 48

Detailed Syllabus:

Unit - I: Numerical Methods **Hours: 06**

Introduction to Numerical Methods, Various methods such as Bisection, False-Position and Newton-Raphson methods

Unit - II: Interpolation **Hours:07**

What is an interpolation, Difference tables, Newton forward and backward Interpolation formula, Lagrange's formula, Newton's Divided Difference Formula?

Unit - III: Numerical Integration and Differentiation **Hours:08**

Numerical Integration, Trapezoidal rule, Differentiating Continuous function (Two point and Three-point formula), Tabulated function and its maxima and minima.

Unit - IV: Solution of Linear Equation **Hours:10**

Solution of Linear Equation, Gauss-Jordan Elimination methods, Gauss - Seidal and Gauss Jacobi Interactive methods

Unit - V: D Solution of Ordinary Differential Equations: **Hours:09**

Solution of Ordinary Differential Equations, Taylor Series and Euler Methods, Runge-Kutta Methods,

Unit - VI: Solution of Partial Differential Equations: Hours:08

Review and examples of partial differential equations, classification of partial differential equations, Difference equation, Laplace's equation and Poisson's equation

Core Books:

1. D. N. Datta: Computer Oriented Numerical Methods, Vikas Pub. House, 2012.
2. S.S. Sastry : Introductory methods of Numerical Analysis, Prentice Hall of India,4th edition 2006

Reference Books:

1. V. Rajaraman Computer Oriented Numerical Methods, 3rd Edition, Prentice-Hall of India Pvt.Ltd, 3rd edition, 2016
2. M. Ray, Har Swarup Sharma, Sanjay Chaudhary: Mathematical Statistics, Ram Prasad and sons, 11th edition
3. B S Grewal: Numerical Methods In Engineering & Science (With Programs In C, C++ And MATLAB), Khanna Publishers, 10th edition, 2014

Web References:

1. https://ece.uwaterloo.ca/~dwharder/NumericalAnalysis/10RootFinding/bisection/example s.html [Bisection Method]
2. https://ece.uwaterloo.ca/~dwharder/NumericalAnalysis/10RootFinding/falseposition/exa mples.html
3. [False Position Method]
4. https://paws.kettering.edu/~ktebeest/math305/simp38b.pdf[Simpson's–1/3 & Simpson's– 3/8]
5. https://hamrocsit.com/note/csc207/solution-of-nonlinear-equations/ [solutions of linear equations]
6. https://sites.google.com/site/nmmat202/file-cabinet [Lecture notes of Numerical methods]

Course Outcomes: Upon successful completion of the course, students will be,

CO1 :	**Able to use Numerical methods in computer technology.**
CO2 :	To learn Interpolation.
CO3 :	To Learn the concepts of Numerical Integration and Numerical Differentiation
CO4 :	To understand linear equations.
CO5 :	To learn techniques to solve ordinary differential and partial differential equations

Course Outcomes Mapping:

Unit No.	Unit Name	Course Outcomes				
		CO1	CO2	CO3	CO4	CO5
1	Introduction to Numerical Methods	√				
2	Interpolation		√			
3	Numerical Integration and Differentiation			√		
4	Solution of Linear Equation				√	
5	Solution of Differential Equations					√
6	Solution of Partial Differential Equations					√

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO1	PSO2
CO1	3	3	2	1	-	2	-	-	-	2	-	2	1
CO2	3	3	2	1	-	2	-	-	-	2	-	2	1
CO3	3	3	2	1	-	2	-	-	-	2	-	2	1
CO4	3	3	2	1	-	2	-	-	-	2	-	2	1
CO5	3	3	2	1	-	2	-	-	-	2	-	2	1

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

CA227: Computer Oriented Management System (100 Marks)

Contact Hours: 4

Pre-requisite: None.

Methodology & Pedagogy: During the Theory sessions, illustrations of working real time information system will be demonstrated. Industrial visits will be arranged in order to demonstrate actual functioning of such systems. Students will be required to carry out a case study of such system.

Outline of the Course:

Unit No.	Title of the Unit	Minimum Number of Hours
1.	Introduction to Information System	07
2.	Transaction Processing Systems	08
3.	Management Information System	09
4.	Decision Support System	09
5.	Knowledge Management & Executive Support Systems	08
6.	Introduction To ERP and Technology	07

Total Hours (Theory): 48
Total: 48

Detailed Syllabus:

Unit - I: Introduction to Information System Hours:07

Data & Information, what is a system, Types of Systems, Types of information, Quality of Information, what is an Information System, Functions of Information System, classification of an Information System, Major Types of Information Systems, Interrelationship of Information Systems.

Unit - II: Transaction Processing Systems Hours:08

Introduction of Transaction Processing System (TPS), Transaction Processing System (TPS) Architecture, Transaction Processing System (TPS) Life cycle, Methods of Transaction Processing System (TPS), Information System Availability Control, Illustrations and live demonstration of TPS such as Banking System, Railway reservation system etc.

Unit - III: Management Information System Hours:09

Introduction to Management: Approaches of Management, Function of Management, System from Functional Perspective, Reports of Management Information System, A business perspective of MIS, Dimensions of MIS, contemporary Approaches to Information System.

Unit - IV: Decision Support System Hours:09

Business value of Improved Decision making, Types of Decision, Decision making Process, The difference between MIS and DSS, Components of DSS, System for Decision Support Group Decision Support System, Business value of GDSS.

Unit - V: Knowledge Management & Executive Support Systems Hours:08

Important Dimensions of knowledge, Organizational learning and Knowledge Management, The Knowledge Management value chain, Overview of different types of Knowledge Management Systems, Characteristics of ESS, The Role of ESS in the Firm, Business value of ESS, Monitoring Corporate Performance, Working Examples of ESS.

Unit - VI: Introduction to ERP and Technology Hours:07

An overview of ERP, Basic ERP Concepts, Risk and Benefits of ERP, ERP and Related Technologies, Business Intelligence, Some case studies on ERP or Information system

Core Books:

1. K. C. Laudon and J. P. Laudon: Management Information Systems, 12th Edition, Pearson Education.
2. Alexis Leon: ERP Demystified, Second Edition, Tata McGraw-Hill.

Reference Books:

1. W.S. Jawadekar: Management Information Systems, 2nd Edition, Tata McGraw-Hill.

Web References:

2. http://www.slideshare.net/NorazilaMat1/laudon-mis12-ppt01-16595885[Function of Information system, A business perspective of MIS, Dimensions of MIS, Contemporary Approaches to Information System]
3. http://www.uh.edu/~mrana/try.htm[Types and Functions of Information system]
4. http://bisom.uncc.edu/courses/info2130/Topics/istypes.htm[Types of Information system]
5. http://kalyan-city.blogspot.com/2011/05/levels-of-management-topmiddleand.html[classification of information system]
6. http://my.safaribooksonline.com/book/management/9780470916803/operationalplanning andcontrol-systems/management_levels_comma_functions_comm[classification of information system]
7. http://ambarwati.dosen.narotama.ac.id/files/2011/05/FIS-2011-w2.pdf[Transaction Processing System]

Course Outcomes: Upon successful completion of the course, the students will able to :

CO1 :	To understand basic of data & information, types of information system, function of information system and its quality.
CO2 :	To understand transaction processing concepts and its working.
CO3 :	Learning Management Information system, its types and its implementation.
CO4 :	To gain understanding of Decision Support Systems.
CO5 :	To learn Knowledge Management, Executive Support Systems, ERP and its technology.

Course Outcomes Mapping:

Unit No.	Unit Name	Course Outcomes				
		Co1	Co2	Co3	Co4	Co5
1	Introduction to Information System	√				
2	Transaction Processing Systems		√			
3	Management Information System			√		
4	Decision Support System				√	
5	Knowledge Management & Executive Support Systems					√
6	Introduction To ERP and Technology					√

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO1	PSO2
CO1	3	2	2	1	1	3	1	3	2	1	2	2	1
CO2	3	2	2	2	2	2	2	2	1	1	2	2	2
CO3	3	3	2	2	3	2	3	3	2	2	2	3	2
CO4	3	3	2	3	3	3	2	2	1	3	2	3	3
CO5	3	3	2	2	3	3	3	2	2	3	3	2	3

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation "-"

CA228: System Analysis and Design

(100 Marks)

Contact Hours: 03

Pre-requisite: None

Methodology & Pedagogy: During theory lectures emphasis will be given on understanding how Information Systems are designed and constructed and how major modifications are made to existing systems, analyse business needs for information and develop an appropriate strategy to solve a particular business problem, understand the issues associated with physical design, utilize some of the tools used by systems analysts in the design of Information Systems, a significant portion of the course will deal with computer aided system engineering (CASE) tools and techniques in the context of a case study.

Outline of the Course:

Unit No.	Title of the Unit	Minimum Number of Hours
1.	Introduction to System Development	05
2.	Classical Systems Development Methodology	07
3.	Introduction to Data flow diagrams and Prototype design	07
4.	Systems Design	06
5.	Quality assurance, maintenance and Audit	06
6.	Designing reliable system, training and conversion	05

Total Hours (Theory): 36

Total Hours: 36

Detail Syllabus:

Unit I: Introduction to System Development **Hours 05**

Concept of system, Basic components, Information systems categories, Business Information Systems, Examples of real-life systems, System analyst – meaning, role and need, Users, need of information system development, Application development portfolio, Integration of portfolio, Controlling and monitoring committees.

Unit II: Classical Systems Development Methodology **Hours 07**

Classical Systems Development Life Cycle (SDLC) method, Requirement determination, Fact finding techniques, Modern methods for determining systems requirements, Joint Application Development Program, Tools for documenting procedures and decisions, Tools for systems development, understanding the format of system requirement specification.

Unit III: Introduction to Data flow diagrams and Prototype design

Hours 07

Structured Analysis Development Strategy, Data Flow Diagrams (DFDs) for analysis and design, Physical and logical DFDs, Purpose of Application Prototyping, Steps in prototype method, Use of prototypes, Tools for prototyping, Strategies for prototype development, misconceptions about prototyping

Unit IV: Systems Design **Hours 06**

Analysis to design transition, Design of output system, Design of input system, Design of security and control, Design of user interface and Design of Database Interaction, design of software, software design and documentation tools.

Unit V: Quality assurance, maintenance and Audit **Hours 06**

Introduction to quality assurance, levels of quality assurance, the testing process with test data: Program testing, Link testing, Full system testing with live data.; Control: audit and security of information systems, why controls are needed, objectives of control, techniques used in control, auditing information systems security of information systems, maintenance, types of maintenance

Unit VI: Designing reliable system, training and conversion **Hours 05**

Introduction to software reliability, approaches of software reliability, user training, training methods and conversion, conversion methods.

Core Books:

1. James A Senn : Analysis and Design of Information System, McGraw Hill International, 2003.
2. Kendall and Kendall : Systems analysis and Design, 5th Edition, Prentice-Hall of India Private Limited, 2003.

Reference Books:

1. V. Rajaraman: Analysis and Design of Information Systems, Prentice-Hall of India Private Limited, 2003.
2. Jeffrey L. Whitten, Lonnie D. Bentely and Kevin C. Dittman : Systems Analysis and Design Methods, Tata McGraw Hill Publishing Co. Ltd., 2001.
3. Tuthill and Leavy : Knowledge Based Systems : Mangers Perspectives : Tab professional and Reference Books, 1991.

Web References:

1. http://lecture-notes-forstudents.blogspot.in/2010/04/system-analysis-and-design-sad.html [For referring SDLC]
2. http://www.slideshare.net/aroravinay/1-introduction-to-ado-infosys[For Introduction of System analysis and design]
3. http://bcastuff.blogspot.in/p/sad-notes.html#.U6uW-ZzWIeo[For brief notes on System analysis]
4. http://www.eis.mdx.ac.uk/staffpages/geetha/bis2030/DFD.html[For DFD notes and examples]
5. http://faculty.washington.edu/ytan/is460/notes/LN11.pdf[For DFD notes and examples]

Course Outcomes: Upon successful completion of the course, students will be,

CO1 :	Understand the basics of system, type of information systems, system analyst-meaning, role and need, portfolio development of the system, the methods or ways to review and select the project.
CO2 :	Understand the concept of SDLC, the fact finding techniques, detailed study of modern methods for collecting the requirements of system to be developed, Decision and procedure documentation and a brief introduction to the SRS document.
CO3 :	Learn the concept of structured analysis, detail study of DFD and its types and the study of data dictionary.
CO4 :	Gain understanding of prototypes, its uses and tools for development, clarification of the misconception about the prototype and categories of tools including the use of automated tools like case tools.
CO5 :	Be able to learn objective of collecting data, methods to collect data for designing the input and output interface, the concept and objective of database design,
CO6 :	Study the designing of reliable and maintainable systems, documentation tool, quality assurance management and need for user training.

Course Outcomes Mapping:

Unit No.	Unit Name	Course Outcomes					
		Co1	Co2	Co3	Co4	Co5	Co6
1.	Introduction to System Development	√					
2.	Classical Systems Development Methodology		√				
3.	Introduction to Data flow diagrams and Prototype design			√			
4.	Systems Design				√		
5.	Quality assurance, maintenance and Audit					√	
6.	Designing reliable system, training and conversion						√

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO1	PSO2
CO1	2	2	3	2	2	1	2	2	2	2	2	3	3
CO2	3	3	3	3	3	3	3	3	3	3	3	3	3
CO3	3	3	3	3	3	3	3	3	3	2	3	3	2
CO4	3	3	3	2	3	2	3	2	2	2	2	3	2
CO5	1	2	2	2	3	3	3	2	1	2	2	3	2
CO6	1	2	2	2	3	3	3	2	1	2	2	3	2

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

CA229: Fundamentals of Data Structures and Algorithms (200 Marks)

Contact Hours: 07

Pre-requisite: CA123: Foundation of Programming, CA129: Fundamentals of OOP's

Methodology & Pedagogy: During the theoretical lectures, the focus will be on understanding the fundamental ideas of data structures and algorithms, as well as how to use and select the most efficient data structures from a variety of options for various applications. Designing, analyzing, and implementing all types of data structures for addressing computational issues using a high-level programming language is emphasized in practical sessions.

Outline of the course:

Unit No.	Title of the Unit	Minimum Number of Hours	
		Theory	Practical
1	Introduction to Data Structures and Algorithms	06	36
2	Linear Data Structures Stack and Queue	10	
3	Dynamic Linear Data Structures	09	
4	Trees Data Structure	08	
5	Graph Data Structure	08	
6	Sorting and Searching	07	

Total Hours (Theory): 48
Total Hours (Lab): 36
Total Hours: 84

Detail Syllabus:

Unit - I: Introduction to Data Structures and Algorithms Hours: 06

Data type, Data object, Data Structure, Abstract Data type, Types of Data Structures, Introduction to Algorithms, Relationship among Data, Data Structures and Algorithms, Analysis of Algorithms, space and time complexity algorithm.

Unit - II: Linear Data Structures Stack and Queue Hours: 10

Linear Data Structures using sequential organization, Concepts of Ordered list, Representation of Stacks using sequential organization, Applications of Stack, Recursion, Application of Recursion, Concepts of Queues, Realization of Queues using sequential organization, Multiqueue, Deque, Priority Queue, Applications of Queue.

Unit - III: Dynamic Data Structures Hours: 09

Linked list, Comparison of sequential and linked organizations, Dynamic Memory Management, Single linked list, Double linked list, Applications of linked list.

Unit - IV: Trees Data Structure Hours: 08

Basic Terminologies, Definition and concepts, Representation of Binary Tree, Operations on Binary Tree, Types of Binary Trees, B Tree, B+ Tree Indexing, Trie Tree Indexing.

Unit - V: Graph Data Structure Hours: 08

Graph Terminology, Representation of Graph, Operations on Graph, Applications of Graph Structure.

Unit - VI: Sorting and Hours: 07

Sorting Notations and Concepts, Sorting Techniques, Sequential Searching, Binary Searching, Search Trees, Hashing: The symbol table, Hashing Functions, Collision Resolution Techniques,

Core Books:

1. Jean-Paul Tremblay, Paul G. Sorenson: An introduction to data structures with applications, 2nd Edition, Tata McGraw Hill Publications, Paperback.
2. Data structures A Programming Approach with C, D.S.Kushwaha and A.K.Misra, PHI
3. Varsha h. Patil, Data Structures Using C++, 1st Edition, Oxford University Press.
4. D. Samanta, Classic Data Structures, 2nd Edition, PHI Publications.

Reference Books:

1. Yashvant Kanethkar : Data structures through C++, 2nd Edition, BPB Publications.
2. Yedidyah Langsam, Moshe J. Augenstein, Aaron M. Tenenbaum : Data structures using C and C++ , 2nd Edition, Eastern economy edition, Pearson Education,2015.
3. Mark Allen Weiss: Data Structures and Algorithm Analysis in C++, 3rd Edition, Pearson Education, 2017.
4. Reema Thareja: Data Structures using C, 1st Edition, Oxford University Press, 2012.

Web References:

1. https://www.cs.auckland.ac.nz/~jmor159/PLDS210/ds_ToC.html [For materials of Data Structures and Algorithms]
2. http://www.cs.usfca.edu/~galles/visualization/ [For Data Structures Visualization]
3. https://www.cs.auckland.ac.nz/~jmor159/PLDS210/mst.html [For Graph Data Structures]
4. http://interactivepython.org/runestone/static/pythonds/Trees/trees.html [For Tree Data Structures]
5. https://www.cs.auckland.ac.nz/~jmor159/PLDS210/niemann/s_man.pdf [For Sorting and Searching Cookbook]
6. http://www.cs.princeton.edu/~rs/AlgsDS07/10Hashing.pdf [For Hashing Materials]
7. http://nptel.ac.in/video.php?subjectId=106102064 [For Data Structures videos]

Course Outcomes: Upon successful completion of the course, students will be,

CO1 :	Design and apply appropriate data structures for solving computing problems
CO2 :	Understands and restates the fundamentals of basic data structures.
CO3 :	Explore and implement linear data structures like link list
CO4 :	Students will familiar with non-linear data structure like tree and graph
CO5 :	Apply Algorithm for solving problems like sorting, searching

Course Outcomes Mapping:

Unit No.	Unit Name	Course Outcomes				
		CO1	CO2	CO3	CO4	CO5
1	Introduction to Data Structures and Algorithms	√				
2	Linear Data Structures Stack and Queue		√			
3	Dynamic Linear Data Structures			√		
4	Trees Data Structure				√	
5	Graph Data Structure				√	
6	Sorting and Searching					√

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO1	PSO2
CO1	3	2	1	-	2	2	2	2	-	-	-	3	1
CO2	2	3	2	2	-	2	-	-	1	-	1	3	1
CO3	2	3	3	2	-	2	-	-	1	-	1	3	2
CO4	2	3	3	2	-	2	-	-	1	-	1	3	2
CO5	3	3	3	2	-	2	-	-	1	-	1	3	3

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

CA230: Fundamentals of Operating Systems
(200 Marks)

Contact Hours: 07

Pre-requisite: None

Methodology & Pedagogy: In order to achieve the course objectives, students will be introduced to the basic operating system concepts and basic functions. During practical sessions, the students will be required to use DOS and UNIX commands to understand the system properties and to write script files.

Outline of the course:

Unit No.	Title of the Unit	Minimum Number of Hours	
		Theory	Practical
1	Operating System Fundamentals	6	36
2	Memory Management	10	
3	Processor Management	10	
4	Device Management	8	
5	File Management	6	
6	Introduction to Shell Script	8	

Total Hours (Theory): 48
Total Hours (Lab): 36
Total Hours: 84

Detailed Syllabus:

Unit - I: Operating System Fundamentals: **Hours: 06**

What is an operating system, Brief history of an operating system, Pyramid structure of an operating system, Operating system hardware and software, Evolution of an operating system, types of operating system, Batch system, Interactive system, Real time system, Hybrid system, Shell, compiler, assembler, interpreter?

Unit - II: Memory Management: **Hours: 10**

Memory management of early system, Fixed partition, Dynamic partition, Re-locatable dynamic partition, Best fit and first fit allocation method, De-allocation, Paged memory allocation, Demand paging allocation, segmented memory allocation, Virtual memory, cache memory, Associative memory, Page replacement policies.

Unit - III: Processor Management: Hours: 10

What is job and Process, Multiprogramming concept, Process scheduler, Job scheduler, various process states, Thread control, Process Control Block (PCB), Processor scheduling policies, FCFS, SJN, SRT, RR, EDF, Multiple level queues

Unit - IV: Device Management: Hours: 08

Types of devices, dedicated device, Virtual device, shared device, I/O devices in cloud, Sequential Access Storage Media, Direct access storage media, Device handler seek strategies, FCFS, SSTF, SCAN and LOOK strategies, Optical disk storage, RAID

Unit - V: File Management: Hours: 06

Fundamentals of File management, File naming conventions, Fixed length and variable length records, Physical file organization, Physical storage allocation, File access methods, File control matrix, File compression

Unit - VI: Introduction to Shell Script: Hours:08

Shell Script, Shell environment, Making Script interactive, Command Line Argument, Evaluating Expression, Operator, Control Statement, Looping statement.

Core Books:

1. Ann McIver McHoes, Ida M Flynn: Understanding Operating System, 7th Edition, Cengage Learning, 2014
2. William Stallings: Operating Systems Internals and Design Principles, 5th Edition, PHI, 2005.
3. Sumitabha Das: Unix concepts & application, 4th Edition, Tata McGraw Hill, 2010.
4. Kenneth Rosen, Douglas Host, James Farber and Richard Rosinski: The Complete Reference, Tata McGraw Hill, 1999.

Reference Books:

1. Silberschatz: Operating System Concepts, 5th Edition, John Wiley & Sons (ASIA) Ltd., 2008.
2. Mark G. Sobell: A Practical Guide to Linux, Pearson Education, 1997.
3. K.J.George, Operating System Concepts and Principles, Sroff Publishers, 2003

Web References:

1. https://study.com/learn/operating-system.html [Operating System Video Lessons]
2. https://www.studytonight.com/operating-system/ [Operating System Tutorial]
3. http://www.tutorialspoint.com/unix/index.htm [Unix Tutorials]
4. http://williamstallings.com/OS/Animation/Animations.html [Animations of Operating Systems Concepts]
5. http://www.ics.uci.edu/~ics143/lectures.html [Lecture notes of OS]

Course Outcomes: Upon successful completion of the course, students will be,

CO1 :	**To understand the fundamental concepts of an operating System.**
CO2 :	To understand memory allocation and de-allocation methods.
CO3 :	To study fundamentals of processor and its scheduling policies.
CO4 :	To study various supported device and file access technologies.
CO5 :	Learning basic shell programming

Course Outcomes Mapping:

Unit No.	**Unit Name**	**Course Outcomes**				
		CO1	**CO2**	**CO3**	**CO4**	**CO5**
1	Operating System Fundamentals	√				
2	Memory Management		√			
3	Processor Management			√		
4	Device Management				√	
5	File Management				√	
6	Introduction to Shell Script					√

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO1	PSO2
CO1	1	1	2	2	-	2	2	-	1	1	-	3	2
CO2	3	2	2	2	-	2	2	-	1	1	-	3	2
CO3	3	2	2	2	-	2	2	-	1	1	-	3	2
CO4	3	2	2	2	-	2	2	-	1	1	-	3	2
CO5	2	2	3	3	-	2	3	-	1	1	-	3	2

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

HS121.02 C: CREATIVITY, PROBLEM SOLVING AND INNOVATION
(100 Marks)

Contact Hours: 03

Pre-requisite: Creative Problem Solving (https://www.coursera.org/learn/creative-problem-solving)

Outline of the Course:

Unit No.	Title of the Unit	Minimum number of hours
1.	Introduction to Creativity, Problem Solving and Innovation	06
2.	Questioning, Learning and Visualization	06
3.	Creative Thinking and Problem Solving	06
4.	Logic, Language and Reasoning	06
5.	Contemporary Issues and Practices in Creativity and Problem Solving	06

Total Hours (Practical): 30
Total Hours: 30

Detailed Syllabus:

Unit - I	**Introduction to Creativity, Problem Solving and Innovation** Definitions of Creativity and Innovation, Need for Problem Solving and Innovation, Scope of Creativity in various Domains, Types and Styles of Thinking, Strategies to develop Creativity, Problem Solving and Innovation skills	**Hours: 06**
Unit - II	**Questioning, Learning and Visualization** Strategy and Methods of Questioning, Asking the Right Questions, Strategy of Learning and its Importance, Sources and Methods of Learning, Purpose and Value of Creativity Education in real life, Visualization strategies - Making thoughts Visible, Mind Mapping and Visualizing Thinking	**Hours: 06**
Unit - III	**Creative Thinking and Problem Solving** Creative Thinking and its need, Strategy of Thinking Fluency, Generating all Possibilities, SCAMPER Technique, Divergent Vs Convergent Thinking, Lateral Vs Vertical Thinking, Fusion of Ideas for Problem Solving, Applying strategies for Problem Solving	**Hours: 06**
Unit - IV	**Logic, Language and Reasoning** Basic Concepts of Logic, Statement Vs Sentence, Premises Vs Conclusion, Concept of an Argument, Functions of Language: Informative, Expressive and Directive, Inductive Vs Deductive Reasoning, Critical Thinking & Creativity, Moral Reasoning	**Hours: 06**
Unit - V	**Contemporary Issues and Practices in Creativity and Problem Solving** Cognitive Research Trust Thinking for Creatively Solving Problems, Case Study on Contemporary Issues and Practices in Creativity and Problem Solving	**Hours: 06**

Text book:

1. R Keith Sawyer, ZigZag, The Surprising Path to Greater Creativity,Jossy-Bass Publication 2013
2. Michael Michalko, Crackling Creativity, The Secrets of Creative Genus, Ten Speed Press 2001

Reference book:

1. Michael Michalko, Thinker Toys, Second Edition, Random House Publication 2006
2. Edward De Beno, De Beno's Thinking Course, Revised Edition, Pearson Publication 1994
3. Edward De Beno, Six Thinking Hats, Revised and Update Edition, Penguin Publication 1999
4. Tony Buzan, How to Mind Map,Thorsons Publication 2002
5. Scott Berkum, The Myths of Innovation, Expended and revised edition, Berkun Publication 2010
6. Tom Kelly and David Kelly, Creative confidence: Unleashing the creative Potential within Us all, William Collins Publication 2013
7. Ira Flatow, The all Laughed, Harper Publication 1992
8. Paul Sloane, Des MacHale& M.A. DiSpezio, The Ultimate Lateral & Critical Thinking Puzzle book,Sterling Publication 2002

Additional Readings

1. Keith Sawer, Group Genius, The Creative Power of Collaboration, Basic Books Publication 2007
2. Edward De Beno, Lateral Thinking, Creativity Step by Step, Penguin Publication 1973
3. Nancy Margulies with Nusa Mall, Mapping Inner Space, Crown House Publication 2002
4. Tom Kelly with Jonathan Littman,The Art of Innovation, Profile Publication 2001
5. Roger Von Oech,A Whack on the Side of the Head. Revised edition, Hachette Publication 1998
6. Roger Von Oech, A Kick in the Seat of the Head, William Morrow 1986
7. Jonah Lehrer,Imagine How Creativity Works,Canongate Books Publication 2012
8. James M Higgins, 101 Creative Problem Solving Techniques, New Management Publication1994
9. Soctt G Isaksen, K Brain Doval, Donald J Treffinger,Creative Approach to Problem Solving, Sage Publication 2000
10. Donald J Treffinger, scott G Isaksen, K Brain stead Dorval Creative Problem Solving An Introduction,Prufrock Press 2006
11. H Scott Fogler& Steven E. LeBlance,Strategies for Creative Problem Solving,Prentice Hall Publication 2008
12. Dave Gray, Sunni Brown and James Macanufo, Game Storming, O'reilly Publication 2010.
13. Howard Gardner, Creating minds,Basic Books Publication 1993
14. MihalyCsikzentmihalyi, Creativity–Flow and Psychology of Discovery and Invention,Harper Publication 1996
15. Martin Gerdner,W. H.,Ahal Insight,Freeman Publication 1978
16. Paul Sloane,Test Your Lateral Thinking IQ,Sterling Publication 1994

Web material:

1. Internet Search based May TED talks and other sources for videos, slide shares, problems, etc.

Course Outcome (COs): At the end of the course, the students will be able to

CO1	Demonstrate creativity in their day to day activities and academic output.
CO2	Solve personal, social and professional problems with a positive and an objective mindset.
CO3	Think creatively and work towards problem solving in a strategic way.
CO4	Initiate new and innovative practices in their chosen field of profession.
CO5	Give logical ideas, opinions, and solutions to problems.
CO6	Think critically over the situation and drawing conclusion.

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO1	PSO2
CO1	-	-	-	-	-	-	-	-	3	-	-	-	2
CO2	-	-	3	-	-	-	-	-	3	-	-	-	2
CO3	-	-	-	-	-	-	-	-	3	-	-	-	2
CO4	-	-	-	-	3	-	-	-	3	-	-	-	2
CO5	-	-	-	-	-	-	-	-	3	-	-	-	2
CO6	-	-	-	-	-	-	-	-	3	-	-	-	3

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation "-"

DETAILED SYLLABUS

FOR

BCA PROGRAMME
(4^{th} SEMESTER)

EFFECTIVE FROM
ACADEMIC YEAR 2023-24

Teaching & Examination Scheme
Bachelor of Computer Applications (B.C.A.) Programme
(Choice Based Credit System)
Effective from Year 2023 - 24
Semester - IV

Course Code	Course Title	Teaching Scheme				Examination Scheme						Total
		Contact Hours			Credit	Theory			Practical			
						Internal		External	Internal		External	
		Theory	Practical	Total		Case Study	Tests		Term work	Tests		
CA231-232	Elective - IV	-	3	3	3	-	-	-	10	20	70	100
CA233	Software Engineering	4	-	4	4	10	20	70	-	-	-	100
CA234	Advanced Database System	-	3	3	3	-	-	-	15	15	70	100
CA235	Open Source Technology	4	3	7	7	10	20	70	15	15	70	200
CA236	Computer Networks	4	-	4	4	10	20	70	-		-	100
HS111.02 C	Human Values and Professional Ethics	-	2	2	2	-		-	30		70	100
	University Elective – II **	-	2	2	2	-	-	-	30		70	100
		12	13	25	25	300			500			800

** Student will take any university elective offered by different institutions of university. CMPICA has decided to offer **Programming the Internet course** for others.

Elective-IV	
Course Code	**Course Title**
CA231	Fundamentals of Embedded Programming
CA232	Foundation of Multimedia

University Elective - II			
S.No.	**Course Code**	**Course Name**	**Department/Faculty**
1	EC282.01	Prototyping Electronics with Arduino	Engineering
2	CE282.01	Web Designing	Engineering
3	CL282.01	Basics of Environmental Impact Assessment	Engineering
4	IT282.01	Internet Technology & Web Design	Engineering
5	ME282.01	Material Science	Engineering
6	PH238.01	Cosmetics in Daily Life	Pharmacy
7	NR261.01	Life Style Diseases & Management	Nursing
8	PT192.01	Occupational Health & Ergonomics	Physiotherapy
9	BM241	Health Care Management	Management
10	PD262	Astrophysics, Space and Cosmos-2	Applied Science
11	EE287	MATLAB Programming	Engineering
12	EE288	Maintenance of Household Apparatus	Engineering

CA231: Fundamentals of Embedded System

(100 Marks)

Contact Hours: 03

Pre-requisite: None

Methodology & Pedagogy: In order to achieve the course objectives, students will be introduced to the basics of embedded system and its structure. Students will be introduced to various microcontrollers, analog and digital electronics components and sensors. Introduction to microcontroller based embedded system design with specifications, development and implementation will be explained. Students will also be introduced to basic programming of an embedded system.

Outline of the course:

Week No	Practical	Description
1 - 2	Introduction and demonstration to Embedded System	Embedded System, Embedded System Basics, Embedded system hardware and Software, Basics of Processor and controllers, Types of controllers, Microcontroller and Microprocessor, Architecture of 8051 and Arduino controller.
3 - 4	Introduction and demonstration to components used in embedded system using its applications	Introduction to bead board. LED, Switch, 7 segment LED display, Resistor, Capacitor, Inductor, Battery, Relay, Integrated Circuit, Potentiometer, DC and AC motor , Multi meter, Soldering iron, Sensor devices , Embedded applications.
5 - 6	Introduction of various software / online tools	Keil software, Tinker cad simulator for arduino controller, Proteus simulator, PCB designing software.
7 - 8	Basics of Embedded C Programming with controller	Embedded C data types, variables, constants, operators, control statements, loops, string, array.
9 - 10	GPIO Programming in embedded C	Interfacing various input / output components like LED, keypad matrix, push button switch, ON/OFF switch, toggle switch, 7 segment display and LCD display.
11 - 12	Basics of Sensor Programming	Interfacing with various sensors like temperature sensor, PIR sensor and ultrasonic sensor.

Total Hours (Practical): 36

Total Hours:36

Core Books:

1. K. Shibu: Introduction to Embedded Systems, Tata McGraw-Hill Education, 2009
2. Michael Margolis, Brian Jepson, Nicholas Robert Weldin: Arduino Cookbook: Recipes to Begin, Expand, and Enhance Your Projects , O'Reilly Media Publisher, 3rd Edition, Kindle Edition, 2020

Reference Books:

1. Muhammad Ali Mazidi, Janice GillispieMazidi, Rolin D. McKinlay: The 8051 Microcontroller and Embedded Systems: Using Assembly and C, Pearson Education India, 2007
2. Massimo Banzi, Michael Shiloh: Getting Started with Arduino: The Open Source Electronics Prototyping Platform ,Make Communication, 3rd Edition, Kindle Edition, 2014

Web References:

1. https://www.tutorialspoint.com/arduino/index.htm [Arduino Fundamentals]
2. https://www.arduino.cc/en/Guide/ [Learning Arduino]
3. https://funduino.de/ [Arduino Fundamentals]
4. https://www.tutorialspoint.com/embedded_systems/embedded_systems_tutorial.pdf [Tutorials for embedded systems]
5. www.tinkercad.com [Arduino Simulator]

Course Outcomes: Upon successful completion of the course, students will be,

CO1 :	To understand fundamentals of embedded system and its components with its applications.
CO2 :	To study modern tools used to simulate embedded systems.
CO3 :	To learn Embedded C Programming.
CO4 :	To Interface GPIO devices.
CO5 :	To interface various sensors.

Course Outcomes Mapping:

Week No.	Practical Title	Course Outcomes				
		CO1	CO2	CO3	CO4	CO5
1	Introduction and demonstration to Embedded system	√				
2	Introduction and demonstration to components used in embedded system using its applications	√				
3	Introduction of various software / online tools		√			
4	Basics of Embedded C Programming with arduino controller			√		
5	GPIO Programming in embedded C				√	
6	Basics of Sensor Programming					√

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	1	3	3	2	-	3	2	1	-	2	2	-	3	2
CO2	1	3	3	3	-	3	2	1	-	2	2	-	3	2
CO3	2	3	3	3	-	3	2	1	-	2	2	-	3	3
CO4	1	3	3	3	-	3	2	1	-	2	2	-	3	3
CO5	1	3	3	3	-	3	2	1	-	2	2	-	3	3

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation "-"

CA232: Foundation of Multimedia
(100 Marks)

Contact Hours: 03

Pre-requisite: Basic Office Automation Skills.

Methodology & Pedagogy: This course is focuses on providing hands-on experience in designing and development of Multimedia components using text, images and videos with the emphasis on job profile of Digital Marketer. Students will understand color, positioning, and other aspects of usage of Multimedia on various social media.

Outline of the course:

Week No	Practical	Description
1 - 2	Concept of colors, sizes, social media platforms	Students will learn about color combinations and some important tools/extensions, popular sizes used in various social media platforms.
3 - 4	Designing social media posts	Students will learn about designing social media post based on "Hook, Story and Offer"
5 - 6	Designing social media posts – carousal	Students will learn about designing social media post for carousal for more attention.
7 - 8	Understanding Video creation and edit	Students will learn about timeline, animation, transition, and editing of video
9 - 10	Creation of Portfolios	Students will learn to create portfolio for showcasing their creations when they start doing their own agencies.
11 - 12	Project based on learning	Students will do a small project based on whatever they have learned up to this.

Total Hours (Practical): 36
Total Hours: 36

Core Books:

1. Tay Vaughan: Multimedia, making it work: Edition 9th: Tata McGraw Hill Publication: 2014.
2. Ze-Nian Li, Mark S. Drew: Fundamentals of Multimedia : PHI: 2005

Reference Books:

1. Russell Brunson: Traffic Secrets: The Underground Playbook for Filling Your Websites and Funnels with Your Dream Customers: Hay House Business: 2020

Web References:

1. https://designschool.canva.com/tutorials/ [Tutorials – Canva]
2. https://help.invideo.io/ [Help – InVideo]
3. https://www.youtube.com/c/canva [Official YouTube channel - Canva]

Course Outcomes: Upon successful completion of the course, students will be,

CO1 :	Be able to identify the basic concepts of multimedia and the web.
CO2 :	Be able to use various multimedia tools.
CO3 :	Be able to design and create social media posts using popular tools for various requirements.
CO4 :	Be able to design and create videos using popular tools for various requirements.
CO5 :	Apply end-user design skills of different requirements in creative industry of digital marketing and able to start own digital marketing agency.

Course Outcomes Mapping:

Week No.	Practical Title	Course Outcomes				
		CO1	CO2	CO3	CO4	CO5
1-2	Concept of colors, sizes, social media platforms	√				√
3-4	Designing social media posts			√		√
5-6	Designing social media posts – carousal			√		√
7-8	Understanding Video creation and edit				√	√
9-10	Creation of Portfolios		√	√	√	√
11-12	Project based on learning	√	√	√	√	√

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO1	PSO2
CO1	2	2	3	3	2	2	1	2	2	2	2	2	1
CO2	3	1	1	3	1	3	2	1	1	2	2	2	2
CO3	2	3	3	2	1	2	2	2	2	2	3	2	3
CO4	2	3	3	2	1	2	2	2	2	2	3	2	3
CO5	2	3	3	3	2	3	3	2	2	3	3	2	3

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation "-"

CA 233: Software Engineering (100 Marks)

Contact Hours: 4 Hours

Pre-requisite: Fundamentals of System Analysis and Design

Methodology & Pedagogy: While learning this course, students can not only understand concepts of software and software engineering but also get familiar with concepts of project management and generation of SRS. Students can also realize significance of entire software development processes with the help of modern tools and techniques used in modeling a software project.

Outline of the course:

Unit No.	Title of the Unit	Minimum Number of Theory Hours
1	Introduction to software and Software Engineering	07
2	Software Process Models	09
3	Managing Software Projects	08
4	Software Requirement analysis and specification	08
5	Software Maintenance and Risk Management	08
6	Modern methods in Software Engineering	08

Total Hours (Theory): 48
Total Hours:48

Detail Syllabus:

Unit I: Introduction to software and Software Engineering Hours 07

Definition of Software, Types of Software, Nature of Software, Software Applications, Definition of Software Engineering, Software Engineering-A Layered Technology, A process framework.

Unit II: Software Process Models Hours 09

Definition of Software Process, Software Process Model - The Waterfall model, Incremental Process Models: The Incremental model, The RAD model Evolutionary Process models: Prototyping, The Spiral model, The Concurrent Development model, A final comment on evolutionary Processes, Component based development, Concept of Product and Process.

Unit III: Managing Software Projects Hours 08

The Management Spectrum, W5HH principles, Concepts of measures, metrics and indicators. Process Metrics and Project Metrics: An overview, Software Measurements: Size oriented and function oriented metrics.

Unit IV: Software Requirement analysis and specification Hours 08

Introduction to system and software requirements, types of software requirements: Functional and non-functional requirements. Requirement Specification (SRS), Requirement Analysis and Requirement Elicitation, Requirement Engineering.

Unit V: Software Maintenance and Risk Management Hours 08

Software risks, risk identification, risk projection, risk mitigation, monitoring and management. Introduction to software maintenance, types of maintenance. Concept of software reengineering, Business Process Reengineering(BPR), BPR life cycle, reverse engineering, Applications of Software Engineering.

Unit VI: Modern methods in Software Engineering Hours 08

Introduction to Agile development: What is Agility? principles and values of Agile development, concept of Extreme Programming, values of XP.

Core Books:

1. Roger S.Pressman, Software Engineering- A practitioner's Approach, McGraw-Hill International Editions
2. Grady Booch, James Rambaugh, Ivar Jacobson: The Unified Modeling Language User
3. Rajib Mall, Fundamentals of software Engineering, Prentice Hall of India.

Reference Books:

1. Ian Sommerville, Software engineering, Pearson education Asia
2. Pankaj Jalote, Software Engineering – A Precise Approach Wiley
3. Merlin Dorfman (Editor), Richard H. Thayer (Editor), Software Engineering
4. Robert C. "Uncle Bob" Martin, Clean Architecture: A Craftsman's Guide to Software Structure and Design
5. Deepak Gaikwad, Viral Thakkar, DevOps Tools from Practitioner's ViewPoint, Wiley

Web References:

1. https://en.wikibooks.org/wiki/Introduction_to_Software_Engineering/Process/Methodology [Introduction to Software Engineering/Process/Methodology]
2. https://www.uml-diagrams.org/ [UMLUnits]
3. https://nptel.ac.in/courses/106/101/106101061/ [Agile Software Development and Extreme Programming and Overview of DevOps]

Course Outcomes: Upon successful completion of the course, students will be,

CO1 :	Able to understand concept of Software and its types.
CO2 :	Get familiarize with software processes and related models.
CO3 :	Able to analyze requirements and mange project
CO4 :	Able to understand concepts of creating Software Requirement Specification
CO5 :	Able to understand risk management, software maintenance and modern Software engineering methods.

Course Outcomes Mapping:

Unit No.	Unit Name	Course Outcomes				
		CO1	CO2	CO3	CO4	CO5
1	Introduction to software and Software Engineering	√				
2	Software Process Models		√			
3	Managing Software Projects			√		
4	Software Requirement analysis and specification				√	
5	Software Maintenance and Risk Management					√
6	Modern methods in Software Engineering					√

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO1	PSO2
CO1	1	2	3	1	3	3	3	1	1	3	1	3	3
CO2	2	2	3	2	3	3	3	1	1	3	3	2	2
CO3	2	3	3	3	3	3	3	2	1	3	3	3	3
CO4	3	2	3	2	3	3	3	3	1	2	2	3	3
CO5	2	3	3	3	2	3	3	1	1	2	1	2	3

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

CA234: Advanced Database System (100 Marks)

Contact Hours: 03

Pre-requisite: Database Fundamentals.

Methodology & Pedagogy: During the practical session student will learn how to create table and perform operation on data and table using different oracle objects. Through procedure and function students can understand the concept of reusability. Student will learn how to handle error using exception and using trigger they can understand how to restrict the operation at particular time or given condition.

Outline of the course:

Week No	Practical	Description
1 - 2	Introduction to PL/SQL:	Basic syntax of PL/SQL Block, Declaration of variable and constants, SELECT INTO
3 - 4	Procedural statements:	Usage of Conditional statements and Looping structure inside the block
5 - 6	Implicit Cursor, Explicit Cursor, Parameterized Cursor:	Usage of cursor and working
7 - 8	Objects of the database: User defined Function and Stored Procedure:	Creations and usage of sub programs
9 - 10	Error Handling using pre-defined exception and user-defined exception:	Runtime errors and handling those errors
11 - 12	Row level and Statement level Trigger, GRANT & REVOKE:	Syntax of trigger and its working, Assign privileges at different level

Total Hours (Practical): 36
Total Hours: 36

Core Books:

1. SQL, PL/SQL – The programming Language Oracle - by Ivan Bayross.
2. Teach Yourself Database Technologies - by Ivan Bayross

Reference Books:

1. Oracle 9i, PL/SQL Programming by Scoot Urban, Oracle Press
2. Practice book on SQL and PL/SQL by Anjali, Amisha, Roopal and Nirav publications

Web References:

1. http://plsql-tutorial.com/index.htm [For PL/SQL]

Course Outcomes: Upon successful completion of the course, students will be,

CO1 :	Be familiar with PL/SQL programming
CO2 :	Be familiar with database objects
CO3 :	Be familiar with retrieval and manipulation of data
CO4 :	Learn the concept of exception handling
CO5 :	Controlling user security

Course Outcomes Mapping:

Week No.	Practical Title	Course Outcomes				
		CO1	CO2	CO3	CO4	CO5
1	Introduction to PL/SQL:	√				
2	Procedural statements:		√			
3	Implicit Cursor, Explicit Cursor, Parameterized Cursor:			√		
4	Objects of the database: User defined Function and Stored Procedure:		√			
5	Error Handling using pre-defined exception and user-defined exception:				√	
6	Row level and Statement level Trigger, GRANT & REVOKE:					√

Course Articulation Matrix:

	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO10	PO11	PO12	PSO 1	PSO 2
CO 1	2	2	3	2	2	1	2	-	-	-	-	-	3	3
CO 2	3	3	3	3	3	3	3	-	-	-	-	-	3	3
CO 3	2	2	3	3	2	2	2	-	-	-	-	-	3	2
CO 4	3	3	3	3	3	3	3	-	-	-	-	-	3	2
CO 5	3	3	3	3	3	3	3	-	-	-	-	-	3	2

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation "-"

CA235: Open Source Technology (200 Marks)

Contact Hours: 07

Pre-requisite: Working knowledge of HTML and basic knowledge of MYSQL

Methodology & Pedagogy: During theory and practical sessions, students able to install & configure PHP and prerequisite software(s). Also, student will be emphasized to develop dynamic web applications.

Outline of the course:

Unit Number	Title of the Unit	Minimum Number of Hours	
		Theory	Practical
1	Introduction to Open Source and Web	08	36
2	PHP Configuration and Basics	07	
3	Array and Functions in PHP	08	
4	Handling Form Data	08	
5	PHP utilities	09	
6	Database Connectivity with MYSQL	08	

Total Hours (Theory): 48
Total Hours (Lab): 36
Total Hours:84

Detail Syllabus:

Unit I: Introduction to Open Source and Web **Hours 08**

Introduction of open source software, Development philosophy of open source software, pros and cons of open source software, open source vs. close source software, Community support in open source, Version Control, Introduction to Webpage and Website, Static and Dynamic Website, Client & Server Side Scripting, Web Hosting: Shared / Cloud / VPS / Dedicated

Unit II: PHP Configuration and Basics **Hours 07**

Introduction to PHP, Installation of Apache, MySQL and PHP (OOP approach), How PHP code is parsed, Embedding PHP and HTML, Executing PHP and viewing in Browser, Data types, Operators, PHP variables: static and global variables, Comments in PHP, Conditional Structure, Looping Structure.

Unit III: Array and Functions in PHP Hours 08

Types of Array, Array Functions, Miscellaneous Functions: define, constant, include, require, header, die, exit, Overview of built in functions of PHP: Date and Time, Math functions, File handling functions, String functions, User Defined Functions.

Unit IV: Handling Form Data Hours 08

HTML Form element & its attributes, Send Form data using GET Method & POST Method, Receive Form data using $_GET, $_POST & $_REQUEST variables, PHP form validations

Unit V: PHP utilities Hours 09

File Uploading: Upload Single and Multiple file using PHP script, Understanding HTTP requests, Exploring and modifying HTTP responses, getting information from web server, Sending mails

Unit VI: Database Connectivity with MYSQL Hours 08

OOPs Concepts, PHP Data Objects (PDO), CRUD operations, Handling Errors, State Management Techniques: Concept of Session, starting session, modifying session variables, Un registering and deleting session variable, Concept of Cookies

Core Books:

1. Vikram Vaswani, PHP: A Beginner's Guide: Indian Edition, First Edition, McGraw Hill, 2009
2. Matt Doyle , Beginning PHP 5.3 , Wrox , 2010

Reference Books:

1. Timothy Boronczyk , Elizabeth Naramore, Jason Gerner, Yann Le Scouarnec, Jeremy Stolz, Michael K. Glass): Beginning PHP6, Apache, MySQL Web Development : Wrox , 2009.
2. Lynn Beighley and Michael Morrison, Head First PHP & MySQL, First Edition, O'Reilly Publication, 2009

Web References:

1. https://github.com/PHPMailer/PHPMailer [PHPMailer Code]
2. https://www.php.net/ [Official website of PHP]
3. https://www.geeksforgeeks.org/php-tutorials/ [Lecture notes of PHP]
4. https://www.w3schools.com/php/default.asp [Lecture notes of PHP]

Course Outcomes: Upon successful completion of the course, students will be,

CO1 :	Gain the skills and knowledge for Open Source and Web Hosting.
CO2 :	To utilize knowledge and skills for basics of PHP and configuration of MYSQL Server.
CO3 :	Learn the array and built-In functions in PHP.
CO4 :	Gain understanding of Form handing and utilities in PHP.
CO5 :	Be able to develop dynamic web based application using PHP and MySQL with state management techniques.

Course Outcomes Mapping:

Unit No.	Unit Name	Course Outcomes				
		CO1	CO2	CO3	CO4	CO5
1	Introduction to Open Source and Web	√				
2	PHP Configuration and Basics		√			
3	Array and Functions in PHP			√		
4	Handling Form Data				√	
5	PHP utilities				√	
6	Database Connectivity with MYSQL					√

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO1	PSO2
CO1	3	3	2	3	-	1	-	1	2	-	-	2	-
CO2	3	3	3	3	-	1	-	1	1	-	-	3	-
CO3	3	3	3	3	-	1	-	1	1	-	-	3	-
CO4	3	3	3	3	-	1	-	1	1	-	-	3	-
CO5	3	3	3	3	2	1	1	1	2	1	1	3	1

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

CA236: Computer Network (100 Marks)

Contact Hours: 04

Pre-requisite: Basic knowledge of computer and its working.

Methodology & Pedagogy: During the lecture sessions, the students will learn about role of computer network in modern communication systems, underlying principles of network communication, need for standardization through network reference model. The teacher will also discuss communication issues at different layers and concepts related to network security.

Outline of the course:

Unit No.	Title of the Unit	Minimum Number of Theory Hours
1	Introduction to computer network	07
2	Basics of network communication	08
3	Network reference model	10
4	Design issues in data link layer	08
5	Design issues in network layer	08
6	Network security	07

Total Hours (Theory): 48
Total Hours:48

Detail Syllabus:

Unit I: Introduction to computer network **Hours 07**

Definition of computer network, aim and objective of computer network: applications of computer network, merits & demerits of computer network, examples of computer network, components of computer network, types of networks, network topologies, network devices: hub, switch, repeater, router, gateway, bridge.

Unit II: Basics of network communication **Hours 08**

Parallel & serial communication, synchronous & asynchronous communication, modes of communication: simple, half-duplex, full-duplex, multiplexing, de-multiplexing: frequency division multiplexing (FDM), time division multiplexing (TDM), communication media: guided media, unguided media, switching techniques: circuit switching, message switching, packet switching.

Unit III: Network reference model **Hours 10**

Need for network reference model, history of TCP/IP and OCI model, layers of OCI and TCP/IP model, comparison between TCP/IP and OSI model, Protocol Hierarchy, Network Protocols: Simple Mail Transfer Protocol (SMTP), File Transfer Protocol (FTP), Multiple Internet Mail Extension (MIME), Post Office Protocol (POP), Telnet, Domain Name Service (DNS), Dynamic

Host Configuration Protocol (DHCP), Address Resolution Protocol (ARP), Reverse Address Resolution Protocol (RARP).

Unit IV: Design issues in data link layer — Hours 08

Services provided to the network layer, framing, error control (error detection and correction code), flow control, data link layer in the internet (Serial Line Internet Protocol, Point-to-Point Protocol).

Unit V: Design issues in network layer — Hours 08

Routing Algorithms, Congestion Control Policies, Concept of Internetworking.

Unit VI: Network security — Hours 07

Introduction to cryptography, symmetric key and public key encryption systems, digital signature and key management concepts, communication security.

Core Books:

1. Andrew S. Tanenbaum: Computer networks, 4th Edition, Pearson Publication: 2019.
2. Behrouz Forouzan: Data communications and networking, 4th Edition, McGraw Hill, 2018.
3. William Stallings: Data and computer communication: 5th Edition, Pearson Publication, 2019.

Reference Books:

1. James F. Kurose, Keith W. Ross: Computer networking – A top down approach: 3rd Edition: Pearson Publication: 2017
2. Narsimha Karumanchi: Elements of computer networking: An Integrated Approach: 1st Edition: Careermonk Publication: 2014
3. Ed Tittel: Computer networking: 1st Edition, Schum's Publication: 2020

Web References:

1. www.firewall.cx/networking-topics.html [General networking concepts]
2. https://help.sap.com/viewer/d0224eca81e249cb821f2cdf45a82ace/6.7.0.0/en-US/8c74877b866910148cc9ba5f39a2fd28.html [Transport layer concepts]
3. https://www.plixer.com/blog/network-layers-explained/ [Network reference layers]

Course Outcomes: Upon successful completion of the course, students will be,

CO1 :	Student will be able to understand role of computer network in digital communication
CO2 :	Student will grasp underlying concepts related with network communication.
CO3 :	Student will understand role of reference models in standardizing communication
CO4 :	Students will learn about intra and inter network communication and technology behind it.
CO5 :	Students will learn about different aspects of network security.

Course Outcomes Mapping:

Unit No.	Unit Name	Course Outcomes				
		CO1	CO2	CO3	CO4	CO5
1	Introduction to computer network	√				
2	Basics of network communication		√			
3	Network reference model			√		
4	Design issues in data link layer				√	
5	Design issues in network layer				√	
6	Network security					√

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO1	PSO2
CO1	2	3	2	1	-	1	1	1	-	-	1	2	-
CO2	2	3	3	2	-	1	1	1	-	-	1	3	1
CO3	3	3	3	2	-	1	1	1	1	-	1	3	3
CO4	1	3	3	2	-	1	1	1	1	-	1	3	2
CO5	3	3	3	2	-	1	1	1	1	-	1	3	2

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

HS111.02 C: HUMAN VALUES & PROFESSIONAL ETHICS (100 Marks)

Contact Hours: 03

Pre-requisite courses: Ethical Leadership through Giving Voice to Values (https://www.coursera.org/learn/uva-darden-giving-voice-to-values?skipBrowseRedirect=true)

Outline of the Course:

Unit. No.	Title of the unit	Minimum number of hours
1.	Introduction to Values and Ethics	05
2.	Elements and Principles of Values	08
3.	Applied Ethics	08
4.	Value, Ethics & Global Issues	05
5.	Contemporary Issues in Values and Ethics	04

Total Hours (Practical): 30
Total Hours: 30

Detailed Syllabus:

Unit – I	**Introduction to Values and Ethics** Need, Relevance and Significance of Values General, Concept and Meaning of Values and Ethics	**Hours: 05**
Unit – II	**Elements and Principles of Values** Universal & Personal Values, Social, Civic & Democratic Value	**Hours: 08**
Unit –III	**Applied Ethics** Universal Code of Ethics, Professional Ethics, Organizational Ethics, Ethical Leadership, Domain Specific Ethics	**Hours: 08**
Unit – IV	**Value, Ethics & Global Issues** Cross-Cultural Issues, Role of Ethics & Values in Sustainability	**Hours: 05**
Unit – V	**Contemporary Issues in Values and Ethics** Case Studies, Presentations, Projects	**Hours: 04**

Course Outcome (COs): At the end of the course, the students will be able to

CO1	Understand the concepts and mechanics of values and ethics.
CO2	Understand the significance of value and ethical inputs in and get motivated to apply them in their life and profession.
CO3	Understand the significance of value and ethical inputs in and get motivated to apply them in social, global and civic issues.
CO4	Develop their responsibility towards society.
CO5	Comprehend their own core values and adhere to those values at their workplace.
CO6	Practice Ethical Leadership.

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO1	PSO2
CO1	-	-	-	-	-	-	-	-	2	-	-	-	-
CO2	-	-	-	-	3	-	-	-	2	-	-	-	-
CO3	-	-	-	-	-	3	-	-	2	-	-	-	-
CO4	-	-	-	-	-	-	-	-	2	2	-	-	-
CO5	-	-	-	-	3	-	-	-	2	-	-	-	-
CO6	-	-	-	-	-	-	-	-	2	-	-	-	-

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation "-"

Reference book:

1. Human Values and Ethics in Workplace, United Nations Settlement Program, 2006. (http://www.unwac.org/new_unwac/pdf/HVWSHE/Human%20Values%20&%20Ethics%20-%20Individual%20Guide.pdf).
2. Ethics for Everyone, Arthur Dorbin, 2009. (http://arthurdobrin.files.wordpress.com/2008/08/ethics-for-everyone.pdf).
3. Values and Ethics for 21st Century, BBVA. (https://www.bbvaopenmind.com/wp- content/uploads/2013/10/Values-and-Ethics-for-the-21st-Century_BBVA.pdf)

Web material:

- www.ethics.org

DETAILED SYLLABUS

FOR

BCA PROGRAMME
(5^{th} SEMESTER)

EFFECTIVE FROM
ACADEMIC YEAR 2023-24

Teaching & Examination Scheme
Bachelor of Computer Applications (B.C.A.) Programme
(Choice Based Credit System)
Effective from Year 2023 - 24
Semester - V

Course Code	Course Title	Teaching Scheme				Examination Scheme						
		Contact Hours			Credit	Theory			Practical			Total
						Internal		External	Internal		External	
		Theory	Practical	Total		Case Study	Tests		Term work	Tests		
CA324-326	Elective-V	4	3	7	7	10	20	70	15	15	70	200
CA327	Object Oriented Programming using JAVA	4	3	7	7	10	20	70	15	15	70	200
CA328	Fundamentals of Visual Programming	4	3	7	7	10	20	70	15	15	70	200
CA329	Introduction to Multi-Paradigms Programming language	-	3	3	3	-	-	-	15	15	70	100
HS131.02 C	Communication and Soft skills	-	2	2	2				30		70	100
		12	14	26	26	300			500			800

Elective-V	
Course Code	**Course Title**
CA324	Document Databases
CA325	Foundation of Data Science
CA326	Advanced Embedded Programming

CA324: Document Databases

(200 Marks)

Contact Hours: 07

Pre-requisite: Should know that Database Management Systems, Database and Data Structure concepts

Methodology & Pedagogy: During theory lectures student will understand that what is difference between SQL and NoSQL i.e. Document Database, the essential concepts that act as the building blocks for many of the NoSQL products. It starts from the fundamentals of NoSQL and graduates to advanced concepts of architecture, storage internals and indexing. During Practical sessions, students will be required to design and create an applications using NoSQL Databases.

Outline of the Course:

Unit No.	Title of the Unit	Minimum Number of Hours	
		Theory	Practical
1	Introduction to Document Database (NoSQL)	07	36
2	Data Models in NoSQL	08	
3	Storage Replica of NoSQL	09	
4	Working with Document Data using MongoDB	08	
5	Executing CRUD operation using MongoDB	08	
6	Managing Database using MongoDB	08	

Total Hours (Theory): 48

Total Hours (Lab): 36

Total Hours: 84

Detail Syllabus:

Unit I: Introduction to Document Database (NoSQL) Hours 07

Introduction to SQL Relational Model, Codd Rules, Overview of Normalization, Overview and History of NoSQL Databases, Taxonomies by Data Model, Overview of No-SQL Storage: Key Value, Document-Oriented, Column-Family and Graph-storage. Difference Between SQL and NoSQL, Techniques and Pattern: CAP Theorem, ACID Vs. BASE properties and Comparison of relational databases to new NoSQL stores.

Unit II: Data Models in NoSQL **Hours 08**

Different Types of Database System to use Document Database by using MongoDB, Cassandra, HBASE, Neo4j use and deployment, Features and Application, Limitation and Challenges to use NoSQL approach, Key-Value and Document Data Models, Column-Family Stores, Aggregate-Data Model. Overview of Replication and Sharding, Overview of Map Reduce on databases. Distribution Models, Single Server, Sharding, Master-Slave Replication, Peer-to-Peer Replication, Combining Sharding and Replication.

Unit III: Storage Replica of NoSQL **Hours 09**

Working with Column-Oriented Databases, Document Store, Key Value Store and Graph data storage. Using Tables and Columns in Relational Databases, Contrasting Column Databases with RDBMS, Document Store Internals, Instructions for Using Collections and Indexes in MongoDB, MongoDB Reliability and Durability, Horizontal Scaling.

Unit IV: Working with Document Data using MongoDB **Hours 08**

Overview of MongoDB, Document Databases, What Is a Document Database? Advantages and Disadvantages of Document Database, Installing MongoDB on Your System, Running MongoDB, Installing Additional Drivers, Introduction to MongoDB key features, Core Server tools, MongoDB through the JavaScript's Shell and others.

Unit V: Executing CRUD operation using MongoDB **Hours 08**

Navigating your databases, viewing available data and collections, inserting data, querying data, using sort, limit and skip functions, retrieving single document, using aggregation commands and constraints, working with condition, updating data, insert and save command, rename a collection, referencing a database

Unit VI: Managing Database using MongoDB **Hours 08**

MongoDB server, backup a single database, single collection, restoring the collection, importing data into Mongo-DB, exporting data from MongoDB, securing with authorization your data, restricting access, protecting your server with authentication.

Core Books:

4. NoSQL Distilled: A Brief Guide to the Emerging World of Polyglot Persistence, Author: Sadalage, P. & Fowler, Publication: Pearson Education.

Reference Books:

1. Redmond, E. &Wilson, Author: Seven Databases in Seven Weeks: A Guide to Modern Databases and the NoSQL Movement Edition: 1st Edition.
2. Getting started with NOSQL by Gaurav Vaish, PACKT.
3. Guy Harrison, Next Generation Database: NoSQL and big data, Apress.
4. Definite Guide to MongoDB by Eelco Plugge, Peter Membrey and Tim Hawkins, Apress.
5. NOSQL theory and Examples by Piotr fulmanski, Simple introduction series: e-book.

Web References:

1. https://www.javatpoint.com/mongodb-tutorial [MongoDB Tutorial - Javatpoint]
2. https://www.tutorialspoint.com/neo4j/neo4j_data_model.htm [Neo4j - Data Model (tutorialspoint.com)]
3. https://appinventiv.com/blog/hbase-vs-cassandra/[HBase vs Cassandra: Which is The Best NoSQL Database (appinventiv.com)]

Course Outcomes: Upon successful completion of the course, students will be,

CO1 :	Able to understand the difference between SQL and NoSQL, needs and what is NoSQL
CO2 :	Able to compare, define and use the four types of NoSQL Databases.
CO3 :	Able to demonstrate an understanding of the detailed architecture, define objects, load the data, and execute the data
CO4 :	Able to implement various column store internals and design Schema, implement CRUD operations, distributed data operations
CO5 :	Able to create and use an Application with Graph Data model

Course Outcomes Mapping:

Unit No.	Unit Name	Course Outcomes				
		CO1	CO2	CO3	CO4	CO5
1	Introduction to Document Database (NoSQL)	√				
2	Data Models in NoSQL		√			
3	Storage Replica of NoSQL			√		
4	Working with Document Data using MongoDB			√		
5	Executing CRUD operation using MongoDB				√	
6	Managing Database using MongoDB					√

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO1	PSO2
CO1	3	-	-	1	-	-	-	-	-	-	-	-	-
CO2	3	-	2	2	-	-	-	3	2	-	-	1	2
CO3	3	3	3	-	1	-	-	-	3	-	-	3	-
CO4	3	-	2	-	2	-	2	2	2	-	3	1	1
CO5	3	3	1	3	3	-	-	3	-	-	1	2	2

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

CA325: Foundation of Data Science (200 Marks)

Contact Hours: 07

Pre-requisite: CA130- Database Fundamentals

Methodology & Pedagogy: During theory sessions topics related to Data Science will be will be covered with suitable examples. During Practical Sessions, students will be required to develop small applications related to reading, cleaning, analysis and visualization. Student will also get exposure of model building using machine learning algorithms. Student will get in-death knowledge of Data Science based real applications using case study approach.

Outline of the course:

Unit No.	Title of the Unit	Minimum Number of Hours	
		Theory	Practical
1	Overview of Data Science	07	36
2	Mathematics and Statistics in Data Science	09	
3	Data Collection and Preparation	08	
4	Data Plotting and Visualization	08	
5	Machine Learning for Data Science	10	
6	Data Science Applications and future trends	06	

Total Hours (Theory): 48

Total Hours (Lab): 36

Total Hours: 84

Detailed Syllabus:

Unit -I: Overview of Data Science **Hours: 07**

Introduction to Data Science, Definition and description of Data Science, history and development of Data Science, Need of Data Science, Data Science Process, Prerequisites for Data Scientist, Components of Data Science, Tools and Skills needed.

Unit -II: Mathematics and Statistics in Data Science **Hours: 09**

Linear Algebra, Calculus, Descriptive and Inferential Statistics, Central Tendency, Measure of Variability, Probability, Implementation using Python.

Unit -III: Data Collection and Preparation **Hours: 08**
Introduction to Data Preprocessing, Data types and forms, Possible data error types, Consistent Data,
Data Cleaning - Filling Missing Values, smoothing noisy data, Detecting and removing outliers, Data Integration, Data transformation- Rescaling, Normalizing data, Standardizing data, Implementation Using Python.

Unit -IV: Data Plotting and Visualization **Hours: 08**
Introduction to Data Visualization, Visual encoding, Basic plots using Python, Data Visualization libraries in Python.

Unit -V: Machine Learning for Data Science **Hours: 10**
Overview of Machine Learning- Supervised, Unsupervised, Semi-Supervised and Reinforcement Learning
Regression Methods – Simple Linear Regression, Multiple Linear Regression, Classification Methods- Logistic Regression, Naïve Bayes, Clustering Method- k-means clustering

Unit -VI: Data Science Applications and future trends **Hours: 06**

Applications- Manufacturing, Personalization, Medical Image Analysis, Internet Search, Genomics, Speech Recognition, Advertising, Financial Fraud Detection, Airline Route Planning, Cyber-threats, transportation, gaming, recommendation system

Core Books:

1. Sanjeev J, Wagh, Manisha S. Bhende, Anuradha D. Thakare : Fundamentals of Data Science: First Edition: CRC Press : 2023
2. Dr. Gypsy Nandi, Dr.Rupamkumar Sharma: Data Science Fundamentals and Practical Approaches: BPB Publications:2020

Reference Books:

1. Parikshit Narenda Mahalle, Gitanjali Rahul Shinde, Priya Dudhale Pise, Jyouti Yogesh Deshmukh:
   Foundation of Data Science for Engineering Problem Solving: Springer:2022
2. K.G.Srinivasa,G.M.Siddesh,ChetanShetty,SowmyaB.J:Statistical Programming in R: First Edition: Oxford Press: 2017.
3. Lillian Pearson : Data Science for Dummies,:John Willey & Sons:2015
4. Rachel Schutt, Carthy O'Neil: Doing Data Science: First Edition:O'Reilly: 2014.

Web References:

1. https://www.cs.cornell.edu/jeh/book.pdf [Foundation of Data Science E-book]
2. https://www.edureka.co/blog/math-and-statistics-for-data-science/ [Essentials Mathematics and Statistics of Data Science]
3. https://www.w3schools.com/datascience/ds_analyze_data.asp [Data Preparation using tools]
4. https://www.kdnuggets.com/2022/06/plotting-data-visualization-data-science.html [Data Plotting and Visualization]
5. https://www.simplilearn.com/10-algorithms-machine-learning-engineers-need-to-know-article [Machine Learning Algorithms]
6. https://www.geeksforgeeks.org/major-applications-of-data-science/ [Applications of Data Science]

Course Outcomes: Upon successful completion of the course, students will be,

CO1 :	Able to understand fundamentals of Data Science.
CO2 :	Able to use mathematics and statistical techniques for Data Science Applications
CO3 :	Able to apply data preparation and visualization techniques.
CO4 :	Able to build model using machine learning techniques.
CO5 :	Able to understand applications and future trends of Data Science.

Course Outcomes Mapping:

Unit No.	Unit Name	Course Outcomes				
		CO1	CO2	CO3	CO4	CO5
1	Overview of Data Science	√				
2	Mathematics and Statistics in Data Science		√			
3	Data Collection and Preparation			√		
4	Data Plotting and Visualization			√		
5	Machine Learning for Data Science				√	
6	Data Science Applications and future trends					√

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO1	PSO2
CO1	1	1	1	1	1	1	-	-	1	-	1	2	-
CO2	3	3	2	3	-	3	-	-	-	-	1	2	-
CO3	2	3	2	3	-	3	-	-	-	-	1	2	-
CO4	3	3	3	3	-	3	-	-	-	-	2	3	-
CO5	1	1	1	-	-	1	-	-	3	-	-	2	-

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

CA326: Advanced Embedded Programming
(200 Marks)

Contact Hours: 07

Pre-requisite: CA231: Fundamentals of Embedded Programming

Methodology & Pedagogy: In order to achieve the course objectives, students will be introduced to advanced concepts of embedded system and its structure. Students will be introduced to various microcontrollers like nodeMCU, Raspberry PI and ESP32 which can be used to design IOT system. Introduction to IOT system design with specifications, development and implementation will be explained. Students will also be introduced to basic IOT programming of an embedded system.

Outline of the course:

Unit No.	Title of the Unit	Minimum Number of Hours	
		Theory	Practical
1	Emerging trends in Embedded system and its applications	06	36
2	Wireless Communications using Arduino	08	
3	Advanced GPIO Programming in embedded	08	
4	Introduction to Raspberry Pi	08	
5	Introduction to Python Programming Language	10	
6	RTOS: Operating System for Embedded System	08	

Total Hours (Theory): 48

Total Hours (Lab): 36

Total Hours: 84

Detail Syllabus:

Unit I: Emerging trends in Embedded system and its applications
Hours 06

Patterns insight from the applications of embedded systems in real life, Multicore in embedded, embedded operating systems. Embedded digital security and surveillance. Convergence embedded systems and applications. Healthcare, Automotive, Entertainment, Localization and internationalization, Real Time Applications of Embedded Systems

Unit II: Wireless Communications using Arduino **Hours 08**

Introduction to Wireless Devices, Interfacing GSM & GPS Module with Arduino, Interfacing Bluetooth Module with Arduino.

Unit III: Advanced GPIO Programming in embedded Hours 08

Character LCD 16x2 interfacing logic and concept, Introduction of LCD command and data signals, LCD based programming, Serial Communication programming

Unit IV: Introduction to Raspberry Pi Hours 08

Introduction to Raspberry Pi, Why Raspberry pi, Features of Raspberry pi, Basic set up and first boot configuration, Different uses of Raspberry pi, Different Versions of Raspberry pi, Vision of Raspberry pi, Different components of the Board.

Unit V: Introduction to Python Programming Language Hours 10

Variables and Numbers, Looping Structures, Conditional Statements, Lists, Tuples and Dictionaries, Type Conversions, Function declaration, calling functions and passing values, Function Returning values. Exiting from functions, Reading/Writing to File.

Unit VI: RTOS: Operating System for Embedded System Hours 08

Inter process communication, RTOS, Classification of RTOS, Difference between GPOS and RTOS, Service provided by RTOS, Features of RTOS. Selection of RTOS, Impacts of RTOS on development of the design, RTOS Architecture, Multitasking, Process life cycle, Task States, Context Switch, Scheduling, Algorithm, shared memory problem.

Core Books:

4. K. Shibu: Introduction to Embedded Systems, Tata McGraw-Hill Education, 2009
5. Dr. Simon Monk: Programming the Raspberry Pi: Getting Started with Python, McGraw Hill Publication.

Reference Books:

1. M. Banzi and M. Shiloh: Getting started with arduino: The open source electronics prototyping platform, 3rd ed. Sebastopol, CA: Maker Media, 2015
2. M. Richardson, S. Wallace: Getting Started with Raspberry Pi, O'Reilly, 2012
3. Massimo Banzi, Michael Shiloh: Getting Started with Arduino: The Open Source Electronics Prototyping Platform, Make Communication, 3rd Edition, Kindle Edition, 2014

Web References:

1. https://www.udemy.com/course/master-the-basics-of-arduino-learn-arduino-step-by-step/
2. https://www.hackerearth.com/blog/developers/arduino-programming-for-beginners/
3. https://www.arduino.cc/en/Tutorial/HomePage
4. https://www.youtube.com/watch?v=UUOChoCbty8[Raspberry Pi - How to start programming with Python]
5. https://www.raspberrypi.org/documentation/usage/python/[Step by Step Programming]

Course Outcomes: Upon successful completion of the course, students will be,

CO1 :	Get awareness about recent trends in embedded system.
CO2 :	Developing Wireless Application using microcontroller
CO3 :	Interface GPIO with microcontroller
CO4 :	To study basics of Raspberry PI and Python Programming
CO5 :	To design RTOS for an embedded system

Course Outcomes Mapping:

Unit No.	Unit Name	Course Outcomes				
		CO1	CO2	CO3	CO4	CO5
1	Emerging trends in Embedded system and its applications	√				
2	Wireless Communications using Arduino		√			
3	Advanced GPIO Programming in embedded			√		
4	Introduction to Raspberry Pi				√	
5	Introduction to Python Programming Language				√	
6	RTOS: Operating System for Embedded System					√

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	3	1	1	1	1	1	1	1	-	-	-	-	1	1
CO2	2	2	1	1	1	1	1	1	-	1	-	-	1	2
CO3	2	2	1	2	1	1	2	1	-	1	2	-	2	2
CO4	2	3	2	2	2	2	2	2	2	2	2	-	2	2
CO5	1	3	3	3	2	2	3	2	2	3	3	-	3	3

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

CA327: Object Oriented Programming using JAVA (200 Marks)

Contact Hours: 07

Pre-requisite: Basic Knowledge of Introduction to Programming in C and OOPs concepts.

Methodology & Pedagogy: During theory lectures illustrations emphasizing the need for Object Oriented Programming concepts will be discussed. Emphasize will be given on Core Java concepts, various Java packages, Multithreading and Collection Frameworks. During Practical sessions, students will be required to develop different Java programs, develop simple to intermediate level applications on Java platform.

Outline of the course:

Unit No.	Title of the Unit	Minimum Number of Hours	
		Theory	**Practical**
1	Introduction to Object Oriented Programming in Java	07	36
2	Fundamental programming structures in java	08	
3	Inheritance, Abstract class, and Interface	09	
4	Packages and Wrapper Classes	08	
5	Exception handling and Multithreading	08	
6	Basics of Collections Framework	08	

Total Hours (Theory): 48

Total Hours (Lab): 36

Total Hours: 84

Detailed Syllabus:

Unit - I: Introduction to Object Oriented Programming in Java **Hours 07**

Programming Paradigms (Procedure Oriented and Object Oriented), Object Oriented Programming concepts, comparison of POP and OOP, Introduction to Java programming language, History of Java, Features of Java, Structure of Java program, creating an application in Java, Compiling and executing applications in java, command line arguments.

Unit - II: Fundamental programming structures in java **Hours 08**

Data types, Variables, Literals, Control & Looping Structure, break, continue and return, Arrays, for-each loop, defining a class in Java, creating objects, Instance variables, methods, and its overloading, use of toString (), method of String class, Constructors and its overloading, this keyword, Objects as Parameters, returning objects, static members, Initializer blocks, Garbage collection and finalize method.

Unit - III: Inheritance, Abstract class, and Interface **Hours 09**

Inheritance and its types, method overriding, final keyword, super keyword, Abstract class and abstract methods, Interfaces, Runtime Polymorphism, Introduction to Anonymous Class.

Unit - IV: Packages and Wrapper Classes **Hours 08**

Defining a Package, finding packages and CLASSPATH, Important of Packages, Access protection, importing packages, Wrapper Classes, Boxing and Unboxing conversions.

Unit - V: Exception handling and Multithreading **Hours 08**

Fundamentals of Exception Handling, Types of Exceptions, keywords used in Exception handling, Multiple catch clauses, Nested try Statements, Java Built in Exceptions, User Defined Exceptions, The Java Thread Model, creating a thread, Creating Multiple Threads, Thread class and the Runnable Interface, Priorities, synchronization, Inter thread communication.

Unit - VI: Basics of Collections Framework **Hours 08**

The Collections Framework Overview, Collection Interfaces- Collections, List, Set, Sorted Set, Deque, The Collection classes- Array List, Linked List, Vector, Hash Set, Tree Set, Priority Queue. Accessing a collection using an Iterator, ListIterator and Enumeration, Map Interfaces and Classes: Map, Map Entry, SortedMap, and NavigableMap, HashMap, LinkedHashMap, TreeMap, Comparable & Comparator Interface.

Core Books:

1. Pravin M. Jain: The class of JAVA, Pearson Education, 2011.
2. Herbert Schildt: The Complete Reference Java J2SE, 5th Edition, TMH Publishing Company Ltd,NewDelhi.

Reference Books:

1. Ivor Horton: Beginning Java JDK, 5th Edition, Wiley Press, 2008.
2. Cay Horstmann and Gary Cornell: Core Java Volume 1, 8th Edition or later, Pearson Education, 2008.
3. Ken Arnold, James Gosling and David Holmes: The Java Programming Language, 4th Edition, Addison Wesley.

Web References:

1. http://ocw.mit.edu/courses/electrical-engineering-and-computer-science/6-092introductiontoprogramming-in-java-january-iap-2010/lecture-notes/ [Lecture Notes in PDF]
2. http://www.javatpoint.com/java-tutorial [Notes]

Course Outcomes: Upon successful completion of the course, students will be,

CO1 :	To understand basic of Object Oriented Concepts, Java Buzz Words and application of Java, execution environment in IDES i.e. NetBeans & Eclipse .
CO2 :	To understand basic syntax and semantics of Java, structure of java.
CO3 :	Learning Inheritance, Abstract class and Interface to understand how the java works ethically .
CO4 :	To gain understanding creating packages and how to import packages , creating application using multithreading & exception handling.
CO5 :	To learn classes and interfaces of collections Framework

Course Outcomes Mapping:

Unit No.	Unit Name	Course Outcomes				
		CO1	**CO2**	**CO3**	**CO4**	**CO5**
1	Introduction to Object Oriented Programming in Java	√	√			
2	Fundamental programming structures in java		√			
3	Inheritance, Abstract class, and Interface			√		
4	Packages and Wrapper Classes			√	√	
5	Exception Handling and Multithreading				√	
6	Basics of Collections Framework					√

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO1	PSO2
CO1	2	2	3	2	2	2	-	-	-	1	-	3	3
CO2	3	3	2	3	3	3	-	-	-	1	-	3	3
CO3	2	2	2	3	2	2	-	-	-	2	-	2	2
CO4	3	3	3	3	2	3	-	-	-	2	-	2	2
CO5	3	3	3	3	3	3	-	-	-	1	-	3	2

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

CA328: Fundamentals of Visual Programming (200 Marks)

Contact Hours: 07

Pre-requisite: Basic knowledge of programming language.

Methodology & Pedagogy: During theory lectures illustrations emphasizing the need for advanced features of .Net framework and to develop windows and web based application using C# language will be given. During Practical sessions, students will be required to develop Web Applications and Desktop Applications using concepts discussed during class.

Outline of the course:

Unit No.	Title of the Unit	Minimum Number of Hours	
		Theory	Practical
1	Introduction to .NET framework	07	36
2	Basics of Object Oriented programming using C#	09	
3	Building desktop application	08	
4	Web programming fundamentals	08	
5	Database connectivity through ADO.NET	08	
6	Building WPF application	08	

Total Hours (Theory): 48

Total Hours (Lab): 36

Total Hours: 84

Detail Syllabus:

Unit I: Introduction to .NET framework **Hours 07**

Introducing .NET Framework, .NET Framework Component, .NET Framework Version Compatibility, Core of .NET Framework: Application Services, Base Class Library and CLR, Types of Applications which can be developed using MS.NET, MS.NET Base Class Library, MS.NET Namespaces, The Common Language Runtime (CLR) , Managed Code , MS.NET Memory Management / Garbage Collection , Common Type System (CTS), Common Language Specification (CLS) , Types of JIT Compilers

Unit II: Basics of Object-Oriented programming using C# **Hours 09**

Overview of Object-Oriented Programming, Principles of Object Orientation programming: Encapsulation, inheritance, polymorphism, Object, Defining Properties, Using Auto- Implemented Properties, Defining Methods, Understanding Constructors, Extending .NET Classes via Inheritance, Defining and Implementing Interfaces, Understanding the Role of Interfaces in .NET.

C# Datatypes & Variables Declaration, Implicit and ExplicitCasting, Checked and Unchecked Blocks – Overflow Checks, Casting between other datatypes, Boxing and Unboxing, Enum and Constant, Operators, Control Statements, Working with Arrays, C# Application Basics: Command line and VS.NET compilation.

Unit III: Building desktop application **Hours 08**

Building Windows Forms Applications, Setting Form Properties, Understanding the Life-cycle of a Form, Using the Windows Forms Designer, MessageBox Class and .config File, Working with Windows Forms Controls: TextBox, Button, Selection, List, Container, Image, ErrorProvider, TooltipProvider, etc. Handling Events: Understanding the Event-Driven Programming Model, Writing Event Handlers, Sharing Event Handlers.

Unit IV: Web programming fundamentals **Hours 08**

Basic of Http Request and Http Response. o Understand form Tag and Comparison between Get and Post. ASP.NET Page Life Cycle. Structure of an ASP.NET Page: ASPX Page, Code behind File, WebConfig and machine config, Rich Web Control,Validation Controls, Concept of custom control.

Unit V: Database connectivity through ADO.NET **Hours 08**

Evolution of ADO.NET, Understanding the ADO.NET Object Model, Connected vs. Disconnected Access, Use ofConnection, Command, DataReader, DataAdapter and Dataset class.

Unit VI: Building WPF application **Hours 08**

Overview of WPF, Features of WPF, Windows Forms vs WPF , Introduction to XAML, WPF Layout: Grid Panel, Stack Panel, Dock Panel, Wrap Panel and Canvas Panel, WPF Controls: TextBlock, PasswordBox, ListBox, Slider,Popup, Menus and Expander.

Core Books:

1. Christian Nagel : Professionals C# and .NET 2021 Edition , Wiley – India, WROX
2. Albahani Josheph: C# 4.0 Pocket Reference, Shroff publication,2010

Reference Books:

1. Jon Skeet, C# in Depth, Managing Publication,2010
2. Drayton David: C# Language Pocket Reference, Shroff publication,2005

Web References:

1. http://csharp.net-informations.com/ [to learn C# and .NET fundamentals]
2. http://www.wpftutorial.net/[tolearn WPF]

Course Outcomes: Upon successful completion of the course, students will be,

CO1 :	Understanding .NET Framework core features and Components.
CO2 :	To develop object oriented programming skills using C#.
CO3 :	To develop windows and web based application using core feature of .NET framework and C#language.
CO4 :	Access and manipulate data in a Microsoft SQL Server database by using MicrosoftADO.NET.
CO5 :	Use modern approach to build windows based application using WPF.

Course Outcomes Mapping:

Week No.	Practical Title	Course Outcomes				
		CO1	CO2	CO3	CO4	CO5
1	Introduction to .NET framework	√				
2	Basics of Object Oriented programming skills using C#		√			
3	Building desktop application			√		
4	Web programming fundamentals			√		
5	Database connectivity through ADO.NET				√	
6	Building WPF application					√

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO1	PSO2
CO1	3	2	2	3	-	2	1	-	-	1	1	2	-
CO2	2	3	2	2	-	3	2	-	-	2	-	1	-
CO3	2	3	3	2	-	2	2	-	-	2	1	2	1
CO4	2	3	3	2	-	3	3	-	-	2	2	2	1
CO5	-	3	3	-	-	-	-	-	-	-	1	-	-

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

CA329: Introduction to Multi-Paradigms Programming language (100 Marks)

Contact Hours: 03

Pre-requisite courses: CA123: Foundation of Programming, CA129: Fundamentals of Object Oriented Programming.

Methodology & Pedagogy: During the practical session student will learn how to write Python Programs and then start implementing various programming constructs. Through the implementations of various datatypes implementation student's will realize significance of high level language. Student will learn how to create user defined function and using them. Under object oriented programming students will learn creation of class, instance and implementation of inheritance.

Outline of the Course:

Week No	Practical	Description
1- 2	Introduction to Python: Running Python, Python Documentation, Python Basics- The print statement, comment, statements and syntax, variable assignments, identifier. Python standard types: Number, String.	Students can learn executing first program of Python. Students start familiar with printing statement in Python, declaring variable and also valid identifier in Python. Then Students will learn Number and String type in Python.
3 - 4	Conditional and looping statements in Python: If - else, if-elif-ilse, conditional expressions, while, for, break, continue, pass.	Usage of Conditional statements and Looping statements.
5 - 6	Python Sequence types: List, Tuple, Dictionary, Set	Implementations of sequence types List, tuple, dictionary and set with built-in functions and various operations.
7 - 8	Functional Programming: Creating function, Calling function, different types of arguments, Scope of variables.	Creating and using functions with different types of arguments and scope of variable.
9 - 10	Object Oriented Programming: Class, Class attributes, Instance, instance attributes, bind	Crating class, instantiating class, defining various members of class. Implementing

	and class methods, abstraction, encapsulation, polymorphism, inheritance.	constructor. Developing programs using OOPS concepts.
11 - 12	Various important package coverages: Numpy and Pandas.	Introducing concept of data frame and its usage: Reading csv file, manipulating data frame and performing various operations on it using Pandas. Working with arrays using numpy .

Total Hours (Practical): 36

Total Hours: 36

Core book:

1. Megnus Lie Hetland: Beginning Python from novice to professional, 2017 edition, Apress.
2. Wesley J. Chun: Core Python Programming, 2nd edition, Pentice Hall,2006.

Reference book:

1. Richard L. Halterman: Learning to Program with Python, Southern Adventist University 2011.
2. Mark Lutz: Programming Python, 4th Edition, O'reilly, 2011. 3.Steve Holden: Python Web Programming, 1st edition,2002.

Web material:

1. https://www.python.org/about/gettingstarted/ [For beginners of Python]
2. htttp://ocw.mit.edu/courses/electrical-engineering-and-computerscience/6-189-agentleintroduction-to-programming-using-pythonjanuary-iap-2008/ [Python Materials]
3. http://people.cs.aau.dk/~normark/prog303/html/notes/paradigms_themesparadigmoverview-section.html [Overview of Programming Paradigms]
4. https://docs.python.org [Python Documentation]

Course Outcome (COs): At the end of the course, the students will be able to

CO1	Able to understand execute python program and implementation of basic number and string types supported by Python.
CO2	Able to learn programming constructs, conditional statements and looping statements.
CO3	Able to implement list, tuple, dictionary and set types supported by Python.
CO4	Able to create functions and implement object oriented programming in Python.
CO5	Able to use packages numpy and pandas in Python.

Course Outcomes Mapping:

Week No.	Practical Title	Course Outcomes				
		CO1	**CO2**	**CO3**	**CO4**	**CO5**
1	Introduction to Python	√				
2	Using various programming constructs		√			
3	Sequence types: List, Tuple, Dictionary, Set			√		
4	Creating and using Functions				√	
5	Object Oriented Programming using Python				√	
6	Implementation using Pandas and Numpy					√

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO1	PSO2
CO1	2	2	3	2	2	1	2	2	2	1	1	3	2
CO2	3	3	3	2	3	3	2	2	2	2	2	3	2
CO3	2	2	3	3	3	2	2	2	2	1	1	3	2
CO4	3	3	3	3	3	3	3	3	3	2	3	3	3
CO5	3	3	3	3	3	3	3	3	3	3	3	3	3
CO6	2	2	3	2	2	1	2	2	2	1	1	3	2

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

HS131.02 C: Communication and Soft skills (100 Marks)

Credits and Hours:

Teaching Scheme	Theory	Practical	Tutorial	Total	Credit
Hours/week	--	30/15	--	30/15	**02**
Marks	**--**	100	--	100	

Pre-requisite courses:

- Communicative English

Outline of the Course:

Sr. No.	Title of the unit	Minimum number of hours
1.	An Introduction to Communication	06
2.	Cross-cultural Communication and Globalization	03
3.	Communication for Career Building	10
4.	Group Dynamics and Soft Skills	05
5.	Effective Presentation Strategies	04
6.	Contemporary Issues in Communication and Soft Skills	02
	Total hours (Theory) :	--
	Total hours (Practical) :	30
	Total hours :	30

Detailed Syllabus:

1.	An Introduction to Communication	06 Hours	20%
	Basics of Communication: Origin, Concept, Process, Levels, Principles and Barriers; Applications of Communication; Rhetoric in Professional Communication; Importance of Ethos, Logos, and Pathos in Communication		
2.	Cross-cultural Communication and Globalization	03 Hours	10%
	Basic Concepts: Culture, Globalization and Cross-cultural Communication; Social and People Skills; Communicating with People of Different Cultures; Conflicts in Cross-cultural Communication and Tactics / techniques to resolve them; Persuasive Communication		
3.	Communication for Career Building	10 Hours	33%
	Cover Letters and Resume; E-mail and Report; Types of Resume; Concept and Rationale of Group Discussion Skills and Aspects assessed in Group Discussion; Concept and Rationale of Personal Interview; Types of Personal Interview; Writing Statement of Purpose		
4.	Group Dynamics and Soft Skills	05 Hours	17%
	An Introduction to Group Dynamics and Soft Skills; Groups and their Structures; Roles and Functions of Members in Groups; Conflict Management; Aptitude and Attitude; Various Intelligences; Developing an Open Mindset		
5.	Effective Presentation Strategies	04 Hours	14%
	Designing Appealing Presentation; Audience Analysis and Supporting Material; Presentation Mechanics and Presentation Process; Managing Yourself during Q and A Session; Fundamentals of Persuasion		
6.	Contemporary Issues in Communication and Soft Skills	02 Hours	06%
	Trends and Practices in Communication, Case Studies		

Recommended Study Material:

- **Text book:**
    1. Koneru, A. Professional Communication, Tata McGrow Hill Education Private Limited
    2. Disanza, J.R. &Legge, N. Business and Professional Communication, Pearson Education
    3. Raman, M & Singh, P. Business Communication, Oxford University Press

- **Reference book:**
    1. Disanza, J.R. &Legge, N. Business and Professional Communication, Pearson Education
    2. Anandamurugan, A. Placement Interviews – Skills for Success, Tata McGrow Hill Education Private Limited

- **Web material:**
    1. https://www.coursera.org/learn/careerdevelopment
    2. https://www.futurelearn.com/courses/writing-applications
    3. https://www.futurelearn.com/courses/workplace-englis

Course Outcome (COs): At the end of the course, the students will be able to

CO1	Gain thorough understanding and proficiency in various Professional Communication Skills.
CO2	Develop awareness and competence in cross-cultural communication in their personal, academic and professional environments.
CO3	Develop business writing and presentation skills to succeed in career.
CO4	Develop soft skills to stand out and take their career to the next level.
CO5	Develop various intelligences and open Mindset to function in multi-disciplinary and cross-cultural work environment.
CO6	Practice new trends in communication in multiple perspectives at personal, professional, and social level.

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO1	PSO2
CO1	-	-	-	-	-	-	-	-	3	-	-	-	2
CO2	-	-	-	-	-	-	-	-	3	-	-	-	2
CO3	-	-	-	-	-	-	-	-	3	-	-	-	2
CO4	-	-	-	-	-	-	-	-	3	-	-	-	2
CO5	-	-	-	-	-	-	2	-	3	-	2	-	2
Co6	-	-	-	-	-	-	-	-	3	-	-	-	2

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

DETAILED SYLLABUS

FOR

BCA PROGRAMME
(6^{th} SEMESTER)

EFFECTIVE FROM
ACADEMIC YEAR 2023-24

Teaching & Examination Scheme
Bachelor of Computer Applications (B.C.A.) Programme
(Choice Based Credit System)
Effective from Year 2023 - 24
Semester - VI

Course Code	Course Title	Teaching Scheme				Examination Scheme						
		Contact Hours			Credit	Theory			Practical			Total
						Internal		External	Internal		External	
		Theory	Practical	Total		Case Study	Tests		Term work	Tests		
CA330	Basics of Mobile Applications	3	3	6	6	10	20	70	15	15	70	200
CA331	Software Testing	-	3	3	3	-	-	-	15	15	70	100
CA332	Mini Project	-	15	15	15	-	-	-	100		200	300
HS132.02 C	Contributor Personality Development	-	2	2	2	-		-	30		70	100
		03	23	26	26	100			600			700

CA330: Basics of Mobile Applications (200 Marks)

Contact Hours: 06

Pre-requisite: Object Oriented Programming using JAVA

Methodology & Pedagogy: During theory lectures illustrations emphasizing the need for basic features of Mobile Computing and Android- the Mobile Application Development platform will be given. During Practical sessions, students will be required to develop Mobile Application using JAVA programming language in Android. Student shall also develop applications with elegant user interface that deal with data storage, documents sharing among applications.

Outline of the course:

Unit No.	Title of the Unit	Minimum Number of Hours	
		Theory	Practical
1	Introduction Android and Android Development tools	07	36
2	User Interface – I	06	
3	User Interface - II	06	
4	Material Design	05	
5	Application Data Handling	07	
6	Web Services	05	

Total Hours (Theory): 36

Total Hours (Lab): 36

Total Hours: 72

Detail Syllabus:

Unit I: Introduction Android and Android Development tools Hours 07

Introduction to Mobile Applications and Android, Basics of Kotlin, setting up development environment, Dalvik Virtual Machine, Android Application Structure, DDMS, Android Manifest File, Gradle, Android permissions, APK, Basic Building blocks – Activities, Services, Broadcast Receivers & Content providers

Unit II: User Interface – I Hours 06

Working with Containers: LinerLayout, RelativeLayout, TableLayout, ScrollView, and FrameLayout, Fragment, Views: Toast, TextView, EditText, Button, ImageButton, CheckBox, ToggleButton, RadioButton, RadioGroup, ProgressBar, AutoComplete TextView, Components for communication -Intents & Intent Filters Intents: Explicit Intents, Implicit Intents, switching between activities and passing data between activities using Intents, Logging

Unit III: User Interface - II **Hours 06**

Calendar: Manage Date and Time, Adapters: ArrayAdapter, BaseAdapter ListView and ListActivity, GridView, Gallery Menus: Option menu, Context menu, Sub menu Tabs and TabActivity Alert Dialogs, Android Toolbar

Unit IV: Material Design **Hours 05**

Android Design Support Library, Vector Drawables, Basics of material design components and principles, Android styles and themes, Animations and Transitions

Unit V: Application Data Handling **Hours 07**

Basics of Storage in Android, Shared Preference: Save key-value data, Store Data in Database, store data in a firebase, save files on Device, Sharing of data

Unit VI: Web Services **Hours 05**

Working with Web Service in Android, Deploying Web Service

Core Books:

1. Wei-Meng Lee: Beginning Android 4 Application Development, Wiley India Pvt Ltd.
2. Reto Meier: Professional Android 2 Application Development, Wrox publication
3. Mark L. Murphy: The Busy Coder's Guide to Android Development

Reference Books:

1. Dawn Griffiths, David Griffiths: Head First Android Development, O'REILLY publication
2. Mark L. Murphy: Beginning Android 2, APRESS publication
3. Mark L. Murphy: The Busy Coder's Guide to Advanced Android Development, Commons Ware, LLC

Web References:

1. https://kotlinlang.org/ [Kotlin Tutorials]
2. https://developer.android.com/guide/ [Android Tutorials]
3. www.vogella.com/tutorials/android.html [Android Practical Tutorials]
4. https://www.androidhive.info/ [Android Practical Tutorials]

Course Outcomes: Upon successful completion of the course, students will be,

CO1 :	Be acquainted with Android OS
CO2 :	Be aware with Android Development Tools
CO3 :	Be able to create proper look and feel of mobile application
CO4 :	Be capable to create interactive mobile application that handle data
CO5 :	Be able to create mobile application that works web service.

Course Outcomes Mapping:

Week No.	Practical Title	Course Outcomes				
		CO1	CO2	CO3	CO4	CO5
1	Introduction Android and Android Development tools	√				
2	User Interface – I		√			
3	User Interface - II			√		
4	Material Design				√	
5	Application Data Handling				√	
6	Web Services					√

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO1	PSO2
CO1	3	2	2	1	3	-	2	1	-	-	2	-	-
CO2	2	-	3	-	2	1	-	-	2	2	3	2	1
CO3	-	3	-	2	-	1	3	2	-	-	-	3	3
CO4	2	1	2	2	2	1	2	-	2	-	2	2	3
CO5	-	2	2	3	1	-	2	2	2	2	-	-	2

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

CA331: Software Testing (100 Marks)

Contact Hours: 3 Hours

Pre-requisite courses: Object oriented programming concepts.

Methodology & Pedagogy: The practical sessions will cover the software testing concepts and practices that support the production of quality software. The practical sessions will cover the application of testing techniques at various levels of testing using manual and automated testing tools.

Outline of the Course:

Week No	Practical	Description
1- 2	Introduction to testing, Checklist preparation	Student will be able to understand role of testing in real world and he/she can prepare checklist for given problem.
3 - 4	Manual Testing concepts: Understanding of Static testing and Dynamic testing	Student will be familiarize with methods of testing which can be applied by inspecting, reviewing, or preparing walkthroughs.
5 - 6	Creating test case from use case scenarios and test scenarios.	Students will going to learn difference between test case and test scenarios. They can also prepare checklists by using UML use case scenarios.
7 - 8	Hand-on exercise on concepts of levels of testing : Unit Testing with JUnit	Student will be familiarize with different concepts of Junit and perform unit testing with using the same.
9 - 10	Testing WebApp using selenium web driver with JAVA/Python	Student will be familiarize with different concepts of Web driver and perform web application testing with using the same.
11 - 12	Understanding of basics Mobile Application testing using Appium and Demonstration of defect tracking tool such as JIRA/ Cucumber etc.	Student will be able to understating basics of mobile application testing using Appium.

Total Hours (Practical): 36

Total Hours: 36

Core book:

1. Sagar Naik, Piyu Tripathy: Software Testing and Quality Assurance, Theory and Practice, Wiley, 2008.
2. Paul C. Jorgensen : Software Testing: A Craftsman's Approach, 4th Edition by , CRC press, Taylor and Francsis Group,2014
3. Roger S Pressman: Software Engineering – A Practitioner's Approach, 7th Edition, McGRAW HILL International Edition, 2010.

Reference book:

1. Mauro Pezze, Michael Young: Software testing and Analysis- Process, Principles and Techniques, Wiley India, 2012.
2. Boris Beizer: Software Testing Techniques: 2nd Edition, Van Nostrand Reinhold, 1990.
3. Daniel Galin: Software Quality Assurance, Pearson Education,2004.
4. Ron Patton: Software Testing, Pearson Education, 2005.

Web material:

1. http://tryqa.com/what-is-software-testing [Basic of Software Testing]
2. https://www.guru99.com/functional-testing.html [Functional Testing]
3. http://www.softwaretestinggenius.com/download/bgstpadmini.pdf [Software Testing Life Cycle]

Course Outcome (COs): At the end of the course, the students will be able to

CO1	Able to understand role, principles of software testing.
CO2	Familiar with the structural techniques of testing.
CO3	Understand the concepts of creating test cases with respect to levels of testing.
CO4	Learn basics of Unit testing.
CO5	Learn basics of web application and mobile application testing.

Course Outcomes Mapping:

Week No.	Practical Title	Course Outcomes				
		CO1	CO2	CO3	CO4	CO5
1	Introduction to checklist preparation	√				
2	Creating test cases		√			
3	Writing test cases			√		
4	Familiar with Unit testing using Junit				√	
5	Understanding Selenium Web driver					√
6	Testing mobile app using Appium					√

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO1	PSO2
CO1	3	2	3	2	2	3	2	2	3	2	2	2	3
CO2	3	2	2	2	3	2	2	1	1	2	1	1	1
CO3	3	2	2	1	2	3	2	1	1	1	2	1	1
CO4	3	2	1	1	2	2	2	2	1	2	1	2	1
CO5	3	3	2	1	2	1	2	1	2	2	1	2	2
CO6	3	2	3	2	2	3	2	2	3	2	2	2	3

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation "-"

CA332: Mini Project

(300 Marks)

Contact Hours: 15

Objective:

The main objective of this mini project is to let the students apply the programming knowledge to a real-world situation/problem and exposed the students with specific programming skills and help in developing a working model in terms of application.

Course Outcomes:

CO1	Student will understand the implementation of concepts of SDLC and Software Engineering.
CO2	The programming concepts they learn during their academics, it will be converted in to the actual implementations.
CO3	Students will be exposed to understand the requirement of proposed software and implement these requirements in terms of programming logic and methods.
CO4	Students must understand the difference between a program and professional application/product/software.
CO5	Students will learn different categories of applications like Desktop application, Web applications, etc.

Guidelines:

Mini Project is in house project development. Every student is required to carry out Mini Project work under the supervision of a guide provided by the placement Coordinator. The guide shall monitor progress of the student continuously. A candidate is required to present the progress of the Mini Project work during the semester as per the schedule provided by the placement Coordinator.

Mini Project proposal should be prepared in consultation with project guide. It should clearly state the objectives and environment of proposed Mini Project to be undertaken. Project documentation must be with the respect to the project only. Project report should strictly follow the points suggested in format of project report. Placement coordinator will provide the format of project report. Student has to submit one copy of Mini Project to the institute. Each Student is required to make a copy of Mini Project in CD and submit along with Mini Project report.

Evaluation:

The project must be evaluated in two aspects:

a. Internal (100 Marks):
    i. Reporting to internal project guide
    ii. Incorporation of suggestions by project guide
    iii. Internal Project viva examination
b. External (200 Marks):
    i. Project Report Preparation & Evaluation
    ii. External Project Viva Examination

Course Code	Course Title	Teaching Scheme		Internal	End Semester Examination		Total
		Contact Hours	Credit	Continuous Evaluation	Report	Presentation & Viva	
CA332	Mini Project	15	15	100	100	100	300

Web References:

1. http://techwhirl.com/writing-software-requirements-specifications/[For effective SRS]
2. http://www.ibm.com/developerworks/websphere/library/techarticles/0306_perks/perks2.html [For best practices of Software Project Development]
3. http://www.uacg.bg/filebank/acadstaff/userfiles/publ_bg_397_SDP_activities_and_steps.pdf [Requirement analysis guidelines]
4. http://www.cs.wustl.edu/~schmidt/PDF/design-principles4.pdf[Software Design Principles and Guidelines]
5. http://www.cse.hcmut.edu.vn/~hiep/KiemthuPhanmem/Tailieuthamkhao/Effective%20Software%2 0Testing%20-2050%20specific%20ways%20to%20improve%20your%20testing.pdf[ForEffective Software Testing]
6. http://www.cs.uics.edu/~jbell/CourseNotes/OO_SoftwareEngineering/SE_Project_Report_Template .pdf[For guidelines to prepare software project report]

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO1	3	3	3	2	3	3	3	3	3	2	2	2	3	2	-
CO2	3	3	3	2	3	3	3	3	3	2	2	2	3	2	-
CO3	3	3	3	2	3	3	3	3	3	2	2	2	3	2	-
CO4	3	3	3	2	3	3	3	3	3	2	2	2	3	2	-
CO5	3	3	3	2	3	3	3	3	3	2	2	2	3	2	-

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation

HS131.02 C: Contributor Personality Development (100 Marks)

Contact Hours: 02

Credits and Hours:

Teaching Scheme	Theory	Practical	Tutorial	Total	Credit
Hours/week	--	30/15	--	30/15	**02**
Marks	**--**	100	--	100	

Pre-requisite courses:

- Communication and Soft Skills

Outline of the Course:

Sr. No.	Title of the unit	Minimum number of hours
1.	Concept of Personality	06
2.	Soft Skills and Personality Development	08
3.	Developing Contributory Personality	06
4.	Life skills and Personality Development	06
5.	Contemporary Issues in CPD	04
	Total hours (Theory) :	--
	Total hours (Practical) :	30
	Total hours :	30

Detailed Syllabus:

1.	Concept of Personality	06 Hours	20%
	Meaning of Personality, Types of Personality, Factors contributing to Personality, Personality Traits, Personality Profiling		
2.	Soft Skills and Personality Development	08 Hours	26%
	Positive Thinking and Mind Set, Leadership, Assertiveness and Negotiation Skills, Self-Management, Interpersonal Skills, Being a Team Player		
3.	Developing Contributory Personality	06 Hours	20%
	Concept of Contributory Personality, Characteristics of a Contributor, The Contributor's Vision of Success & Career, The Scope of Contribution in a field, Embarking on the Journey to Contributor ship, Developing Contributor Personality, Reviewing Some Contributors Personalities		
4.	Life skills and Personality Development	06 Hours	20%
	Concept of life skills, Self-awareness, Empathy, Decision Making, Problem Solving		
5.	Contemporary Issues in CPD	04 Hours	14%
	Contemporary Trends and Practices in Contributory Personality Development, Case Study & Presentations		

Recommended Study Material:

❖ **Text book:**

1. Personality Development & Soft Skills, Oxford University Press
2. Soft Skills, Bookboon
3. Personality Development, Swami Vivekananda; Advaita Ashrama

❖ **Reference book:**

1. Contributor Personality Program Workbook (Volume 1,2),
2. Contributor Personality Program ActivGuide, Illumine Knowledge Pvt. Ltd

❖ **Web material:**

1. https://www.coursera.org/learn/wharton-succcess
2. https://www.coursera.org/learn/personality-types-at-work
3. https://www.coursera.org/learn/self-awareness

Course Outcome (COs): At the end of the course, the students will be able to

CO1	Identify one's individual personality strengths and challenges.
CO2	Develop more assertive and optimist attitude towards work and life.
CO3	Develop quintessential soft skills to groom one's personality.
CO4	Identify traits of contributor personality.
CO5	Contribute to self, society, nation, and globe.
CO6	Develop skills of global citizenship to perform societal responsibilities.

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO1	PSO2
CO1	-	-	-	-	-	-	-	-	2	-	-	-	-
CO2	-	-	-	-	-	-	-	-	2	2	-	-	3
CO3	-	-	-	-	-	-	-	-	2	-	-	-	-
CO4	-	-	-	-	-	-	-	-	2	-	-	-	-
CO5	-	-	-	-	-	-	3	-	2	-	-	-	-
Co6	-	-	-	-	-	-	-	-	2	-	3	-	-

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation "-"

DETAILED SYLLABUS

FOR

BCA PROGRAMME
(7^{th} SEMESTER)

EFFECTIVE FROM
ACADEMIC YEAR 2023-24

Teaching & Examination Scheme
Bachelor of Computer Applications (B.C.A.) Programme
(Choice Based Credit System)
Effective from Year 2023 - 24
Semester - VII

Course Code	Course Title	Teaching Scheme				Examination Scheme						
		Contact Hours			Credit	Theory			Practical			Total
						Internal		External	Internal		External	
		Theory	Practical	Total		Case Study	Tests		Term work	Tests		
CA333	Online MOOC Course on Current Trends of Information Technology #	-	-	-	03	-	-	-	-		-	-
CA334	Minor Project	-	24	24	24	-	-	-	200		400	600
		-	24	24	27	100			600			700

Students can select any MOOC course of their choice related to current trends of Information Technology with at least 30 hours. 03 credits will be transferred in the result of students based on this course.

CA334: Minor Project

(600 Marks)

Contact Hours: 24

Objective:

The main objective of this Minor project is to let the students apply the programming knowledge to a real-world situation/problem and exposed the students with specific programming skills and help in developing a working model in terms of application.

Course Outcomes:

CO1	Student will understand the implementation of concepts of SDLC and Software Engineering.
CO2	The programming concepts they learn during their academics, it will be converted in to the actual implementations.
CO3	Students will be exposed to understand the requirement of proposed software and implement these requirements in terms of programming logic and methods.
CO4	Students must understand the difference between a program and professional application/product/software.
CO5	Students will learn different categories of applications like Desktop application, Web applications, etc.

Guidelines:

Minor Project is in house project development. Every student is required to carry out Minor Project work under the supervision of a guide provided by the placement Coordinator. The guide shall monitor progress of the student continuously. A candidate is required to present the progress of the Minor Project work during the semester as per the schedule provided by the placement Coordinator.

Minor Project proposal should be prepared in consultation with project guide. It should clearly state the objectives and environment of proposed Minor Project to be undertaken. Project documentation must be with the respect to the project only. Project report should strictly follow the points suggested in format of project report. Placement coordinator will provide the format of project report. Student has to submit one copy of Minor Project to the institute. Each Student is required to make a copy of Minor Project in CD and submit along with Minor Project report.

Evaluation:

The project must be evaluated in two aspects:

a. Internal (200 Marks):
    i. Reporting to internal project guide
    ii. Incorporation of suggestions by project guide
    iii. Internal Project viva examination
b. External (400 Marks)
    i. Project Report Preparation & Evaluation
    ii. External Project Viva Examination

Course Code	Course Title	Teaching Scheme		Internal	End Semester Examination		Total
		Contact Hours	Credit	Continuous Evaluation	Report	Presentation & Viva	
CA334	Minor Project	24	24	200	200	200	600

Web References:

1. http://techwhirl.com/writing-software-requirements-specifications/[For effective SRS]
2. http://www.ibm.com/developerworks/websphere/library/techarticles/0306_perks/perks2.html [For best practices of Software Project Development]
3. http://www.uacg.bg/filebank/acadstaff/userfiles/publ_bg_397_SDP_activities_and_steps.pdf [Requirement analysis guidelines]
4. http://www.cs.wustl.edu/~schmidt/PDF/design-principles4.pdf[Software Design Principles and Guidelines]
5. http://www.cse.hcmut.edu.vn/~hiep/KiemthuPhanmem/Tailieuthamkhao/Effective%20Software%2 0Testing%20-2050%20specific%20ways%20to%20improve%20your%20testing.pdf [For Effective

Software Testing]

6. http://www.cs.uics.edu/~jbell/CourseNotes/OO_SoftwareEngineering/SE_Project_Report_Template .pdf [For guidelines to prepare software project report]

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO1	3	3	3	2	3	3	3	3	3	2	2	2	3	2	-
CO2	3	3	3	2	3	3	3	3	3	2	2	2	3	2	-
CO3	3	3	3	2	3	3	3	3	3	2	2	2	3	2	-
CO4	3	3	3	2	3	3	3	3	3	2	2	2	3	2	-
CO5	3	3	3	2	3	3	3	3	3	2	2	2	3	2	-

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation

DETAILED SYLLABUS

FOR

BCA PROGRAMME
(8th SEMESTER)

EFFECTIVE FROM
ACADEMIC YEAR 2023-24

Teaching & Examination Scheme
Bachelor of Computer Applications (B.C.A.) Programme
(Choice Based Credit System)
Effective from Year 2023 - 24
Semester - VIII

Course Code	Course Title	Teaching Scheme				Examination Scheme						
		Contact Hours			Credit	Theory			Practical			Total
						Internal		External	Internal		External	
		Theory	Practical	Total		Case Study	Tests		Term work	Tests		
CA335	Dissertation	-	24	24	24	-	-	-	200		400	600
		-	24	24	24	-			600			600

CA335: Dissertation

(100 Marks)

Contact Hours: 24 Hours

Objective: To allow students to undertake a substantial in-depth study of a specific topic in Computer Science

Guidelines:

- Decide the topic of dissertation with consultation of your supervisor.
- It is mandatory to report to supervisor two times in month.
- Decide the content and flow of dissertation with the consultation of your supervisor
- Before submission of dissertation report, it is expected from student to publish/present at least one research paper in National/International conference/journal of repute.
- Submit two spiral binding copies of your dissertation duly signed by your supervisor and Head of the department.
- The dissertation evaluation criteria are as follows:
    - Research topic and familiarity with the theory
    - Literature Survey
    - Research Methodologies used
    - Discussion and Conclusions
    - Structure and Presentation

Web material:

1. https://dominikrys.com/posts/cs-dissertation-tips/ [Dissertation Tips]
2. https://researchguides.njit.edu/compsci/litreview [Literature Review Guidelines]
3. https://www.ic.unicamp.br/~wainer/cursos/2s2006/epistemico/ramesh.pdf[Research in Computer Science]
4. https://www.indeed.com/career-advice/career-development/how-to-write-a-conclusion-for-a-research-paper [How to write conclusions]

Course Outcome (COs): At the end of the course, the students will be able to

CO1	Able to identify and investigate a research problem in computer science
CO2	Able to apply an appropriate research design and associated methods rigorously
CO3	Able to draw appropriate conclusions and indicate the significance of findings
CO4	Able to report the research in scholarly fashion.

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO1	PSO2
CO1	3	3	1	1	3	3	2	2	2	3	2	3	3
CO2	3	3	3	2	3	3	2	2	2	3	2	3	3
CO3	2	3	1	3	3	3	2	2	2	3	2	3	3
CO4	-	1	-	2	3	3	2	2	2	3	2	3	3

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”